



UNIVERSITEIT VAN AMSTERDAM

Queueing Theory

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INTRODUCTION. Waiting is inherently related to many situations that we encounter in practice. Some examples of such practical situations are production systems, transportation and stocking systems, communication systems, and health care systems. Queueing models are beneficial for the design of these systems in terms of layout, capacities and control. In this course, we focus on the mathematical analysis of several elementary queueing models. Specifically, as waiting primarily occurs due to randomness in the arrival and service processes of customers, queueing theory is embedded in the field of applied probability. We focus mainly on methods for analysing queueing models and on their applications. A principal reference for the course is *Queueing Systems* by Adan and Resing [1].

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Basic Concepts from Probability Theory

SECTION 1.1

| Random Variables

Random variables are denoted by capitals, X, Y , etc. The *expected value* or *mean* of X is denoted by $\mathbb{E}[X]$ and its *variance* by

$$\sigma^2(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2,$$

where $\sigma(X)$ is the *standard deviation* of X , and $\mathbb{E}[X^2]$ denotes the *second moment*.

An important quantity is the *coefficient of variation* of the positive random variable X defined as

$$c_X = \frac{\sigma(X)}{\mathbb{E}[X]},$$

which is a (dimensionless) measure of the variability of the random variable X .

- Discrete random variables (e.g., number of customers): *probability mass function* (PMF) $\mathbb{P}(X = n) = p(n)$ for $n = 0, 1, 2, \dots$, and

$$\mathbb{E}[X] = \sum_{n=0}^{\infty} np(n).$$

- Continuous random variables (e.g., waiting time of customers): *cumulative distribution function* (CDF) $F(x) = \mathbb{P}(X \leq x)$, *probability density function* (PDF) $f(x) = F'(x)$, and

$$\mathbb{E}[X] = \int_{x=0}^{\infty} x dF(x), \quad \mathbb{E}[X^2] = \int_{x=0}^{\infty} x^2 dF(x).$$

SECTION 1.2

| Generating Function

Let X be a non-negative discrete random variable, then the *generating function* is given by

$$P_X(z) = \mathbb{E}[z^X] = \sum_{n=0}^{\infty} p_n z^n.$$

Note that $|P_X(z)| \leq 1$ for all $|z| \leq 1$.

PROPOSITION 1.1. (i) $P_X(0) = p(0) = p_0$, and $P_X(1) = 1$.

(ii)

$$P'_X(z) = \sum_{n=0}^{\infty} n p_n z^{n-1}, \quad P'_X(1) = \sum_{n=0}^{\infty} n p_n = \mathbb{E}[X],$$

and, more generally,

$$P_X^{(k)}(1) = \mathbb{E}[X(X-1) \cdots (X-k+1)].$$

(iii) For the generating function of the sum $Z = X + Y$ of two independent discrete random variables X and Y , it holds that

$$P_Z(z) = \mathbb{E}[z^Z] = \mathbb{E}[z^{X+Y}] = \mathbb{E}[z^X] \mathbb{E}[z^Y] = P_X(z) P_Y(z).$$

When Z equals X with probability q and Y with probability $1 - q$, in other words,

$$Z = IX + (1 - I)Y, \quad \text{where } I = \begin{cases} 1, & q \\ 0, & 1 - q \end{cases},$$

then

$$\begin{aligned} P_Z(z) &= \mathbb{E}[z^{IX+(1-I)Y}] \\ &= \mathbb{E}[z^{IX+(1-I)Y} \mid I = 1] \mathbb{P}(I = 1) + \mathbb{E}[z^{IX+(1-I)Y} \mid I = 0] \mathbb{P}(I = 0) \\ &= q \mathbb{E}[z^X] + (1 - q) \mathbb{E}[z^Y] = q P_X(z) + (1 - q) P_Y(z). \end{aligned}$$

SECTION 1.3

| Laplace-Stieltjes Transform

The *Laplace-Stieltjes transform* (LST) $\tilde{X}(s)$ of a non-negative random variable X with distribution function $F(\cdot)$ and density $f(\cdot)$, is defined as

$$\tilde{X} = \mathbb{E}[e^{-sX}] = \int_{x=0}^{\infty} e^{-sx} dF(x) = \int_{x=0}^{\infty} e^{-sx} f(x) dx, \quad s \geq 0.$$

PROPOSITION 1.2. (i) $\tilde{X}(0) = 1$.

(ii)

$$\tilde{X}'(s) = - \int_{x=0}^{\infty} x e^{-sx} f(x) dx, \quad \tilde{X}'(0) = - \int_{x=0}^{\infty} x f(x) dx = -\mathbb{E}[X],$$

and, more generally,

$$\tilde{X}^{(k)}(0) = (-1)^k \mathbb{E}[X^k].$$

(iii) For the transform of the sum $Z = X + Y$ of two independent random variables X and Y , it holds that

$$\tilde{Z}(s) = \tilde{X}(s)\tilde{Y}(s).$$

When $Z = IX + (1 - I)Y$, where $I = \begin{cases} 1, & q \\ 0, & 1 - q \end{cases}$, then

$$\tilde{Z}(s) = q\tilde{X}(s) + (1 - q)\tilde{Y}(s).$$

EXAMPLE 1.3. Let N be a discrete random variable with generating function $P_N(z)$, let X_i be *i.i.d.* random variables with LST $\tilde{X}(s)$, and let $Y = \sum_{i=1}^N X_i$ be a random sum with LST $\tilde{Y}(s)$. Then

$$\begin{aligned} \tilde{Y}(s) &= \sum_{n=0}^{\infty} \mathbb{P}(N = n) \mathbb{E}[e^{-sY} \mid N = n] \\ &= \sum_{n=0}^{\infty} \mathbb{P}(N = n) \mathbb{E}\left[e^{-s \sum_{i=1}^n X_i}\right] \\ &= \sum_{n=0}^{\infty} \mathbb{P}(N = n) \left(\tilde{X}(s)\right)^n = P_N(\tilde{X}(s)). \end{aligned}$$

Waiting times in queueing models are random sums of random variables.

SECTION 1.4

| Useful Probability Distributions

Geometric Distribution

A *geometric* random variable X with parameter p has PMF

$$\mathbb{P}(X = n) = (1 - p)p^n, \quad n = 0, 1, 2, \dots$$

For this distribution we have

$$P_X(z) = \sum_{n=0}^{\infty} (1 - p)p^n z^n = (1 - p) \sum_{n=0}^{\infty} (pz)^n = \frac{1 - p}{1 - pz}.$$

Recall here that

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1 - a}, \quad |a| < 1.$$

Poisson Distribution

A *Poisson* random variable X with parameter μ has PMF

$$\mathbb{P}(X = n) = \frac{\mu^n}{n!} e^{-\mu}, \quad n = 0, 1, 2, \dots$$

For this distribution we have

$$P_X(z) = \sum_{n=0}^{\infty} \frac{\mu^n}{n!} e^{-\mu} z^n = e^{-\mu} \sum_{n=0}^{\infty} \frac{(\mu z)^n}{n!} = e^{-\mu} e^{\mu z} = e^{-\mu(1-z)}.$$

Recall here that

$$\sum_{n=0}^{\infty} \frac{b^n}{n!} = e^b.$$

Exponential Distribution

An exponential random variable X with parameter μ has PDF and CDF

$$f(t) = \mu e^{-\mu t}, \quad F(t) = 1 - e^{-\mu t}, \quad t \geq 0.$$

For this distribution we have

$$\tilde{X}(s) = \int_{t=0}^{\infty} e^{-st} \mu e^{-\mu t} dt = \mu \int_{t=0}^{\infty} e^{-(\mu+s)t} dt = \frac{\mu}{\mu + s}.$$

PROPOSITION 1.4. (i) The memoryless property: for all $x \geq 0$ and $t \geq 0$,

$$\mathbb{P}(X > x + t \mid x > t) = \frac{\mathbb{P}(X > x + t, x > t)}{\mathbb{P}(x > t)} = \frac{e^{-\mu(t+x)}}{e^{-\mu t}} = e^{-\mu x} = \mathbb{P}(X > x).$$

(ii)

$$\mathbb{E}[X] = \frac{1}{\mu}, \quad \mathbb{E}[X^2] = \frac{2}{\mu^2}, \quad \sigma_X^2 = \frac{1}{\mu^2}, \quad c_X = 1.$$

- (iii) If $X_1, \dots, X_n$ are independent exponential random variables with parameters $\mu_1, \dots, \mu_n$ respectively, then $\min(X_1, \dots, X_n)$ is again an exponential random variable with parameter $\mu_1, \dots, \mu_n$ and the probability that X_i is the smallest one is given by $\frac{\mu_i}{\mu_1 + \dots + \mu_n}$, for $i = 1, \dots, n$.

Exponential random variables are often used to model *interarrival times* in queueing models.

Erlang Distribution

An *Erlang* random variable X with parameters k and μ has CDF

$$F(t) = 1 - \sum_{j=0}^{k-1} \frac{(\mu t)^j}{j!} e^{-\mu t}, \quad t \geq 0,$$

where X is the sum of k independent random variables $X_1, \dots, X_k$ having a common exponential distribution with mean μ^{-1} . For this distribution we have

$$\tilde{X}(s) = \left(\frac{\mu}{\mu + s} \right)^k, \quad \mathbb{E}[X] = \frac{k}{\mu}, \quad \sigma^2(X) = \frac{k}{\mu^2}, \quad c_X^2 = \frac{1}{k} \leq 1.$$

Hyperexponential Distribution

A *hyperexponential* random variable X with parameters p_i and μ_i for $i = 1, \dots, k$ (notation $H_k(p_1, \dots, p_k; \mu_1, \dots, \mu_k)$) has CDF

$$F(t) = \sum_{i=1}^k p_i (1 - e^{-\mu_i t}), \quad t \geq 0,$$

where $X = X_i \sim \exp(\mu_i)$ independent with probability p_i . For this distribution we have

$$\tilde{X}(s) = \sum_{i=1}^k \frac{p_i \mu_i}{\mu_i + s}, \quad \mathbb{E}[X] = \sum_{i=1}^k \frac{p_i}{\mu_i}, \quad c_X \geq 1.$$

Phase-Type Distribution

The exponential distribution, Erlang distribution and hyperexponential distribution are all examples of a so-called *phase-type distribution*. Random variable X has a phase-type distribution if X describes the time until absorption of a Markov process with an absorbing state, which is appealing for numerical methods.

Other examples of phase-type distributions are *Coxian* distributions, notation C_k , and mixtures of Erlang distributions with the same scale parameters (notation $E_{k-1,k}$ if we take a mixture of an E_{k-1} and an E_k random variable).

SECTION 1.5

| Poisson Process

Let $N(t)$ be the number of arrivals in $[0, t]$ for a *Poisson process* with rate λ , i.e., the time between successive arrivals is exponentially distributed with parameter λ and independent of the past. Then $N(t)$ has a *Poisson distribution* with parameter λt , so

$$\mathbb{P}(N(t) = k) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}, \quad k = 0, 1, 2, \dots,$$

where $\mathbb{E}[N(t)] = \lambda t$. Let $Y_1, Y_2, \dots \sim \exp(\lambda)$, independent. Then

$$\begin{aligned} \mathbb{P}(N(t) \geq k) &= \mathbb{P}(Y_1 + \dots + Y_k \leq t) \\ &= \mathbb{P}(\text{Erlang}(k, \lambda) \leq t) \\ &= 1 - \mathbb{P}(N(t) \leq k-1) = 1 - \sum_{j=0}^{k-1} \frac{(\lambda t)^j}{j!} e^{-\lambda t}. \end{aligned}$$

PROPOSITION 1.5. (i) *Merging (Superposition):* Suppose that $N_1(t)$ and $N_2(t)$ are two independent Poisson processes with respective rates λ_1 and λ_2 . Then the sum $N_1(t) + N_2(t)$ of the two processes is again a Poisson process with rate $\lambda_1 + \lambda_2$.

(ii) *Splitting:* Suppose that $N(t)$ is a Poisson process with rate λ and that each arrival is marked with probability p independent of all other arrivals. Let $N_1(t)$ and $N_2(t)$ denote respectively the number of marked and unmarked arrivals in $[0, t]$. Then $N_1(t)$ and $N_2(t)$ are both Poisson processes with respective rates λp and $\lambda(1-p)$. These two processes are independent.

Proof.

□

| Exercises

EXERCISE 1.1. Let $X_1, \dots, X_n$ be independent exponential random variables with mean $\mathbb{E}[X_i] = \frac{1}{\mu_i}$, $i = 1, \dots, n$. Define

$$Y_n = \min(X_1, \dots, X_n), \quad Z_n = \max(X_1, \dots, X_n).$$

- (i) Determine the distributions of Y_n and Z_n .
- (ii) Show that the probability that X_i is the smallest one among $X_1, \dots, X_n$ is equal to $\frac{\mu_i}{\mu_1 + \dots + \mu_n}$, $i = 1, \dots, n$.

EXERCISE 1.2. Let $X_1, X_2, \dots$ be independent exponential random variables with mean $\frac{1}{\mu}$ and let N be a discrete random variable with

$$\mathbb{P}(N = k) = (1-p)p^{k-1}, \quad k = 1, 2, \dots,$$

where $0 \leq p < 1$ (i.e., N is a shifted geometric random variable). Show that S defined as

$$S = \sum_{n=1}^N X_n$$

is again exponentially distributed with parameter $(1 - p)\mu$.

EXERCISE 1.3. Show that the coefficient of variation of a hyperexponential distribution is greater than or equal to 1.

EXERCISE 1.4 (Poisson Process). Suppose that arrivals occur at $T_1, T_2, \dots$. The interarrival times $A_n = T_n - T_{n-1}$ are independent and have common exponential distribution with mean $\frac{1}{\lambda}$, where $T_0 = 0$ by convention. Let $N(t)$ denote the number of arrivals in $[0, t]$ and define for $n = 0, 1, 2, \dots$,

$$p_n(t) = \mathbb{P}(N(t) = n), \quad t > 0.$$

(i) Determine $p_0(t)$.

(ii) Show that for $n = 1, 2, \dots$

$$p'_n(t) = -\lambda p_n(t) + \lambda p_{n-1}(t), \quad t > 0,$$

with initial condition $p_n(0) = 0$.

(iii) Solve the preceding differential equations for $n = 1, 2, \dots$

EXERCISE 1.5 (Poisson Process). Suppose that arrivals occur at $T_1, T_2, \dots$. The interarrival times $A_n = T_n - T_{n-1}$ are independent and have common exponential distribution with mean $\frac{1}{\lambda}$, where $T_0 = 0$ by convention. Let $N(t)$ denote the number of arrivals in $[0, t]$ and define for $n = 0, 1, 2, \dots$,

$$p_n(t) = \mathbb{P}(N(t) = n), \quad t > 0.$$

(i) Determine $p_0(t)$.

(ii) Show that for $n = 1, 2, \dots$

$$p_n(t) = \int_0^t p_{n-1}(t-x)\lambda e^{-\lambda x} dx, \quad t > 0.$$

(iii) Solve the preceding integral equations for $n = 1, 2, \dots$

EXERCISE 1.6 (Fitting a Distribution). Suppose that processing a job on a certain machine takes 4 minutes on average with a standard deviation of 3 minutes. Show that if we model the processing time as a mixture of an Erlang-1 (exponential) distribution and an Erlang-2 distribution with density

$$f(t) = p\mu e^{-\mu t} + (1-p)\mu^2 t e^{-\mu t},$$

the parameters p and μ can be chosen in such a way that this distribution matches the mean and standard deviation of the processing times on the machine.

EXERCISE 1.7. Consider a random variable X with a Coxian-2 distribution with parameters μ_1 and μ_2 and branching probability p_1 .

(i) Show that $c_X^2 \geq 0.5$.

(ii) Show that if $\mu_1 < \mu_2$, then this Coxian-2 distribution is identical to the Coxian-2 distribution with parameters $\hat{\mu}_1$, $\hat{\mu}_2$ and $\hat{p}_1$ where

$$\hat{\mu}_1 = \mu_2, \quad \hat{\mu}_2 = \mu_1, \quad \hat{p}_1 = 1 - (1 - p_1) \frac{\mu_1}{\mu_2}.$$

Part (ii) implies that for any Coxian-2 distribution we may assume without loss of generality that $\mu_1 \geq \mu_2$.

EXERCISE 1.8. Let X and Y be exponentials with parameters μ and λ , respectively. Suppose that $\lambda < \mu$. Let Z be equal to X with probability $\frac{\lambda}{\mu}$ and equal to $X + Y$ with probability $1 - \frac{\lambda}{\mu}$. Show that Z is an exponential with parameter λ .

EXERCISE 1.9. Consider a H_2 distribution with parameters $\mu_1 > \mu_2$ and branching probabilities q_1 and q_2 , respectively. Show that the C_2 distribution with parameters μ_1 and μ_2 and branching probability p_1 given by

$$p_1 = 1 - \frac{(q_1\mu_1 + q_2\mu_2)}{\mu_1},$$

is equivalent to the H_2 distribution.

EXERCISE 1.10 (Poisson Distribution). Let $X_1, \dots, X_n$ be independent Poisson random variables with means $\mu_1, \dots, \mu_n$, respectively. Show that the sum $X_1 + \dots + X_n$ is Poisson distributed with mean $\mu_1 + \dots + \mu_n$.

Queueing Models and Fundamental Relations

SECTION 2.1

| Queueing Models and Kendall's Notation

A queueing model is characterised by: arrival process of customers; service times of customers; service capacity; waiting room; service discipline; behaviour of customers.

Kendall's notation for queueing models: $\cdot/\cdot/\cdot$

- First position: interarrival time of customers.
- Second position: service time of customers.
- Third position: number of servers/service capacity.
- Optional Fourth position: waiting room/buffer size; default is ∞ .
- Optional: service discipline; default is *FCFS*: First-Come First-Served.

First two positions can, for example, be *M*: memoryless (exponential); *G*: general; *D*: deterministic; *E_r*: Erlang; *H*: hyperexponential.

SECTION 2.2

| Occupation Rate

Consider a single-server queue with *infinite* capacity of the waiting room. Let the arrival rate be $\lambda = \frac{1}{\mathbb{E}[A]}$ and the mean service time be $\mathbb{E}[B]$. Then the system is *stable*, i.e., the queue length does not go to infinity in the long run, if we have

$$\rho = \lambda \mathbb{E}[B] = \frac{\mathbb{E}[B]}{\mathbb{E}[A]} < 1.$$

Here ρ is called *occupation rate* or *server utilisation*, because it is the fraction of time the server is working. Queueing systems with *finite* capacity of the waiting room are always stable.

SECTION 2.3

| Performance Measures

Viewpoint customer:

- W_n : waiting time of the n -th customer in the system;
- S_n : sojourn time of the n -th customer in the system, $S_n = W_n + B_n$, where B_n denotes the service time.

Viewpoint system:

- $L^q(t)$: number of customers waiting at time t ;
- $L(t)$: total number of customers in the system at time t , $L(t) = L^q(t) + L^s(t)$.

Viewpoint server:

- ρ : occupation rate.
- $V(t)$: amount of work in the system at time t ;
- BP : duration of the busy period of the server.

SECTION 2.4

| Little's Law

Little's law gives a very important relation between $\mathbb{E}[L]$, the mean number of customers in the system, $\mathbb{E}[S]$, the mean sojourn time, and λ , the average number of customers entering the system per unit time. *Little's law* states that

$$\mathbb{E}[L] = \lambda \mathbb{E}[S].$$

Can be seen as *system perspective*. This differs from—and holds more generally than—*customer perspective*. For Poisson arrivals and exponential service times

$$\mathbb{E}[S] = \frac{\mathbb{E}[L]}{\mu} + \frac{1}{\mu},$$

combined with

$$\mathbb{E}[S] = \rho \mathbb{E}[S] + \frac{1}{\mu},$$

we have

$$\mathbb{E}[S] = \frac{1}{\mu(1 - \rho)}.$$

Proof.

$$\mathbb{E}[L] = \frac{1}{t} \int_{x=0}^t L(x) dx \approx \frac{1}{t} \sum_{i=1}^{N(t)} S_i = \frac{N(t)}{t} \cdot \frac{1}{N(t)} \sum_{i=1}^{N(t)} S_i = \lambda \mathbb{E}[S],$$

where $N(t)$ denotes the number of arrivals in $(0, t)$. For $t \rightarrow \infty$, this leads to $\mathbb{E}[L] = \lambda \mathbb{E}[S]$. □

EXAMPLE 2.1. Suppose that each customer in the system pays £1 per time unit. There are two options to do this.

- (i) Let customers pay continuously in time. Average reward earned by system is $\mathbb{E}[L]$ per time unit.
- (ii) Let customers pay £1 per time unit for their residence when they leave. The average number of customers leaving per time unit is λ ; hence, $\lambda \mathbb{E}[S]$ per time unit.

In *Little's law* the word *system* can be interpreted in different ways. It can be applied to:

- The queue and those in service $\mathbb{E}[L] = \lambda \mathbb{E}[S]$.
- The waiting room $\mathbb{E}[L^q] = \lambda \mathbb{E}[W]$.
- The server $\mathbb{E}[L^s] = \lambda \mathbb{E}[B] = \rho$ (in the single-server case $\mathbb{E}[L^s] = \rho \cdot 1 + (1 - \rho) \cdot 0$).

Remark. *Little's law* requires caution in models with finite capacity of the waiting room, where customers might get blocked/lost.

SECTION 2.5

| PASTA Property

PROPOSITION 2.2 (Poisson Arrivals See Time Averages (PASTA)).

Distribution of system state at Poisson arrival epochs and arbitrary epochs is the same.

EXAMPLE 2.3. If we don't have Poisson arrivals, consider $G/G/1$ queue with $A \in [3, 4]$, $B \in [1, 2]$. In this case, an arriving customer will always find no customers.

$$L^a = 0, \quad L = \begin{cases} 0, & 1 - \rho \\ 1, & \rho \end{cases}.$$

| Exercises

EXERCISE 2.1. In a gas station there is one gas pump. Cars arrive at the gas station according to a Poisson process. The arrival rate is 20 cars per hour. An arriving car finding n cars at the station immediately leaves with probability $q_n = \frac{n}{4}$, and joins the queue with probability $1 - q_n$, $n = 0, 1, 2, 3, 4$. Cars are served in order of arrival. The service time (i.e., the time needed for pumping and paying) is exponential. The mean service time is 3 minutes.

- (i) Determine the stationary distribution of the number of cars at the gas station.
- (ii) Determine the mean number of cars at the gas station.
- (iii) Determine the mean sojourn time (waiting time plus service time) of cars deciding to take gas at the station.
- (iv) Determine the mean sojourn time and the mean waiting time of all cars arriving at the gas station.

SOLUTION. As time unit we take 1 minute, then the arrival rate $\lambda = \frac{20}{60} = \frac{1}{3}$, and the service rate $\mu = \frac{1}{3}$.

- (i) Solve the (global) balance equations

$$\lambda(1 - q_n)p_n = \mu p_{n+1}, \quad n = 0, 1, 2, 3,$$

where $\lambda = \mu = \frac{1}{3}$, together with the normalisation equation $\sum_{n=0}^4 p_n = 1$. This gives

$$p_0 = \frac{32}{103}, \quad p_1 = \frac{32}{103}, \quad p_2 = \frac{24}{103}, \quad p_3 = \frac{12}{103}, \quad p_4 = \frac{3}{103}.$$

- (ii)

$$\mathbb{E}[L] = \sum_{n=0}^4 np_n = \frac{128}{103} \approx 1.24.$$

- (iii) Some of the cars leave immediately, so we cannot apply *Little's law* using the arrival rate λ , and we need to compute the effective arrival rate for the cars taking gas:

$$\lambda' = \sum_{n=0}^4 \lambda(1 - q_n)p_n = \frac{71}{103}\lambda.$$

Now we can apply *Little's law*:

$$\mathbb{E}[S] = \frac{\mathbb{E}[L]}{\lambda'} = \frac{384}{71} \approx 5.41.$$

- (iv) The cars taking no gas don't wait, so when using *Little's law* with the overall arrival rate λ we compute the average response time of all cars arriving at the station:

$$\mathbb{E}[S] = \frac{\mathbb{E}[L]}{\lambda} = \frac{384}{103} \approx 3.73.$$

And the mean waiting time is derived from the mean sojourn time by subtracting the mean service time:

$$\mathbb{E}[W] = \frac{171}{103} \approx 1.66.$$

□

$M/M/1$ Queue

In this chapter we will analyse the model with exponential interarrival times with mean λ^{-1} , exponential service times with mean μ^{-1} and a single server. Customers are served in order of arrival. We require the *stability condition* that

$$\rho = \frac{\lambda}{\mu} < 1.$$

SECTION 3.1

| Time-Dependent Behaviour

Let $p_n(t) = \mathbb{P}(L(t) = n)$ denote the probability that at time t there are n customers in the system, $n = 0, 1, \dots$. The resulting Markov process (a birth-death process) is quite complicated, and its analysis is not directly relevant here.

SECTION 3.2

| Limiting Behaviour

We will consider the limiting behaviour of the system

$$p_n = \lim_{t \rightarrow \infty} \mathbb{P}(L(t) = n).$$

It follows that the limiting or equilibrium probabilities p_n satisfy the balance equations

$$\begin{aligned} \lambda p_0 &= \mu p_1, \\ (\lambda + \mu)p_n &= \lambda p_{n-1} + \mu p_{n+1}, \quad n = 1, 2, \dots, \\ \sum_{n=0}^{\infty} p_n &= 1. \end{aligned} \quad (\text{normalisation equation})$$

We will show how to do this using *recursion*.¹

$$\lambda p_{n-1} = \mu p_n \implies p_n = \frac{\lambda}{\mu} p_{n-1} = \rho p_{n-1} = \rho^2 p_{n-2} = \dots = \rho^n p_0.$$

The normalisation condition

$$\sum_{n=0}^{\infty} p_n = 1 \iff \sum_{n=0}^{\infty} \rho^n p_0 = 1 \iff p_0 \sum_{n=0}^{\infty} \rho^n = 1 \iff p_0 \frac{1}{1-\rho} = 1 \iff p_0 = 1 - \rho.$$

We conclude

$$p_n = (1 - \rho)\rho^n, \quad n = 0, 1, \dots \sim \text{geo}(\rho).$$

¹ There are several ways to determine the solution of the system of balance equations and the normalisation condition: direct approach, recursion, generating function approach and global balance principle. The other methods are overkill here, but they will be useful in more involved models later.

SECTION 3.3

| Mean Performance Measures

From the equilibrium probabilities we can derive expressions for the mean number of customers in the system and the mean time spent in the system. For the first one we get

$$\mathbb{E}[L] = \sum_{n=0}^{\infty} n \mathbb{P}(L = n) = \sum_{n=0}^{\infty} n p_n = \sum_{n=0}^{\infty} n(1 - \rho) \rho^n = \frac{\rho}{1 - \rho}.$$

By applying *Little's law*, we get the *mean sojourn time*:

$$\mathbb{E}[S] = \frac{1}{\lambda} \mathbb{E}[L] = \frac{1}{\lambda} \cdot \frac{\rho}{1 - \rho} = \frac{1}{\mu(1 - \rho)}.$$

Remark. In fact, $\mathbb{E}[L]$ and $\mathbb{E}[S]$ can also be determined directly, i.e., without knowing the probabilities p_n , by combining *Little's law* and the *PASTA* property:

$$\mathbb{E}[S] = \frac{\mathbb{E}[L]}{\mu} + \frac{1}{\mu} = \frac{\lambda \mathbb{E}[S]}{\mu} + \frac{1}{\mu} = \rho \mathbb{E}[S] + \frac{1}{\mu}.$$

Then

$$(1 - \rho) \mathbb{E}[S] = \frac{1}{\mu} \implies \mathbb{E}[S] = \frac{1}{\mu(1 - \rho)}.$$

This approach is called the *mean value approach*.

The *mean number of customers in the queue*, $\mathbb{E}[L^q]$, can be obtained from $\mathbb{E}[L]$ by subtracting the mean number of customers in service, so

$$\mathbb{E}[L^q] = \mathbb{E}[L] - \mathbb{E}[L^s] = \frac{\rho}{1 - \rho} - \rho = \frac{\rho^2}{1 - \rho}.$$

The *mean waiting time*, $\mathbb{E}[W]$, follows from $\mathbb{E}[S]$ by subtracting the mean service time:

$$\mathbb{E}[W] = \mathbb{E}[S] - \frac{1}{\mu} = \frac{1}{\mu(1 - \rho)} - \frac{1}{\mu} = \frac{\rho}{\mu(1 - \rho)}.$$

Remark. Or by applying *Little's law*:

$$\mathbb{E}[W] = \frac{1}{\lambda} \mathbb{E}[L^q] = \frac{1}{\lambda} \cdot \frac{\rho^2}{1 - \rho} = \frac{\rho}{\mu(1 - \rho)}.$$

SECTION 3.4

| Distribution of the Sojourn Time and the Waiting Time

Denote by L^a the number of customers in the system just before the arrival of a customer and let B_k be the service time of the k -th customer. Of course, the customer in service has a residual service time instead of an ordinary service time. But these are the same, since the exponential service time distribution is memoryless. So the random variables B_k are independent and exponentially distributed with mean $\frac{1}{\mu}$. Then we have

$$S = \sum_{k=1}^{L^a+1} B_k = \sum_{k=1}^{L+1} B_k,$$

where $\sum_{k=1}^{n+1} B_k \sim \text{Erlang}(n+1, \mu)$. By conditioning on L^a and using that L^a and B_k are independent it follows that

$$\mathbb{P}\left(\sum_{k=1}^{n+1} B_k > t\right) = e^{-\mu t} \sum_{k=0}^n \frac{(\mu t)^k}{k!}.$$

The problem is to find the probability that an arriving customer finds n customers in the system. PASTA property states that the fraction of arriving customers who find n customers in the system is equal to the fraction of time there are n customers in the system, so

$$\mathbb{P}(L = n) = p_n = (1 - \rho)\rho^n.$$

Then

$$\begin{aligned} \mathbb{P}(S > t) &= \mathbb{P}\left(\sum_{k=1}^{L+1} B_k > t\right) \\ &= \sum_{n=0}^{\infty} \mathbb{P}\left(\sum_{k=1}^{n+1} B_k > t\right) \mathbb{P}(L = n) \\ &= \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{(\mu t)^k}{k!} e^{-\mu t} (1 - \rho) \rho^n \\ &= e^{-\mu t} (1 - \rho) \sum_{k=0}^{\infty} \frac{(\mu t)^k}{k!} \sum_{n=k}^{\infty} \rho^n \\ &= e^{-\mu t} (1 - \rho) \sum_{k=0}^{\infty} \frac{(\mu t)^k}{k!} \cdot \frac{\rho^k}{1 - \rho} \\ &= e^{-\mu t} e^{-\mu t} \sum_{k=0}^{\infty} \frac{(\mu \rho t)^k}{k!} \\ &= e^{-\mu t} e^{\mu \rho t} = e^{-\mu(1-\rho)t}, \quad t \geq 0. \end{aligned}$$

Remark. Or somewhat simpler calculation to obtain LST.

$$\begin{aligned} \tilde{S}(s) &= \mathbb{E}[e^{-sS}] \\ &= \mathbb{E}\left[e^{-s(\sum_{k=1}^{L+1} B_k)}\right] \end{aligned}$$

$$\begin{aligned}
&= \sum_{n=0}^{\infty} \mathbb{E} \left[e^{-s(B_1 + \dots + B_{n+1})} \right] \mathbb{P}(L = n) \\
&= \sum_{n=0}^{\infty} \left(\frac{\mu}{\mu + s} \right)^{n+1} (1 - \rho) \rho^n \\
&= \frac{\mu(1 - \rho)}{\mu + s} \sum_{n=0}^{\infty} \left(\frac{\rho\mu}{\mu + s} \right)^n \\
&= \frac{\mu(1 - \rho)}{\mu + s} \cdot \frac{1}{1 - \frac{\rho\mu}{\mu + s}} \\
&= \frac{\mu(1 - \rho)}{\mu(1 - \rho) + s} \sim \exp(\mu(1 - \rho)).
\end{aligned}$$

To find the distribution of the waiting time $W = \sum_{k=1}^L B_k$, note that $S = W + B$, where the random variable B is the service time. Since W and B are independent, it follows that

$$\tilde{S}(s) = \tilde{W}(s)\tilde{B}(s) = \tilde{W}(s) \frac{\mu}{\mu + s},$$

and thus,

$$\begin{aligned}
\tilde{W}(s) &= \frac{\mu + s}{\mu} \cdot \frac{\mu(1 - \rho)}{\mu(1 - \rho) + s} \\
&= \frac{(\mu + s)(1 - \rho)}{\mu(1 - \rho) + s} \\
&= (1 - \rho) + \rho \frac{\mu(1 - \rho)}{\mu(1 - \rho) + s} = q\tilde{X}(s) + (1 - q)\tilde{Y}(s).
\end{aligned}$$

From the transform of W we conclude that

$$W = \begin{cases} 0, & 1 - \rho \\ e^{\mu(1 - \rho)}, & \rho \end{cases}.$$

Hence

$$\mathbb{P}(W > t) = \rho e^{-\mu(1 - \rho)t}.$$

SECTION 3.5

| Priorities

Type 1 and type 2 customers arrive according to independent Poisson processes with rates λ_1 and λ_2 , respectively. The service times of all customers are exponentially distributed with the same mean $\frac{1}{\mu}$. We assume that

$$\rho = \rho_1 + \rho_2 = \frac{\lambda_1 + \lambda_2}{\mu} < 1,$$

where $\rho_i = \frac{\lambda_i}{\mu}$, i.e., the occupation rate due to type i customers. Type 1 customers are treated with priority over type 2 customers.

Preemptive-Resume Priority

In the *preemptive-resume priority* rule, type 1 customers have absolute priority over type 2 customers. Absolute priority means that when a type 2 customer is in service and a type 1 customer arrives, the type 2 service is interrupted and the server proceeds with the type 1 customer. Once there are no more type 1 customers in the system, the server resumes the service of the type 2 customer at the point where it was interrupted.

Remark. In case of exponential service, by memoryless property, preemptive resume $\sim$ preemptive repeat.

PROPOSITION 3.1. (i) Type-1 customers do not “feel” presence of type-2 customers.

$$\begin{aligned} \mathbb{E}[L_1] &= \frac{\rho_1}{1 - \rho_1}, & \mathbb{E}[S_1] &= \frac{1}{\mu(1 - \rho_1)}. \\ \mathbb{P}(L_1 = n) &= (1 - \rho_1)\rho_1^n, & \mathbb{P}(S_1 > t) &= e^{-\mu(1 - \rho_1)t}. \end{aligned}$$

(ii) Total number of customers is not affected by priorities.

$$\begin{aligned} \mathbb{E}[L_1] + \mathbb{E}[L_2] &= \frac{\rho_1 + \rho_2}{1 - \rho_1 - \rho_2} \\ \implies \mathbb{E}[L_2] &= \frac{\rho_1 + \rho_2}{1 - \rho_1 - \rho_2} - \frac{\rho_1}{1 - \rho_1} = \frac{\rho_2}{(1 - \rho_1)(1 - \rho_1 - \rho_2)}. \end{aligned}$$

By Little’s law, applied to type-2 customers only:

$$\mathbb{E}[S_2] = \frac{\mathbb{E}[L_2]}{\lambda_2} = \frac{1}{\mu(1 - \rho_1)(1 - \rho_1 - \rho_2)}.$$

Non-Preemptive Priority

We now consider the situation that type 1 customers have nearly absolute priority over type 2 jobs. This priority rule is therefore called *non-preemptive*. In this case, high-priority customers do suffer from low-priority customers, because they are not allowed to interrupt low-priority customers. Priorities still do not influence total number of customers, due to identical service time distributions of both types of customers.

We again will use the *mean-value approach* to find mean numbers of customers and mean sojourn times. However, we have to adapt the arrival relation for high-priority customers to

find $\mathbb{E}[L_1]$ and $\mathbb{E}[S_1]$ in this case. For the mean sojourn time of type 1 customers we find

$$\mathbb{E}[S_1] = \frac{\mathbb{E}[L_1]}{\mu} + \frac{\rho_2}{\mu} + \frac{1}{\mu}.$$

Together with *Little's law*, applied to type-1 customers only,

$$\mathbb{E}[L_1] = \lambda \mathbb{E}[S_1],$$

we obtain

$$\mathbb{E}[L_1] = \frac{(1 + \rho_2)\rho_1}{1 - \rho_1}, \quad \mathbb{E}[S_1] = \frac{1 + \rho_2}{\mu(1 - \rho_1)}.$$

We saw that second property still holds

$$\mathbb{E}[L_2] = \frac{\rho_1 + \rho_2}{1 - \rho_1 - \rho_2} - \frac{(1 + \rho_2)\rho_1}{1 - \rho_1} = \frac{(1 - \rho_1(1 - \rho_1 - \rho_2))\rho_2}{(1 - \rho_1)(1 - \rho_1 - \rho_2)}.$$

Finally, *Little's law* applied to type-2 customers only, will again give

$$\mathbb{E}[S_2] = \frac{\mathbb{E}[L_2]}{\lambda_2} = \frac{1 - \rho_1(1 - \rho_1 - \rho_2)}{\mu(1 - \rho_1)(1 - \rho_1 - \rho_2)}.$$

SECTION 3.6

| Production Inventory Model

Model setup: items are produced and stored in an inventory system. Base stock policy: keep producing until safety stock of S items is achieved. Two options when order arrives and stock is empty: backorder, i.e., order is placed in queue, produced items go to oldest backorder (if any); lose sale.

Base-Stock Policy with Backorders

Set of states: number of items on stock minus number of backorders $\{S, S-1, \dots, 1, 0, -1, \dots\}$.

$$p_n = \lim_{t \rightarrow \infty} \mathbb{P}(L(t) = n).$$

For stability, we require $\rho = \frac{\lambda}{\mu} < 1$. Similarity with M/M/1 queue:

$$p_n = (1 - \rho)\rho^{S-n}, \quad n = S, S-1, \dots$$

This distribution can be used to derive equilibrium distribution of number of items on stock, I , and number of backorders, B :

$$\begin{aligned} \mathbb{P}(I = 0) &= \sum_{n=0}^{\infty} p_{-n} = \rho^S, \\ \mathbb{P}(I = n) &= p_n = (1 - \rho)\rho^{S-n}, \quad n = 1, 2, \dots, S, \\ \mathbb{P}(B = 0) &= \sum_{n=0}^S p_n = 1 - \rho^{S+1}, \\ \mathbb{P}(B = n) &= p_{-n} = (1 - \rho)\rho^{S+n}, \quad n = 1, 2, \dots \end{aligned}$$

Each item in stock has holding cost of h units per time unit; each backordered item has penalty cost of b units per time unit. Goal: strike balance between cost of maintaining inventory (increase S), and cost of outstanding orders. Define total average cost per time unit by

$$\gamma_C(S) = h\mathbb{E}[I] + b\mathbb{E}[B].$$

From equilibrium distribution, we obtain

$$\begin{aligned} \mathbb{E}[I] &= \sum_{n=0}^S n\mathbb{P}(I = n) = S - \frac{\rho}{1 - \rho}(1 - \rho^S), \\ \mathbb{E}[B] &= \sum_{n=0}^{\infty} n\mathbb{P}(B = n) = \frac{\rho^{S+1}}{1 - \rho}, \\ \gamma_C(S) &= h\mathbb{E}[I] + b\mathbb{E}[B] = h\left(S - \frac{\rho}{1 - \rho}\right) + (h + b)\frac{\rho^{S+1}}{1 - \rho}. \end{aligned}$$

We want to find the optimal value of the base-stock level S^{opt} , so that the total average cost per unit of time is minimised.

Base-Stock Policy with Lost Sales

No backordering, when $I = 0$ orders are immediately lost. Set of states is finite $\{S, S-1, \dots, 0\}$, and no stability requirement. Similarity with $M/M/1/S$ queue. Using global balance principle

$$\lambda p_{n+1} = \mu p_n \implies p_n = \rho p_{n+1} = \dots = \rho^{S-n} p_S, \quad n = 0, 1, \dots, S,$$

and the normalisation condition

$$\sum_{n=0}^S \rho^{S-n} p_S = 1 \implies p_S = \frac{1 - \rho}{1 - \rho^{S+1}},$$

we conclude

$$p_n = \rho^{S-n} \frac{1 - \rho}{1 - \rho^{S+1}}, \quad n = 0, 1, \dots, S.$$

Mean number of items on stock is

$$\mathbb{E}[I] = \sum_{n=0}^S n p_n = \frac{1 + S}{1 - \rho^{S+1}} - \frac{1}{1 - \rho}.$$

Each item in stock has holding cost of h units per time unit; each lost sale imposes penalty cost of ℓ units. Goal: strike balance between cost of maintaining inventory (increase S), and cost of lost sales. Define total average cost per time unit by

$$\gamma_C(S) = h\mathbb{E}[I] + \ell\lambda p_0.$$

From previously found expressions

$$\gamma_C(S) = h\mathbb{E}[I] + \ell\lambda p_0 = h \left(\frac{1 + S}{1 - \rho^{S+1}} - \frac{1}{1 - \rho} \right) + \ell\lambda p_0,$$

again we want to find the optimal value of the base-stock level S^{opt} , so that the total average cost per unit of time is minimised.

SECTION 3.7

| Busy Period

Due to the memoryless property of the exponential distribution, an *idle period* (IP) is exponentially distributed with mean $\frac{1}{\lambda}$. Then we determine the mean and the distribution of a *busy period* (BP).

It is clear that the mean BP divided by the mean cycle length is equal to the fraction of time the server is working, so

$$\frac{\mathbb{E}[BP]}{\mathbb{E}[BP] + \mathbb{E}[IP]} = \rho, \quad \mathbb{E}[IP] = \frac{1}{\lambda}.$$

Hence,

$$\mathbb{E}[BP] = \underbrace{\frac{1}{\mu}}_{\text{service time first customer}} + \underbrace{\rho \mathbb{E}[BP]}_{\text{work that arrives}} = \frac{1}{\mu(1-\rho)} = \mathbb{E}[S].$$

Consider a more general problem. Let the random variable C_n be the time until the system is empty again (residual BP) if there are now n customers present in the system. Clearly, C_1 is the length of a BP, since a BP starts when the first customer after an idle period arrives and it ends when the system is empty again. Then

$$C_n = \sum_{i=1}^n X_i, \quad X_1, X_2, \dots \stackrel{i.i.d.}{\sim} C_1.$$

The random variables C_n satisfy the following recursion relation:

$$C_n = Y + \begin{cases} C_{n+1}, & \frac{\lambda}{\lambda+\mu} \\ C_{n-1}, & \frac{\mu}{\lambda+\mu} \end{cases},$$

where Y is the time until first event (arrival or departure $\sim \exp(\mu + \lambda)$). From this relation we get for the LST $\tilde{C}_n(s)$ of C_n that

$$\tilde{C}_n(s) = \tilde{Y}(s) \left(\frac{\lambda}{\lambda + \mu} \tilde{C}_{n+1}(s) + \frac{\mu}{\lambda + \mu} \tilde{C}_{n-1}(s) \right),$$

where

$$\tilde{Y}(s) = \frac{\lambda + \mu}{\lambda + \mu + s}.$$

And thus, after rewriting,

$$(\lambda + \mu + s)\tilde{C}_n(s) = \lambda \tilde{C}_{n+1}(s) + \mu \tilde{C}_{n-1}(s).$$

This is a second-order recurrence relation with constant coefficients, solution of the form

$$\tilde{C}_n(s) = c_1 x_1^n(s) + c_2 x_2^n(s),$$

where $x_1(s)$ and $x_2(s)$ are the roots of the quadratic equation

$$(\lambda + \mu + s)x = \lambda x^2 + \mu x^0$$

satisfying $0 < x_1(s) < 1 < x_2(s)$. Solution to this quadratic equation

$$x_1(s), x_2(s) = \frac{\lambda + \mu + s \pm \sqrt{(\lambda + \mu + s)^2 - 4\lambda\mu}}{2\lambda}.$$

Since $0 \leq \tilde{C}_n(s) \leq 1$ it follows that $c_2 = 0$. The coefficient c_1 follows from the fact that $C_0 = 0$ and hence $\tilde{C}_0(s) = 1$, yielding $c_1 = 1$. Hence we obtain

$$\tilde{C}_n(s) = x_1^n(s)$$

and in particular, for the LST $\tilde{BP}(s)$ of the BP, we find

$$\tilde{BP}(s) = \tilde{C}_1(s) = x_1(s) = \frac{\lambda + \mu + s \pm \sqrt{(\lambda + \mu + s)^2 - 4\lambda\mu}}{2\lambda}.$$

| Exercises

EXERCISE 3.1 (Bulk Arrivals). In a work station orders arrive according to a Poisson arrival process with arrival rate λ . An order consists of N independent jobs. The distribution of N is given by

$$\mathbb{P}(N = k) = (1 - p)p^{k-1}$$

with $k = 1, 2, \dots$ and $0 \leq p < 1$. Each job requires an exponentially distributed amount of processing time with mean $\frac{1}{\mu}$.

- (i) Derive the distribution of the total processing time of an order.
- (ii) Determine the distribution of the number of orders in the system.

EXERCISE 3.2 (Variable Production Rate). Consider a work station where jobs arrive according to a Poisson process with arrival rate λ . The jobs have an exponentially distributed service time with mean $\frac{1}{\mu}$. So the service completion rate (the rate at which jobs depart from the system) is equal to μ . If the queue length drops below the threshold Q_L , the service completion rate is lowered to μ_L . If the queue length reaches Q_H , where $Q_H > Q_L$, the service rate is increased to μ_H (L stands for low, H for high). Determine the queue length distribution and the mean time spent in the system.

EXERCISE 3.3. A repair man fixes broken televisions. The repair time is exponentially distributed with a mean of 30 minutes. Broken televisions arrive at his repair shop according to a Poisson stream, on average 10 broken televisions per day (8 hours).

- (i) What is the fraction of time that the repair man has no work to do?
- (ii) How many televisions are, on average, at his repair shop?
- (iii) What is the mean throughput time (waiting time plus repair time) of a television?

SOLUTION. As time unit we take 1 hour. The arrival rate is $\lambda = \frac{5}{4}$, and the service rate is $\mu = 2$. Then the occupation rate $\rho = \frac{5}{8}$.

- (i) The fraction of time that the repair man has no work to do is $1 - \rho = \frac{3}{8}$.
- (ii)

$$\mathbb{E}[L] = \frac{\rho}{1 - \rho} = \frac{5}{3}.$$

- (iii)

$$\mathbb{E}[S] = \frac{1}{\lambda} \mathbb{E}[L] = \frac{4}{3}.$$

□

EXERCISE 3.4. In a gas station there is one gas pump. Cars arrive at the gas station according to a Poisson process. The arrival rate is 20 cars per hour. Cars are served in order of arrival. The service time (i.e. the time needed for pumping and paying) is exponentially distributed. The mean service time is 2 minutes.

- (i) Determine the distribution, mean and variance of the number of cars at the gas station.
- (ii) Determine the distribution of the sojourn time and the waiting time.
- (iii) What is the fraction of cars that has to wait longer than 2 minutes?

An arriving car finding 2 cars at the station immediately leaves.

- (iv) Determine the distribution, mean and variance of the number of cars at the gas station.
- (v) Determine the mean sojourn time and the mean waiting time of all cars (including the ones that immediately leave the gas station).

SOLUTION. As time unit we take 1 minute. The arrival rate is $\lambda = \frac{1}{3}$, and the service rate is $\mu = \frac{1}{2}$. Then the occupation rate $\rho = \frac{2}{3}$.

(i)

$$\mathbb{P}(L = n) = (1 - \rho)\rho^n = \frac{1}{3} \left(\frac{2}{3}\right)^n, \quad n = 0, 1, 2, \dots$$

Then

$$\mathbb{E}[L] = \frac{\rho}{1 - \rho} = 2, \quad \sigma^2(L) =$$

- (ii)
- (iii)
- (iv)
- (v)

□

EXERCISE 3.5. A gas station has two pumps, one for gas and the other for LPG. For each pump customers arrive according to a Poisson process. The average arrival rates are 20 customers per hour for gas and 5 customers per hour for LPG. The service times are exponential. For both pumps the mean service time is 2 minutes.

- (i) Determine the distribution of the number of customers at the gas pump, and at the LPG pump.
- (ii) Determine the distribution of the total number of customers at the gas station.

EXERCISE 3.6. Consider an M/M/1 queue with two types of customers. The mean service time of all customers is 5 minutes. The arrival rate of type 1 customers is 4 customers per hour and for type 2 customers it is 5 customers per hour. Type 1 customers are treated with priority over type 2 customers.

- (i) Determine the mean sojourn time of type 1 and 2 customers under the preemptive-resume priority rule.

- (ii) Determine the mean sojourn time of type 1 and 2 customers under the non-preemptive priority rule.

SOLUTION. The service rate $\mu = \frac{1}{5}$, and the arrival rates $\lambda_1 = \frac{1}{15}$, $\lambda_2 = \frac{1}{12}$. Then the occupation rates $\rho_1 = \frac{1}{3}$, $\rho_2 = \frac{5}{12}$.

(i)

$$\mathbb{E}[S_1] = \frac{1}{\mu(1 - \rho_1)} = 7.5, \quad \mathbb{E}[S_2] = \frac{1}{\mu(1 - \rho_1)(1 - \rho_1 - \rho_2)} = 30.$$

(ii)

$$\mathbb{E}[S_1] = \frac{1 + \rho_2}{\mu(1 - \rho_1)} = 10.625, \quad \mathbb{E}[S_2] = \frac{1 - \rho_1(1 - \rho_1 - \rho_2)}{\mu(1 - \rho_1)(1 - \rho_1 - \rho_2)} = 27.5.$$

□

EXERCISE 3.7. Consider an $M/M/1$ queue with an arrival rate of 60 customers per hour and a mean service time of 45 seconds. A period during which there are 5 or more customers in the system is called crowded; when there are fewer than 5 customers, it is called quiet. What is the mean number of crowded periods per day (8 hours) and how long do they last on average?

EXERCISE 3.8. Consider a machine where jobs arrive according to a Poisson stream with a rate of 20 jobs per hour. The processing times are exponentially distributed with a mean of $\frac{1}{\mu}$ hours. The processing cost is 16μ dollars per hour, and the waiting cost is 20 dollars per order per hour. Determine the processing speed μ minimising the average cost per hour.

- EXERCISE 3.9. (i) Consider the base-stock policy with backorders. What is the expected waiting time for the delivery of an order, given that it can not be fulfilled directly from stock?
- (ii) Consider the base-stock policy with lost sales. What is the probability that two successive orders will be lost?

$M/M/c$ Queue

In this chapter we will analyse the model with exponential interarrival times with mean $\frac{1}{\lambda}$, exponential service times with mean $\frac{1}{\mu}$ and c parallel identical servers. Customers are served in order of arrival. We suppose that the occupation rate per server

$$\rho = \frac{\lambda}{c\mu} < 1.$$

SECTION 4.1

| Equilibrium Probabilities

Let $L(t)$ be Markov process, more specifically birth-death process. Equilibrium probabilities

$$p_n = \lim_{t \rightarrow \infty} \mathbb{P}(L(t) = n)$$

satisfy *balance equations*:

$$\lambda p_{n-1} = \begin{cases} n\mu p_n, & n = 1, 2, \dots, c \\ c\mu p_n, & n = c, c+1, \dots \end{cases} = \min(n, c)\mu p_n, \quad n = 1, 2, \dots$$

Iterating gives

$$\begin{aligned} p_n &= \frac{\lambda}{n\mu} p_{n-1} \\ &= \frac{\lambda}{n\mu} \cdot \frac{\lambda}{(n-1)\mu} p_{n-2} \\ &= \dots = \frac{\lambda^n}{n!\mu^n} p_0 = \frac{(c\rho)^n}{n!} p_0, \quad n \leq c, \end{aligned}$$

and

$$\begin{aligned} p_n &= \frac{\lambda}{c\mu} p_{n-1} \\ &= \frac{\lambda}{c\mu} \cdot \frac{\lambda}{c\mu} p_{n-2} \\ &= \left(\frac{\lambda}{c\mu} \right)^{n-c} p_c \\ &= \rho^{n-c} p_c = \rho^{n-c} \frac{(c\rho)^c}{c!} p_0 = \frac{\rho^n c^c}{c!} p_0, \quad n \geq c. \end{aligned}$$

Probability p_0 follows from *normalisation condition*, yielding

$$\sum_{n=0}^{\infty} p_n = \sum_{n=0}^{c-1} \frac{(c\rho)^n}{n!} p_0 + \sum_{n=c}^{\infty} \rho^{n-c} \frac{(c\rho)^c}{c!} p_0 = 1.$$

Then

$$p_0 = \left(\sum_{n=0}^{c-1} \frac{(c\rho)^n}{n!} + \frac{(c\rho)^c}{c!} \sum_{n=c}^{\infty} \rho^{n-c} \right)^{-1} = \left(\sum_{n=0}^{c-1} \frac{(c\rho)^n}{n!} + \frac{(c\rho)^c}{c!} \frac{1}{1-\rho} \right)^{-1}.$$

The probability of waiting Π_W is given by:

$$\begin{aligned} \Pi_W &= \mathbb{P}(W > 0) \\ &= \mathbb{P}(L \geq c) \\ &= \sum_{n=c}^{\infty} p_n \\ &= \sum_{n=c}^{\infty} \rho^{n-c} p_c \\ &= \frac{p_c}{1-\rho} \\ &= \frac{(c\rho)^c}{c!} \cdot \frac{p_0}{1-\rho} \\ &= \frac{(c\rho)^c}{c!} \left((1-\rho) \sum_{n=0}^{c-1} \frac{(c\rho)^n}{n!} + \frac{(c\rho)^c}{c!} \right)^{-1}. \end{aligned}$$

SECTION 4.2

| Mean Queue Length and Mean Waiting Time

From the equilibrium probabilities we directly obtain for the *mean queue length*:

$$\begin{aligned}
 \mathbb{E}[L^q] &= \sum_{n=c}^{\infty} (n - c) p_n \\
 &= \sum_{m=0}^{\infty} m p_{c+m} \\
 &= \sum_{m=0}^{\infty} m \rho^m p_c \\
 &= \frac{p_c}{1 - \rho} \sum_{m=0}^{\infty} m (1 - \rho) \rho^m \\
 &= \frac{p_c}{1 - \rho} \cdot \frac{\rho}{(1 - \rho)^2} = \Pi_W \frac{\rho}{1 - \rho}.
 \end{aligned}$$

We derive the *mean waiting time* via *Little's law*:

$$\mathbb{E}[W] = \frac{1}{\lambda} \mathbb{E}[L^q] = \frac{1}{\lambda} \Pi_W \frac{\rho}{1 - \rho} = \Pi_W \frac{1}{c\mu(1 - \rho)}.$$

Remark. These formulas can also be found by using the mean value technique. If not all servers are busy on arrival the waiting time is zero. If all servers are busy and there are zero or more customers waiting, then a new arriving customers first has to wait until the first departure and then continues to wait for as many departures as there were customers waiting upon arrival. An interdeparture time is the minimum of c exponential (residual) service times with mean $\frac{1}{\mu}$, and thus it is exponential with mean $\frac{1}{c\mu}$. So we obtain the arrival relation

$$\mathbb{E}[W] = \Pi_W \frac{1}{c\mu} + \mathbb{E}[L^q] \frac{1}{c\mu}.$$

Together with Little's law $\mathbb{E}[L^q] = \lambda \mathbb{E}[W]$, we have

$$\mathbb{E}[W] = \Pi_W \frac{1}{c\mu} + \mathbb{E}[W] \rho.$$

This yields

$$\mathbb{E}[W] = \Pi_W \frac{1}{c\mu(1 - \rho)}, \quad \mathbb{E}[L^q] = \Pi_W \frac{\rho}{1 - \rho}.$$

SECTION 4.3

| Distribution of the Waiting Time and the Sojourn Time

Waiting time may be expressed as:

$$W = \begin{cases} 0, & L^a \leq c-1 \\ \sum_{k=1}^{L^a-c+1} D_k, & L^a \geq c, \quad D_1, D_2, \dots \stackrel{i.i.d.}{\sim} \exp(c\mu) \end{cases}.$$

Then by PASTA property, we find

$$W = \sum_{k=1}^{\max(L-c+1, 0)} D_k.$$

There are three different methods to determine $\mathbb{P}(W > t)$.

(i) By conditioning on queue length found upon arrival:

$$\begin{aligned} \mathbb{P}(W > t) &= \sum_{n=0}^{\infty} \mathbb{P}\left(\sum_{k=1}^{\max(L-c+1, 0)} D_k > t \mid L = n\right) \mathbb{P}(L = n) \\ &= \sum_{n=c}^{\infty} \mathbb{P}\left(\sum_{k=1}^{n-c+1} D_k > t\right) p_n \\ &= \sum_{n=0}^{\infty} \mathbb{P}\left(\sum_{k=1}^{n+1} D_k > t\right) p_{c+n} \\ &= \sum_{n=0}^{\infty} \sum_{k=0}^n \mathbb{P}\left(\sum_{k=1}^{n+1} D_k > t\right) p_{c+n} \\ &= \sum_{n=0}^{\infty} \left(\sum_{k=0}^n \frac{(c\mu t)^k}{k!} e^{-c\mu t}\right) \rho^n p_c \\ &= \sum_{k=0}^{\infty} \frac{(c\mu \rho t)^k}{k!} e^{-c\mu t} \\ &= \frac{p_c}{1-\rho} \sum_{k=0}^{\infty} \frac{(c\mu \rho t)^k}{k!} e^{-c\mu t} \\ &= \Pi_W e^{-c\mu(1-\rho)t}, \quad t \geq 0 \end{aligned}$$

(ii) By using LST:

$$\begin{aligned} \mathbb{E}[e^{-sW}] &= \sum_{n=0}^{\infty} \mathbb{E}\left[e^{-s \sum_{k=1}^{\max(L-c+1, 0)} D_k} \mid L = n\right] P(L = n) \\ &= \sum_{n=0}^{c-1} p_n + \sum_{n=c}^{\infty} \mathbb{E}\left[e^{-s \sum_{k=1}^{n-c+1} D_k}\right] p_n \\ &= \sum_{n=0}^{c-1} p_n + \sum_{n=0}^{\infty} \mathbb{E}\left[e^{-s \sum_{k=1}^{n+1} D_k}\right] p_{c+n} \\ &= \sum_{n=0}^{c-1} p_n + \sum_{n=0}^{\infty} \left(\frac{c\mu}{c\mu + s}\right)^{n+1} \rho^n p_c \end{aligned}$$

$$\begin{aligned}
&= \sum_{n=0}^{c-1} p_n + \frac{p_c c \mu}{c \mu + s} \sum_{n=0}^{\infty} \left(\frac{\rho c \mu}{c \mu + s} \right)^n \\
&= \sum_{n=0}^{c-1} p_n + \frac{p_c c \mu}{1 - \rho} \cdot \frac{1}{c \mu (1 - \rho) + s} \\
&= 1 - \Pi_W + \Pi_W \frac{c \mu (1 - \rho)}{c \mu (1 - \rho) + s}.
\end{aligned}$$

This yields

$$W = \begin{cases} 0, & 1 - \Pi_W \\ \exp(c \mu (1 - \rho)), & \Pi_W \end{cases},$$

and for conditional waiting time,

$$\mathbb{P}(W > t \mid W > 0) = \frac{\mathbb{P}(W > t)}{\mathbb{P}(W > 0)} = e^{-c \mu (1 - \rho) t}, \quad t \geq 0.$$

(iii) By using [Exercise 1.2](#), combined with

$$\mathbb{P}(L = c + m \mid L \geq c) = (1 - \rho) \rho^m \sim \text{geo}(\rho).$$

To determine the distribution of the sojourn time we condition on the length of the service time, so

$$\begin{aligned}
\mathbb{P}(S > t) &= \mathbb{P}(W + B > t) \\
&= \int_{x=0}^{\infty} P(W + x > t) \mu e^{-\mu x} dx \\
&= \int_{x=0}^t P(W > t - x) \mu e^{-\mu x} dx + \int_{x=t}^{\infty} \mu e^{-\mu x} dx \\
&= \int_{x=0}^t \Pi_W e^{-c \mu (1 - \rho)(t - x)} \mu e^{-\mu x} dx + e^{-\mu t} \\
&= \Pi_W \int_{x=0}^t e^{-c \mu (1 - \rho)(t - x)} \mu e^{-\mu x} dx + e^{-\mu t} \\
&= \frac{\Pi_W}{1 - c(1 - \rho)} (e^{-c \mu (1 - \rho)t} - e^{-\mu t}) + e^{-\mu t} \\
&= \frac{\Pi_W}{1 - c(1 - \rho)} e^{-c \mu (1 - \rho)t} + \left(1 - \frac{\Pi_W}{1 - c(1 - \rho)} \right) e^{-\mu t},
\end{aligned}$$

where we assumed $c(1 - \rho) \neq 1$. In case $c(1 - \rho) = 1$, we find

$$P(S > t) = (\mu \Pi_W t + 1) e^{-\mu t}.$$

Exercises

EXERCISE 4.1 (A Fast and A Slow Machine). Consider two parallel machines with a common buffer where jobs arrive according to a Poisson stream with rate λ . The processing times are exponentially distributed with mean $\frac{1}{\mu_1}$ on machine 1 and $\frac{1}{\mu_2}$ on machine 2 ($\mu_1 > \mu_2$). Jobs are processed in order of arrival. A job arriving when both machines are idle is assigned to the fast machine. We assume that

$$\rho = \frac{\lambda}{\mu_1 + \mu_2}$$

is less than one.

- (i) Determine the distribution of the number of jobs in the system.
- (ii) Use this distribution to derive the mean number of jobs in the system.
- (iii) When is it better to not use the slower machine at all?
- (iv) Calculate for the following two cases the mean number of jobs in the system with and without the slow machine.
 - $\lambda = 2, \mu_1 = 5, \mu_2 = 1$.
 - $\lambda = 3, \mu_1 = 5, \mu_2 = 1$.

Note. It is shown that one should not remove the slow machine if $r > 0.5$ where $r = \frac{\mu_2}{\mu_1}$. When $0 \leq r < 0.5$ the slow machine should be removed (and the resulting system is stable) whenever $\rho \leq \rho_c$, where

$$\rho_c = \frac{2 + r^2 - \sqrt{(2 + r^2)^2 + 4(1 + r^2)(2r - 1)(1 + r)}}{2(1 + r^2)}.$$

EXERCISE 4.2. One is planning to build new telephone boxes near the railway station. The question is how many boxes are needed. Measurements showed that approximately 80 persons per hour want to make a phone call. The duration of a call is approximately exponentially distributed with mean 1 minute. How many boxes are needed such that the mean waiting time is less than 2 minutes?

EXERCISE 4.3. An insurance company has a call center handling questions of customers. Nearly 40 calls per hour have to be handled. The time needed to help a customer is exponentially distributed with mean 3 minutes. How many operators are needed such that only 5% of the customers has to wait longer than 2 minutes?

EXERCISE 4.4. In a dairy barn there are two water troughs (i.e. drinking places). From each trough only one cow can drink at the same time. When both troughs are occupied new arriving cows wait patiently for their turn. It takes an exponential time to drink with mean 3 minutes. Cows arrive at the water troughs according to a Poisson process with rate 20 cows per hour.

- (i) Determine the probability that there are i cows at the water troughs (waiting or drinking), $i = 0, 1, 2, \dots$.
- (ii) Determine the mean number of cows waiting at the troughs and the mean waiting time.

- (iii) What is the fraction of cows finding both troughs occupied on arrival?
- (iv) How many troughs are needed such that at most 10% of the cows find all troughs occupied on arrival?

EXERCISE 4.5. A computer consists of three processors. Their main task is to execute jobs from users. These jobs arrive according to a Poisson process with rate 15 jobs per minute. The execution time is exponentially distributed with mean 10 seconds. When a processor completes a job and there are no other jobs waiting to be executed, the processor starts to execute maintenance jobs. These jobs are always available and they take an exponential time with mean 5 seconds. But as soon as a job from a user arrives, the processor interrupts the execution of the maintenance job and starts to execute the new job. The execution of the maintenance job will be resumed later (at the point where it was interrupted).

- (i) What is the mean number of processors busy with executing jobs from users?
- (ii) How many maintenance jobs are on average completed per minute?
- (iii) What is the probability that a job from a user has to wait?
- (iv) Determine the mean waiting time of a job from a user.

$M/E_r/1$ Queue

SECTION 5.1

| State Description

We consider a single-server queue. Customers arrive according to a Poisson process with rate λ and they are treated in order of arrival. The service times are Erlang- r distributed with mean $\frac{r}{\mu}$. For stability we require that the occupation rate

$$\rho := \lambda \frac{r}{\mu} < 1.$$

The natural way to describe the state of a nonempty system is by the pair (k, l) where k denotes the number of customers in the system and l the remaining number of service phases of the customer in service. An alternative way to describe the state is by counting the total number of uncompleted phases of work in the system: $n = (k - 1)r + l$ if $k \geq 1$.

SECTION 5.2

| Equilibrium Distribution

Let the random variable $L^f(t)$ denote the number of remaining phases of work in the system at time t . Let p_n be the equilibrium probability of n phases of work in the system:

$$p_n = \lim_{t \rightarrow \infty} \mathbb{P}(L^f(t) = n).$$

Let $L(t)$ be the number of customers in the system at time t is not a Markov process. Let q_i be the probability of i customers in the system:

$$\begin{aligned} q_i &= \lim_{t \rightarrow \infty} \mathbb{P}(L(t) = i) \\ &= \lim_{t \rightarrow \infty} \mathbb{P}\left((i-1)r + 1 \leq L^f(t) \leq i \cdot r\right) \\ &= \sum_{n=(i-1)r+1}^{i \cdot r} \lim_{t \rightarrow \infty} \mathbb{P}(L^f(t) = n) \\ &= \sum_{n=(i-1)r+1}^{i \cdot r} p_n, \quad i \geq 1 \end{aligned}$$

It is easy to see that $q_0 = p_0 = 1 - \rho$. We obtain the following set of equilibrium equations:

$$\lambda p_0 = \mu p_1, \quad (5.1)$$

$$(\lambda + \mu)p_n = \mu p_{n+1}, \quad n = 1, \dots, r-1, \quad (5.2)$$

$$(\lambda + \mu)p_n = \lambda p_{n-r} + \mu p_{n+1}, \quad n = r, r+1, r+2, \dots \quad (5.3)$$

Recursive method seen in $M/M/1$ queue no longer works, $(r+1)$ -th order recurrence relation. We first look for solutions of interior equation (5.3) of the form

$$p_n = x^n, \quad n = 0, 1, 2, \dots$$

Substituting $p_n = x^n$ in interior equation (5.3):

$$\begin{aligned} (\lambda + \mu) \frac{x^n}{x^{n-r}} &= \lambda \frac{x^{n-r}}{x^{n-r}} + \mu \frac{x^{n+1}}{x^{n-r}}, \\ \implies (\lambda + \mu)x^r &= \lambda + \mu x^{r+1}. \end{aligned} \quad (5.4)$$

If stability condition is satisfied, then this r -th order polynomial equation has root $x = 1$ (irrelevant because of normalisation condition); or exactly r distinct roots x_k with $|x_k| < 1$, say $x_1, \dots, x_r$ (*Roucé's theorem*). Substitute

$$p_n = \sum_{k=1}^r c_k x_k^n$$

in $r-1$ boundary equations (5.2), determine $c_1, \dots, c_r$, from (5.2) and normalisation condition $\sum_{n=0}^{\infty} p_n = 1$. Other boundary equation (5.1) is then automatically satisfied. It can be shown that this set of equations has a unique solution, given by

$$c_k = \frac{1 - \rho}{\prod_{j \neq k} \left(1 - \frac{x_j}{x_k}\right)}, \quad k = 1, \dots, r.$$

The important conclusion is that for the $M/E_r/1$ queue the equilibrium probabilities p_n can be expressed as a mixture of r geometric distributions:

$$p_n = \sum_{k=1}^r \frac{c_k}{1-x_k} (1-x_k) x_k^n, \quad n = 0, 1, 2, \dots$$

From the distribution of the number of phases in the system we can easily find the distribution of the number of customers in the system.

$$\begin{aligned} q_i &= \sum_{n=(i-1)r+1}^{ir} p_n \\ &= \sum_{n=(i-1)r+1}^{ir} \sum_{k=1}^r c_k x_k^n \\ &= \sum_{k=1}^r c_k \sum_{n=(i-1)r+1}^{ir} x_k^n \\ &= \sum_{k=1}^r c_k \left(x_k^{-(r-1)} + x_k^{-(r-2)} + \dots + 1 \right) (x_k^r)^i. \end{aligned}$$

Hence, the queue length probabilities can also be expressed as a mixture of r geometric distributions.

SECTION 5.3

| Mean Waiting Time

From the PASTA property, we find

$$\mathbb{E}[W] = \mathbb{E}[L^f] \frac{1}{\mu}.$$

From equilibrium distribution we derive

$$\begin{aligned} \mathbb{E}[L^f] &= \sum_{n=0}^{\infty} n p_n \\ &= \sum_{n=0}^{\infty} n \sum_{k=1}^r c_k x_k^n \\ &= \sum_{k=1}^r \frac{c_k}{1 - x_k} \\ &= \sum_{k=1}^r \frac{c_k}{1 - x_k} \sum_{n=0}^{\infty} n (1 - x_k) x_k^n \\ &= \sum_{k=1}^r \frac{c_k}{1 - x_k} \cdot \frac{x_k}{(1 - x_k)^2} \\ &= \sum_{k=1}^r \frac{c_k x_k}{(1 - x_k)^2}. \end{aligned}$$

Remark. Alternatively, we could also have used $\mathbb{E}[L^q] = \sum_{i=1}^{\infty} (i-1)q_i$, together with Little's law $\mathbb{E}[W] = \frac{1}{\lambda} \mathbb{E}[L^q]$.

An arriving customer has to wait for the customers in the queue and, if the server is busy, for the one in service. According to the PASTA property the mean number of customers waiting in the queue is

$$\begin{aligned} \mathbb{E}[W] &= \mathbb{E}[L^q] \frac{r}{\mu} + \rho \mathbb{E}[R] \\ &= \rho \mathbb{E}[W] + \rho \mathbb{E}[R], & (\text{Little's law } \mathbb{E}[L^q] = \lambda \mathbb{E}[W]) \\ \implies \mathbb{E}[W] &= \frac{\rho}{1 - \rho} \cdot \mathbb{E}[R], \end{aligned}$$

where R denotes the *residual service time* of customer in service. Then

$$\begin{aligned} \mathbb{E}[R] &= \frac{1}{r} \cdot \frac{r}{\mu} + \frac{1}{r} \cdot \frac{r-1}{\mu} + \cdots + \frac{1}{r} \cdot \frac{1}{\mu} \\ &= \frac{1}{r} \cdot \frac{1}{\mu} (1 + 2 + \cdots + r) \\ &= \frac{1}{r} \cdot \frac{1}{\mu} \frac{r(r+1)}{2} = \frac{r+1}{2} \cdot \frac{1}{\mu}. \end{aligned}$$

SECTION 5.4

| Distribution of the Waiting Time

By the PASTA property, the waiting time can be expressed as

$$W = \sum_{i=1}^{L^f} B_i, \quad B_1, B_2, \dots \stackrel{i.i.d}{\sim} \exp(\mu).$$

We follow similar approach as seen for M/M/1 queue. By conditioning on L^f ,

$$\begin{aligned} \mathbb{P}(W > t) &= \sum_{n=1}^{\infty} \mathbb{P}\left(\sum_{i=1}^n B_i > t\right) p_n \\ &= \sum_{n=1}^{\infty} \sum_{i=0}^{n-1} \frac{(\mu t)^i}{i!} e^{-\mu t} \sum_{k=1}^r c_k x_k^n \\ &= \sum_{k=1}^r c_k \sum_{i=0}^{\infty} \sum_{n=i+1}^{\infty} \frac{(\mu t)^i}{i!} e^{-\mu t} x_k^n \\ &= \sum_{k=1}^r c_k x_k \sum_{i=0}^{\infty} \frac{(\mu x_k t)^i}{i!} e^{-\mu t} \\ &= \sum_{k=1}^r \frac{c_k x_k}{1 - x_k} e^{-\mu(1-x_k)t}, \quad t \geq 0. \end{aligned}$$

Remark. This distribution is a generalisation of the one for the M/M/1 model (namely, not one, but a mixture of exponential distributions).

| Exercises

EXERCISE 5.1 (Rouché's Theorem). Let the bounded region D have as its boundary a contour C . Let the functions $f(z)$ and $g(z)$ be analytic both in D and on C , and assume that $|f(z)| < |g(z)|$ on C . Then $f(z) + g(z)$ has in D the same number of zeros as $g(z)$, all zeros counted according to their multiplicity.

- (i) Use *Rouché's theorem* to prove that the polynomial equation (5.4) has exactly r (possibly complex) roots x with $|x| < 1$.

Hint: take $f(z) = \mu z^{r+1}$, $g(z) = -(\lambda + \mu)z^r + \lambda$, and take as contour C the circle with centre 0 and radius $1 - \varepsilon$ with ε small and positive.

- (ii) Show that all roots are simple.

Hint: show that there are no z for which both $f(z) + g(z)$ and $f'(z) + g'(z)$ vanish at the same time.

EXERCISE 5.2 (Generating Function Approach). Consider the M/E_r/1 queue with arrival rate λ and mean service time $\frac{r}{\mu}$. Let $\mathbb{P}(z)$ be the generating function of the probabilities p_n , so

$$\mathbb{P}(z) = \sum_{n=0}^{\infty} p_n z^n, \quad |z| \leq 1.$$

- (i) Show, by multiplying the equilibrium equations (5.1)-(5.3) with z^n and adding over all states n , that

$$\mathbb{P}(z)(\lambda + \mu) - p_0\mu = \mathbb{P}(z)\lambda z^r + (\mathbb{P}(z) - p_0)\mu z^{-1}.$$

- (ii) Show that $\mathbb{P}(z)$ is given by

$$\mathbb{P}(z) = \frac{(1 - \rho)\mu}{\mu - \lambda(z^r + z^{r-1} + \dots + z)}. \quad (5.5)$$

- (iii) Show, by partial fraction decomposition of $\mathbb{P}(z)$, that the probabilities p_n can be written as

$$p_n = \sum_{k=1}^r c_k \left(\frac{1}{z_k} \right)^n, \quad n = 0, 1, 2, \dots,$$

where $z_1, \dots, z_r$ are the zeros of the denominator in (5.5).

EXERCISE 5.3. Orders arrive according to a Poisson process, so the interarrival times are independent and exponentially distributed. Each order has to pass, stage by stage, through a series of r independent production stages. Each stage takes an exponentially distributed time. The total production time, the workload, of the order is the sum of r independent, identically distributed random variables. So the distribution of the workload is the r -stage Erlang distribution.

- (i) The state of this system (i.e., the work station together with its queue) can be characterised by the number of orders in the system and by the number of not yet completed stages for the order in the work station. Describe the flow diagram and give the set of equations for the equilibrium state probabilities.
- (ii) The state of the system at a certain time can also be described by the total number of production stages that have to be completed before all the orders in the system are ready. Give the flow diagram and formulate the equations for the equilibrium state probabilities.
- (iii) Define

$$p_n = \mathbb{P}(\text{total number of orders in the system} = n),$$

$$q_k = \mathbb{P}(\text{total number of production stages in the system} = k).$$

Give the relation between these two probabilities.

EXERCISE 5.4. Orders arrive in so called bulks. Each bulk consists of r independent orders. The bulks themselves arrive according to a Poisson process. The sequence in which orders of one bulk are processed in the work station is unimportant. All orders of a bulk have to wait until all orders of bulks that have arrived earlier are completed. The workload of each order is exponentially distributed. The state of the system is simply characterized by the number of orders in the system. Describe the flow diagram for this situation. Compare the result with that of the previous exercise, part (ii).

EXERCISE 5.5. Jobs arrive at a machine according to a Poisson process with a rate of 16 jobs per hour. Each job consists of 2 tasks. Each task has an exponentially distributed processing time with a mean of 1 minute. Jobs are processed in order of arrival.

- (i) Determine the distribution of the number of uncompleted tasks in the system.
- (ii) Determine the distribution of the number of jobs in the system.
- (iii) What is the mean number of jobs in the system?
- (iv) What is the mean waiting time of a job?

EXERCISE 5.6. Consider an $M/E_2/1/2$ queue with arrival rate 1 and mean service time 2. At time $t = 0$ the system is empty. Determine the mean time till the first customer is rejected on arrival.

EXERCISE 5.7. Consider the $M/E_2/1$ queue with an arrival rate of 4 customers per hour and a mean service time of 8 minutes.

- (i) Determine the distribution of the waiting time.
- (ii) What is the fraction of customers that has to wait longer than 5 minutes?

EXERCISE 5.8. Customers arrive in groups at a single-server queue. These groups arrive according to a Poisson process with a rate of 3 groups per hour. With probability $\frac{1}{3}$ a group consists of 2 customers, and with probability $\frac{2}{3}$ it consists of 1 customer only. All customers have an exponential service time with a mean of 6 minutes. Groups are served in order of arrival. Within a group, customers are served in random order.

- (i) Determine the distribution of the number of customers in the system.
- (ii) Determine the mean sojourn time of an arbitrary customer.

EXERCISE 5.9. In a repair shop of an airline company defective airplane engines are repaired. Defects occur according to a Poisson with a rate of 1 defective engine per 2 weeks. The mean repair time of an engine is $\frac{2}{3}$ week. The repair time distribution can be well approximated by an Erlang-2 distribution. In the repair shop only one engine can be in repair at the same time.

- (i) Show that q_n , the probability that there are n engines in the repair shop, is given by

$$q_n = \frac{6}{5} \left(\frac{1}{4} \right)^n - \frac{8}{15} \left(\frac{1}{5} \right)^n.$$

- (ii) Determine the mean sojourn time (waiting time plus repair time) of an engine in the repair shop.
- (iii) The airline company has several spare engines in a depot. When a defect occurs, the defective engine is immediately replaced by a spare engine (if there is one available) and the defective engine is sent to the repair shop. After repair the engine is as good as new, and it is transported to the depot. When a defect occurs and no spare engine is available, the airplane has to stay on the ground and it has to wait till a spare engine becomes available. Determine the minimal number of spare engines needed such that for 99% of the defects there is a spare engine available.

EXERCISE 5.10. Jobs arrive according to a Poisson process at a machine. The arrival rate is 25 jobs per hour. With probability $\frac{2}{5}$ a job consists of 1 task, and with probability $\frac{3}{5}$ it consists of 2 tasks. Tasks have an exponentially distributed processing time with a mean of 1 minute. Jobs are processed in order of arrival.

- (i) Determine the distribution of the number of uncompleted tasks at the machine.
- (ii) Determine the mean waiting time of a job.
- (iii) Determine the mean number of jobs in the system.

EXERCISE 5.11. Customers arrive in groups at a server. The group size is 2 and groups arrive according to a Poisson stream with a rate of 2 groups per hour. Customers are served one by one, and they require an exponential service time with a mean of 5 minutes.

- (i) Determine the distribution of the number of customers in the system.
- (ii) What is the mean waiting time of the first customer in a group?
- (iii) And what is the mean waiting time of the second one?

M/G/1 Queue

SECTION 6.1

| State Description

In the $M/G/1$ queue customers arrive according to a Poisson process with rate λ and they are treated in order of arrival. The service times are *i.i.d.* with distribution function $F_B(\cdot)$ and density $f_B(\cdot)$. For stability we have to require that the *occupation rate*

$$\rho = \lambda \mathbb{E}[B] < 1.$$

Let $X(t)$ be the amount of received (or remaining) service of customer in service (if any), and $\{L(t), X(t)\}_{t \geq 0}$ be a Markov process. Continuous nature of $X(t)$ severely complicates analysis.

Denote by L_k^d the number of customers in the system right after departure of k -th customer. We will determine the limiting distribution:

$$d_n = \lim_{k \rightarrow \infty} \mathbb{P}(L_k^d = n).$$

The probability d_n can be interpreted as the fraction of customers who leave behind n customers. But in fact we are more interested in the limiting distribution p_n defined as

$$p_n = \lim_{t \rightarrow \infty} \mathbb{P}(L(t) = n),$$

where $L(t)$ is the number of customers in the system at time t . The probability p_n can be interpreted as the fraction of time there are n customers in the system. Another distribution is the limiting distribution of the number of customers in the system seen by an arriving customer, i.e.,

$$a_n = \lim_{k \rightarrow \infty} \mathbb{P}(L_k^a = n),$$

where L_k^a is the number of customers in the system just before the k -th arriving customer.

From the PASTA property, we already know that $a_n = p_n$ for all n . In general $L_k^a \neq L_k^d$, but $a_n = d_n$, using up/downcrossing argument. We will focus analysis on $\{L_k^d\}_{k \geq 1}$ and d_n .

SECTION 6.2

| Departure Distribution

Let A_k be the customers arriving during service time of customer k . Thus we have

$$L_{k+1}^d = \begin{cases} L_k^d - 1 + A_{k+1}, & L_k^d > 0 \\ A_{k+1}, & L_k^d = 0 \end{cases} = \max(L_k^d - 1, 0) + A_{k+1}.$$

The sequence $\{L_k^d\}_{k=0}^\infty$ is discrete-time Markov chain with transition probabilities

$$p_{ij} = \begin{cases} 0, & j < i - 1 \\ \mathbb{P}(A = j - i + 1), & i > 0 \text{ and } j \geq i - 1 \\ \mathbb{P}(A = j), & i = 0 \end{cases}.$$

The matrix P of transition probabilities takes the form

$$P = (p_{ij}) = \begin{pmatrix} \alpha_0 & \alpha_1 & \alpha_2 & \alpha_3 & \cdots \\ \alpha_0 & \alpha_1 & \alpha_2 & \alpha_3 & \cdots \\ 0 & \alpha_0 & \alpha_1 & \alpha_2 & \cdots \\ 0 & 0 & \alpha_0 & \alpha_1 & \cdots \\ 0 & 0 & 0 & \alpha_0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix},$$

where

$$\begin{aligned} \alpha_n &= \mathbb{P}(A = n) \\ &= \mathbb{P}(\text{Poisson}(\lambda B) = n) \\ &= \int_{t=0}^{\infty} \mathbb{P}(A_{k+1} = n \mid B = t) d\mathbb{P}(B < t) \\ &= \int_{t=0}^{\infty} \frac{(\lambda t)^n}{n!} e^{-\lambda t} d\mathbb{P}(B < t) \\ &= \int_{t=0}^{\infty} \frac{(\lambda t)^n}{n!} e^{-\lambda t} f_B(t) dt. \end{aligned}$$

Introduce probability generating function

$$\begin{aligned} \mathbb{P}_A(z) &= \sum_{n=0}^{\infty} \alpha_n z^n \\ &= \sum_{n=0}^{\infty} \int_{t=0}^{\infty} \frac{(\lambda t)^n}{n!} e^{-\lambda t} f_B(t) dt z^n \\ &= \int_{t=0}^{\infty} \sum_{n=0}^{\infty} \frac{(\lambda t z)^n}{n!} e^{-\lambda t} f_B(t) dt \\ &= \int_{t=0}^{\infty} e^{\lambda t(1-z)} f_B(t) dt = \widetilde{B}(\lambda(1-z)). \end{aligned}$$

The limiting probabilities d_n satisfy the equilibrium equations:

$$d_n = \sum_{m=0}^{n+1} d_m p_{mn}$$

$$\begin{aligned}
&= d_0 p_{0n} + d_1 p_{1n} + \cdots + d_n p_{nn} + d_{n+1} p_{n+1,n} \\
&= d_0 \alpha_n + d_1 \alpha_{n-1} + d_2 \alpha_{n-2} + \cdots + \alpha_1 d_n + \alpha_0 d_{n+1} \\
&= \sum_{k=0}^n \alpha_k d_{n+1-k} + d_0 \alpha_n, \quad n = 0, 1, \dots
\end{aligned}$$

Introduce the probability generating function and recall

$$P_L^d(z) = \sum_{n=0}^{\infty} d_n z^n, \quad d_n = \sum_{k=0}^n \alpha_k d_{n+1-k} + d_0 \alpha_n.$$

Multiplying by z^n and summing over all n leads to

$$\begin{aligned}
\mathbb{P}_L^d(z) &= \sum_{n=0}^{\infty} d_n z^n \\
&= \sum_{n=0}^{\infty} \left(\sum_{k=0}^n d_{n+1-k} \alpha_k + d_0 \alpha_n \right) z^n \\
&= z^{-1} \sum_{n=0}^{\infty} \sum_{k=0}^n d_{n+1-k} z^{n+1-k} \alpha_k z^k + \sum_{n=0}^{\infty} d_0 \alpha_n z^n \\
&= z^{-1} \sum_{k=0}^{\infty} \sum_{n=k}^{\infty} d_{n+1-k} z^{n+1-k} \alpha_k z^k + d_0 \mathbb{P}_A(z) \\
&= z^{-1} \sum_{k=0}^{\infty} \alpha_k z^k \sum_{n=k}^{\infty} d_{n+1-k} z^{n+1-k} + d_0 \mathbb{P}_A(z) \\
&= z^{-1} \sum_{k=0}^{\infty} \alpha_k z^k \sum_{m=1}^{\infty} d_m z^m + d_0 \mathbb{P}_A(z) \\
&= z^{-1} \mathbb{P}_A(z) (\mathbb{P}_L^d(z) - d_0) + d_0 \mathbb{P}_A(z) \\
&= \left[z^{-1} (\mathbb{P}_L^d(z) - d_0) + d_0 \right] \mathbb{P}_A(z).
\end{aligned}$$

Rearranging terms

$$\begin{aligned}
&\implies \mathbb{P}_{L^d}(z) \left[1 - \frac{1}{z} \mathbb{P}_A(z) \right] = d_0 \mathbb{P}_A(z) \left[1 - \frac{1}{z} \right] \\
&\implies \mathbb{P}_{L^d}(z) [z - \mathbb{P}_A(z)] = d_0 \mathbb{P}_A(z) [z - 1]
\end{aligned}$$

Hence we find the *Pollaczek-Khinchin formula*:

$$P_{L^d}(z) = \frac{d_0 P_A(z)(z-1)}{z - P_A(z)} = \frac{(1-\rho) \tilde{B}(\lambda(1-z))(z-1)}{z - \tilde{B}(\lambda(1-z))}.$$

From the *Pollaczek-Khinchin formula* we can derive the results:

$$\begin{aligned}
\mathbb{E}[L] &= \mathbb{E}[L^d] = \left. \frac{d}{dz} \mathbb{P}_{L^d}(z) \right|_{z=1} \stackrel{L'Hopital}{=} \frac{\lambda^2 \mathbb{E}[B^2]}{2(1-\rho)} + \rho, \\
\mathbb{E}[L^q] &= \frac{\lambda^2 \mathbb{E}[B^2]}{2(1-\rho)},
\end{aligned}$$

$$\mathbb{E}[W] \stackrel{\text{Little's law}}{=} \frac{\lambda \mathbb{E}[B^2]}{2(1-\rho)} = \frac{\lambda \mathbb{E}[B] \mathbb{E}[B]}{2(1-\rho)} \cdot \frac{\mathbb{E}[B^2]}{\mathbb{E}[B]^2} = \underbrace{\frac{\rho}{1-\rho}}_{\text{impact load}} \underbrace{\frac{c_B^2 + 1}{2}}_{\substack{\text{variability} \\ \text{service time}}} \underbrace{\mathbb{E}[B]}_{\text{scale}},$$

$$\mathbb{E}[S] = \frac{\lambda \mathbb{E}[B^2]}{2(1-\rho)} + \mathbb{E}[B].$$

Remark. Alternative derivation of Pollaczek-Khinchin formula: recall recursive relation

$$L_{k+1}^d = \max(L_k^d - 1, 0) + A_{k+1}.$$

This yields as $k \rightarrow \infty$,

$$L^d \stackrel{s.t.}{=} \max(L^d - 1, 0) + A.$$

Directly using this relation, we find

$$\begin{aligned} \mathbb{P}_{L^d}(z) &= \mathbb{E}\left[z^{\max(L^d-1,0)+A}\right] \\ &= \mathbb{E}\left[z^{\max(L^d-1,0)} z^A\right] \\ &= \mathbb{E}\left[z^{\max(L^d-1,0)}\right] \mathbb{E}[z^A] \\ &= \mathbb{E}\left[z^{\max(L^d-1,0)}\right] \mathbb{P}_A(z). \end{aligned}$$

Conditioning on queue length at departure epochs, we find

$$\begin{aligned} \mathbb{E}\left[z^{\max(L^d-1,0)}\right] &= \sum_{n=0}^{\infty} \mathbb{E}\left[z^{\max(L^d-1,0)} \mid L^d = n\right] \mathbb{P}(L^d = n) \\ &= \sum_{n=0}^{\infty} z^{\max(n-1,0)} d_n \\ &= d_0 + \sum_{n=1}^{\infty} d_n z^{n-1} \\ &= d_0 + \frac{1}{z} \sum_{n=1}^{\infty} d_n z^n \\ &= d_0 + \frac{1}{z} [\mathbb{P}_{L^d}(z) - d_0] \\ &= \frac{1}{z} \mathbb{P}_{L^d}(z) + d_0 \left(1 - \frac{1}{z}\right). \end{aligned}$$

In conclusion, we find

$$P_L^d(z) = \left[\frac{1}{z} P_L^d(z) - d_0 \left(1 - \frac{1}{z}\right) \right] P_A(z).$$

Rearranging terms, we obtain

$$\begin{aligned} \mathbb{P}_{L^d}(z) \left[1 - \frac{1}{z} \mathbb{P}_A(z) \right] &= d_0 \mathbb{P}_A(z) \left(1 - \frac{1}{z} \right) \\ \implies \mathbb{P}_{L^d}(z) &= \frac{d_0 \mathbb{P}_A(z)(z-1)}{z - \mathbb{P}_A(z)}. \end{aligned}$$

In practice, LST and PGF are often used to:

- (i) Determine moments, e.g., for discrete random variable X ,

$$\mathbb{E}[X^k] = (-1)^k \frac{d^k \mathbb{P}_X(z)}{dz^k} \Big|_{z=1}.$$

- (ii) Invert to obtain density/distribution: recognise transform; recover analytically, yielding formal expression; numerical inversion.

EXAMPLE 6.1 ($M/M/1$ Queue). Here we have

$$\tilde{B}(s) = \frac{\mu}{\mu + s},$$

so that

$$\begin{aligned} \mathbb{P}_{L^d}(z) &= \frac{(1-\rho) \frac{\mu}{\mu+\lambda-\lambda z} (1-z)}{\frac{\mu}{\mu+\lambda-\lambda z} - z} \\ &= \frac{(\mu-\lambda)(1-z)}{\mu(1-z) - \lambda z(1-z)} \\ &= \frac{\mu-\lambda}{\mu-\lambda z} \\ &= \frac{1-\rho}{1-\rho z} = \sum_{n=0}^{\infty} (1-\rho) \rho^n z^n = \sum_{n=0}^{\infty} d_n z^n. \end{aligned}$$

Hence,

$$d_n = (1-\rho) \rho^n, \quad n = 0, 1, 2, \dots$$

EXAMPLE 6.2 ($M/E_2/1$ Queue). Mean $\frac{2}{\mu}$ and $\rho = \frac{1}{3}$. Then

$$\tilde{B}(s) = \left(\frac{\mu}{\mu + s} \right)^2.$$

So

$$\begin{aligned} \mathbb{P}_{L^d}(z) &= \frac{(1-\rho) \left(\frac{\mu}{\mu+\lambda-\lambda z} \right)^2 (1-z)}{\left(\frac{\mu}{\mu+\lambda-\lambda z} \right)^2 - z} \\ &= \frac{(1-\rho) \mu^2 (1-z)}{\mu^2 - z(\mu + \lambda - \lambda z)^2} \\ &= \frac{(1-\rho)(1-z)}{1-z \left(1 + \frac{\rho(1-z)}{2} \right)^2} \\ &= \frac{1-\rho}{1-\rho z - \frac{\rho^2 z(1-z)}{4}}. \end{aligned}$$

For $\rho = \frac{1}{3}$, we then find

$$\begin{aligned} \mathbb{P}_{L^d}(z) &= \frac{\frac{2}{3}}{1 - \frac{z}{3} - \frac{z(1-z)}{36}} \\ &= \frac{24}{36 - 13z + z^2} \end{aligned}$$

$$\begin{aligned}
&= \frac{24}{(4-z)(9-z)} \\
&= \frac{\frac{24}{5}}{4-z} - \frac{\frac{24}{5}}{9-z} \\
&= \frac{\frac{6}{5}}{1-\frac{z}{4}} - \frac{\frac{8}{15}}{1-\frac{z}{9}} \\
&= \frac{6}{5} \sum_{n=0}^{\infty} \left(\frac{1}{4}z\right)^n - \frac{8}{15} \sum_{n=0}^{\infty} \left(\frac{1}{9}z\right)^n \\
&= \sum_{n=0}^{\infty} d_n z^n.
\end{aligned}$$

Hence,

$$d_n = p_n = \frac{6}{5} \left(\frac{1}{4}\right)^n - \frac{8}{15} \left(\frac{1}{9}\right)^n, \quad n = 0, 1, 2, \dots$$

EXAMPLE 6.3 ($M/H_2/1$ Queue). Arrival rate $\lambda = 1$. Hyperexponentially distributed service time with parameters $p_1 = 1 - p_2 = \frac{1}{4}$ and $\mu_1 = 1, \mu_2 = 2$, i.e.,

$$B = \begin{cases} \exp(1), & \text{w.p. } \frac{1}{4} \\ \exp(2), & \text{w.p. } \frac{3}{4} \end{cases}.$$

Then $\mathbb{E}[B] = \frac{1}{4} \cdot 1 + \frac{3}{4} \cdot \frac{1}{2} = \frac{5}{8}$,

$$\widetilde{B}(s) = \frac{1}{4} \cdot \frac{1}{1+s} + \frac{3}{4} \cdot \frac{2}{2+s} = \frac{1}{4} \cdot \frac{8+7s}{(1+s)(2+s)}.$$

Substitution into Pollaczek-Khinchin formula yields

$$\begin{aligned}
P_L^d(z) &= \frac{\frac{3}{8} \frac{1}{4} \frac{15-7z}{(2-z)(3-z)} (1-z)}{\frac{1}{4} \frac{15-7z}{(2-z)(3-z)} - z} \\
&= \frac{3}{8} \cdot \frac{(15-7z)(1-z)}{(15-7z) - 4z(2-z)(3-z)} \\
&= \frac{3}{8} \cdot \frac{15-7z}{(3-2z)(5-2z)} \\
&= \frac{3}{8} \cdot \frac{9/4}{3-2z} + \frac{3}{8} \cdot \frac{5/4}{5-2z} \\
&= \frac{9/32}{1-\frac{2z}{3}} + \frac{3/32}{1-\frac{2z}{5}} \\
&= \frac{9}{32} \sum_{n=0}^{\infty} \left(\frac{2}{3}z\right)^n + \frac{3}{32} \sum_{n=0}^{\infty} \left(\frac{2}{5}z\right)^n \\
&= \sum_{n=0}^{\infty} d_n z^n.
\end{aligned}$$

So

$$d_n = \frac{9}{32} \left(\frac{2}{3}\right)^n + \frac{3}{32} \left(\frac{2}{5}\right)^n, \quad n = 0, 1, 2, \dots$$

SECTION 6.3

| Distribution of the Sojourn Time

Denote the total time spent in the system for this customer by the random variable S by

$$S = \sum_{i=1}^{L^q} B_i + L^s \cdot R,$$

where R denotes the remaining service time of customer in service. In considering a first-come first-served system, L_k^d denotes the number of customers right after departure of k -th customer, i.e., the number of arrivals during sojourn time of k -th customer. We observe following relation

$$\begin{aligned} & \mathbb{P}(L^d = n \text{ right after departure} \mid \text{sojourn time } S = t) \\ &= \mathbb{P}(A = n \text{ arrivals during sojourn time of length } t) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}. \end{aligned}$$

Conditioning on sojourn time

$$\begin{aligned} d_n &= P(L^d = n) \\ &= \int_{t=0}^{\infty} P(L^d = n \mid S = t) dP(S < t) \\ &= \int_{t=0}^{\infty} P(A(0, t) = n) dP(S < t) \\ &= \int_{t=0}^{\infty} e^{-\lambda t} \frac{(\lambda t)^n}{n!} dP(S < t). \end{aligned}$$

We have now expressed d_n into sojourn time distribution. Using this relation, we calculate PGF of $P_{L^d}(z)$:

$$\begin{aligned} P_{L^d}(z) &= \sum_{n=0}^{\infty} d_n z^n \\ &= \sum_{n=0}^{\infty} \int_{t=0}^{\infty} e^{-\lambda t} \frac{(\lambda t)^n}{n!} dP(S < t) z^n \\ &= \int_{t=0}^{\infty} e^{-\lambda t} \sum_{n=0}^{\infty} \frac{(\lambda t z)^n}{n!} dP(S < t) \\ &= \int_{t=0}^{\infty} e^{-\lambda t(1-z)} dP(S < t) = \tilde{S}(\lambda(1-z)) \end{aligned}$$

Change of variable $s = \lambda(1-z)$, yields *second Pollaczek-Khinchin formula*:

$$\begin{aligned} \tilde{S}(s) &= P_{L^d}\left(1 - \frac{s}{\lambda}\right) \\ &= \frac{(1-\rho)\tilde{B}(s)s}{\lambda\tilde{B}(s) + s - \lambda} \\ &= \frac{(1-\rho)\tilde{B}(s)s}{s - \lambda(1 - \tilde{B}(s))}. \end{aligned}$$

From this relation, we can also find back *Little's law*:

$$\frac{d}{ds} \tilde{S}(s) = \frac{d}{ds} P_{L^d} \left(1 - \frac{s}{\lambda} \right) = P_{L^{d'}} \left(1 - \frac{s}{\lambda} \right) \cdot \left(-\frac{1}{\lambda} \right).$$

At $s = 0$:

$$-\mathbb{E}[S] = \mathbb{E}[L] \left(-\frac{1}{\lambda} \right) \implies \mathbb{E}[S] = \frac{1}{\lambda} \mathbb{E}[L].$$

EXAMPLE 6.4 ($M/H_2/1$ Queue). As considered previously

SECTION 6.4

| Distribution of the Waiting Time

Waiting-time distribution can be derived from sojourn time distribution as follows. Since $S = W + B$,

$$\widetilde{S}(s) = \widetilde{W}(s)\widetilde{B}(s).$$

This yields the *third Pollaczek-Khinchin formula*:

$$\widetilde{W}(s) = \frac{\widetilde{S}(s)}{\widetilde{B}(s)} = \frac{(1 - \rho)s}{\lambda \widetilde{B}(s) + s - \lambda} = \frac{(1 - \rho)s}{s - \lambda(1 - \widetilde{B}(s))}.$$

SECTION 6.5

| Lindley's Equation

Denote by A_n the interarrival time between the n -th and $n + 1$ -th customer arriving at the system. Further let W_n and B_n denote the waiting time and the service time, respectively, for the n -th customer. *Lindley's equation* states

$$W_{n+1} = \max(W_n + B_n - A_n, 0).$$

This yields as $n \rightarrow \infty$,

$$W \stackrel{s.t.}{=} \max(W + B - A, 0) = \max(S - A, 0).$$

Noting that $A \sim \exp(\lambda)$ and $A \perp S$, we get

$$\begin{aligned} \tilde{W}(s) &= E(e^{-sW}) \\ &= E(e^{-s \max(S-A, 0)}) \\ &= \int_{x=0}^{\infty} \int_{y=0}^{\infty} e^{-s \max(x-y, 0)} f_S(x) \lambda e^{-\lambda y} dx dy \\ &= \int_{x=0}^{\infty} \int_{y=0}^x e^{-s(x-y)} f_S(x) \lambda e^{-\lambda y} dx dy + \int_{x=0}^{\infty} \int_{y=x}^{\infty} f_S(x) \lambda e^{-\lambda y} dx dy \\ &= \frac{\lambda}{s - \lambda} \int_{x=0}^{\infty} (e^{-\lambda x} - e^{-sx}) f_S(x) dx + \int_{x=0}^{\infty} e^{-\lambda x} f_S(x) dx \\ &= \frac{\lambda}{s - \lambda} (\tilde{S}(\lambda) - \tilde{S}(s)) + \tilde{S}(\lambda) \\ &= \frac{s}{s - \lambda} \tilde{S}(\lambda) - \frac{\lambda}{s - \lambda} \tilde{S}(s). \end{aligned}$$

Using that

$$\tilde{S}(s) = \tilde{W}(s) \tilde{B}(s),$$

we get by rearranging terms

$$\tilde{W}(s) = \frac{\tilde{S}(\lambda)s}{\lambda \tilde{B}(s) + s - \lambda} = \frac{(1 - \rho)s}{\lambda \tilde{B}(s) + s - \lambda}$$

because of the fact that

$$\begin{aligned} \tilde{S}(\lambda) &= \int_{x=0}^{\infty} e^{-\lambda x} f_S(x) dx \\ &= \int_{x=0}^{\infty} P(A > x) f_S(x) dx \\ &= P(A > S) = P(W = 0) = 1 - \rho. \end{aligned}$$

SECTION 6.6

| Mean Waiting Time

The *mean-value approach* use arrival relation

$$\mathbb{E}[W] = \mathbb{E}[L^q] \cdot \mathbb{E}[B] + \rho \mathbb{E}[R],$$

where R denotes residual service time of customer in service. Together with *Little's law*, $\mathbb{E}[L^q] = \lambda \mathbb{E}[W]$, this yields

$$\mathbb{E}[W] = \frac{\rho}{1 - \rho} \mathbb{E}[R].$$

Mean residual service time $\mathbb{E}[R]$ turns out to be

$$\mathbb{E}[R] = \frac{\mathbb{E}[B^2]}{2\mathbb{E}[B]} = \frac{\frac{1}{2}\mathbb{E}[B^2]}{\mathbb{E}[B]^2} \mathbb{E}[B] = \frac{1}{2} (c_B^2 + 1) \mathbb{E}[B].$$

Hence,

$$\mathbb{E}[W] = \frac{\rho}{1 - \rho} \cdot \frac{\mathbb{E}[B^2]}{2\mathbb{E}[B]} = \frac{\lambda \mathbb{E}[B^2]}{2(1 - \rho)}.$$

SECTION 6.7

| Distribution of the Busy Period

Similar derivation as in M/M/1 queue (does not depend on exponential distribution of B).

$$E(BP) = \frac{\mathbb{E}[B]}{1 - \rho}.$$

Recall

$$\mathbb{E}[S] = \frac{\rho}{2(1 - \rho)} \mathbb{E}[B](1 + c_B^2) + \mathbb{E}[B] = \frac{2 - \rho + \rho c_B^2}{2(1 - \rho)} \mathbb{E}[B].$$

Now suppose $E(BP) = \mathbb{E}[S]$, then

$$2 - \rho + \rho c_B^2 = 2 \iff \rho = \rho c_B^2 \iff c_B^2 = 1 \iff B \sim \exp().$$

Let B be service time of first customer in busy period. Let $N(B)$ be number of customers arriving during B . We will use previous expression to derive LST of BP :

$$\widetilde{BP}(s) = E(e^{-sBP}) = \int_{t=0}^{\infty} E(e^{-sBP} \mid B = t) f_B(t) dt.$$

Working out inner expectation by also conditioning on $N(B)$:

$$\begin{aligned} E(e^{-sBP} \mid B = t) &= \sum_{n=0}^{\infty} E(e^{-sBP} \mid B = t, N(B) = n) P(N(B) = n \mid B = t) \\ &= \sum_{n=0}^{\infty} E(e^{-s(t + BP_1 + \dots + BP_{N(B)})} \mid B = t, N(B) = n) \frac{(\lambda t)^n}{n!} e^{-\lambda t} \\ &= \sum_{n=0}^{\infty} E(e^{-s(t + \sum_{i=1}^n BP_i)}) \frac{(\lambda t)^n}{n!} e^{-\lambda t} \\ &= \sum_{n=0}^{\infty} e^{-st} (E(e^{-sBP}))^n \frac{(\lambda t)^n}{n!} e^{-\lambda t} \\ &= e^{-(\lambda + s)t} \sum_{n=0}^{\infty} \frac{(\lambda t \widetilde{BP}(s))^n}{n!} \\ &= e^{-(\lambda + s)t} e^{\lambda t \widetilde{BP}(s)} = e^{-(s + \lambda(1 - \widetilde{BP}(s)))t}. \end{aligned}$$

Finally, we conclude

$$\begin{aligned} \widetilde{BP}(s) &= E(e^{-sBP}) \\ &= \int_{t=0}^{\infty} E(e^{-sBP} \mid B = t) f_B(t) dt \\ &= \int_{t=0}^{\infty} e^{-(s + \lambda(1 - \widetilde{BP}(s)))t} f_B(t) dt \\ &= \widetilde{B}(s + \lambda(1 - \widetilde{BP}(s))). \end{aligned}$$

In general, it is difficult to find even $\widetilde{BP}(s)$, let alone transform it back. However, this allows us to find moments of BP by taking derivatives:

$$\widetilde{BP}'(s) = \widetilde{B}'(s + \lambda(1 - \widetilde{BP}(s))) \cdot (1 - \lambda \widetilde{BP}'(s)).$$

Setting $s = 0$:

$$\begin{aligned} -E(BP) &= \widetilde{BP}'(0) = \widetilde{B}'(0) \cdot (1 - \lambda \widetilde{BP}'(0)) = -\mathbb{E}[B](1 + \lambda E(BP)). \\ \implies E(BP) &= \mathbb{E}[B] + \rho E(BP) \implies E(BP) = \frac{\mathbb{E}[B]}{1 - \rho}. \end{aligned}$$

Similar procedure yields

$$E(BP^2) = \frac{\mathbb{E}[B^2]}{(1 - \rho)^3}.$$

Let M denotes the number of customers arriving during busy period; let $N(B)$ be the number of customers arriving during B ,

$$M = 1 + \sum_{i=1}^{N(B)} M_i.$$

We calculate probability generating function of M . First

$$\begin{aligned} E(z^M \mid B = t) &= \sum_{n=0}^{\infty} E(z^M \mid N(B) = n, B = t) P(N(B) = n \mid B = t) \\ &= \sum_{n=0}^{\infty} E(z^{1+\sum_{i=1}^n M_i}) e^{-\lambda t} \frac{(\lambda t)^n}{n!} \\ &= \sum_{n=0}^{\infty} z(P_M(z))^n e^{-\lambda t} \frac{(\lambda t)^n}{n!} \\ &= z e^{-\lambda t} e^{\lambda t P_M(z)} = z e^{-\lambda t(1-P_M(z))}. \end{aligned}$$

We conclude

$$\begin{aligned} P_M(z) &= E(z^M) \\ &= \int_{t=0}^{\infty} E(z^M \mid B = t) f_B(t) dt \\ &= \int_{t=0}^{\infty} z e^{-\lambda t(1-P_M(z))} f_B(t) dt = z \widetilde{B}(\lambda(1 - P_M(z))). \end{aligned}$$

Again, this allows us to find moments by taking derivatives:

$$P'_M(z) = \widetilde{B}'(\lambda(1 - P_M(z))) + z \widetilde{B}'(\lambda(1 - P_M(z))) \cdot (-\lambda P'_M(z)).$$

Substituting $z = 1$ yields

$$E(M) = \widetilde{B}(0) + \widetilde{B}'(0) \cdot (-\lambda E(M)) = 1 + \lambda \mathbb{E}[B] E(M).$$

Hence

$$E(M) = \frac{1}{1 - \rho}, \quad \frac{1}{E(M)} = 1 - \rho = P(W = 0).$$

Exercises

EXERCISE 6.1 (Post office). At a post office customers arrive according to a Poisson process with a rate of 30 customers per hour. A quarter of the customers wants to cash a cheque. Their service time is exponentially distributed with a mean of 2 minutes. The other customers want to buy stamps and their service times are exponentially distributed with a mean of 1 minute. In the post office there is only one server.

- (i) Determine the generating function $P_L(z)$ of the number of customers in the system.
- (ii) Determine the distribution of the number of customers in the system.
- (iii) Determine the mean number of customers in the system.
- (iv) Determine $\tilde{S}(s)$, the LST of the sojourn time.
- (v) Determine the mean and the distribution of the sojourn time.
- (vi) Determine the mean busy period duration.
- (vii) Determine the mean number of customers in the system and the mean sojourn time in case all customers have an exponentially distributed service time with a mean of 75 seconds.

EXERCISE 6.2.

EXERCISE 6.3 (Post office). At a post office customers arrive according to a Poisson process with a rate of 60 customers per hour. Half of the customers have a service time that is the sum of a fixed time of 15 seconds and an exponentially distributed time with a mean of 15 seconds. The other half have an exponentially distributed service time with a mean of 1 minute. Determine the mean waiting time and the mean number of customers waiting in the queue.

SOLUTION. As time unit we take 1 minute, so $\lambda = 1$. Let X be exponential with parameter 4 and Y be exponential with parameter 1. Then,

$$\begin{aligned}\mathbb{E}[B] &= \frac{1}{2} \cdot E\left[\frac{1}{4} + X\right] + \frac{1}{2} \cdot \mathbb{E}[Y] = \frac{3}{4}, \\ \mathbb{E}[B^2] &= \frac{1}{2} \cdot E\left[\left(\frac{1}{4} + X\right)^2\right] + \frac{1}{2} \cdot E(Y^2) = \frac{1}{2} \cdot \frac{5}{16} + \frac{1}{2} \cdot 2 = \frac{37}{32},\end{aligned}$$

and so

$$\mathbb{E}[R] = \frac{\mathbb{E}[B^2]}{2\mathbb{E}[B]} = \frac{\frac{37}{32}}{2 \cdot \frac{3}{4}} = \frac{37}{48}.$$

Hence,

$$\begin{aligned}\mathbb{E}[W] &= \frac{\rho}{1-\rho} \mathbb{E}[R] = \frac{\frac{3}{4}}{1-\frac{3}{4}} \cdot \frac{37}{48} = \frac{37}{16}, \\ \mathbb{E}[L^q] &= \frac{\lambda^2 \mathbb{E}[B^2]}{2(1-\rho)} = \frac{\frac{37}{32}}{2(1-\frac{3}{4})} = \frac{37}{16}.\end{aligned}$$

□

EXERCISE 6.4 (Two Phase Production).

EXERCISE 6.5. Consider an $M/G/1$ queue, where the server successfully completes a service time with probability p . If a service time is not completed successfully, it has to be repeated until it is successful. Determine the mean sojourn time of a customer in the following two cases:

- (i) The repeated service times are identical.
- (ii) The repeated service times are independent, and thus (possibly) different.

EXERCISE 6.6.

EXERCISE 6.7.

EXERCISE 6.8 (Robotic Dairy Barn). In a robotic dairy barn cows are automatically milked by a robot. The cows are lured into the robot by a feeder with nice food that can only be reached by first passing through the robot. When a cow is in the robot, the robot first detects whether the cow has to be milked. If so, then the cow will be milked, and otherwise, the cow can immediately leave the robot and walk to the feeder. A visit to the robot with (resp. without) milking takes an exponential time with a mean of 6 (resp. 3) minutes. Cows arrive at the robot according to a Poisson process with a rate of 10 cows per hour, a quarter of which will be milked.

- (i) Show that the LST of the service time in minutes of an arbitrary cow at the robot is given by

$$\tilde{B}(s) = \frac{1}{4} \cdot \frac{4 + 21s}{(1 + 6s)(1 + 3s)}.$$

- (ii) Show that the LST of the waiting time in minutes of an arbitrary cow in front of the robot is given by

$$\tilde{W}(s) = \frac{3}{8} + \frac{9}{16} \cdot \frac{1}{1 + 12s} + \frac{1}{16} \cdot \frac{1}{1 + 4s}.$$

- (iii) Determine the fraction of cows for which the waiting time in front of the robot is less than 3 minutes.
- (iv) Determine the mean waiting time in front of the robot.

SOLUTION. The time unit is 1 minute: $\lambda = \frac{1}{6}$.

- (i)

$$\tilde{B}(s) = \frac{1}{4} \cdot \frac{1}{1 + 6s} + \frac{3}{4} \cdot \frac{1}{1 + 3s} = \frac{4 + 21s}{4(1 + 6s)(1 + 3s)}.$$

- (ii)

$$\mathbb{E}[B] = \frac{1}{4} \cdot 6 + \frac{3}{4} \cdot 3 = \frac{15}{4}.$$

Then $\rho = \lambda \mathbb{E}[B] = \frac{5}{8}$.

$$\tilde{W}(s) = \frac{(1 - \rho)s}{\lambda \tilde{B}(s) + s - \lambda}$$

$$\begin{aligned}
&= \frac{(1 - \frac{5}{8})s}{\frac{1}{6} \cdot \frac{4+21s}{4(1+6s)(1+3s)} + s - \frac{1}{6}} \\
&= \frac{3}{8} + \frac{9}{16} \cdot \frac{1}{1+12s} + \frac{1}{16} \cdot \frac{1}{1+4s}.
\end{aligned}$$

(iii) The distribution function of the waiting time is given by

$$F_W(x) = \frac{3}{8} + \frac{9}{16}(1 - e^{-12x}) + \frac{1}{16}(1 - e^{-4x}).$$

Hence, the fraction of cows for which the waiting time is less than 3 minutes equals

$$F_W(3) = \frac{3}{8} + \frac{9}{16}(1 - e^{-36}) + \frac{1}{16}(1 - e^{-12}) \approx 0.532.$$

(iv)

$$\begin{aligned}
\mathbb{E}[B^2] &= \frac{1}{4} \cdot \frac{2}{(\frac{1}{6})^2} + \frac{3}{4} \cdot \frac{2}{(\frac{1}{3})^2} = \frac{63}{2}, \\
\mathbb{E}[R] &= \frac{\mathbb{E}[B^2]}{2\mathbb{E}[B]} = \frac{\frac{63}{2}}{2 \cdot \frac{15}{4}} = \frac{21}{5}, \\
\mathbb{E}[W] &= \frac{\rho}{1-\rho} \mathbb{E}[R] = \frac{\frac{5}{8}}{1-\frac{5}{8}} \cdot \frac{21}{5} = 7.
\end{aligned}$$

□

EXERCISE 6.9. Consider a machine processing parts. These parts arrive according to a Poisson stream with a rate of 1 product per hour. The machine processes one part at a time. Each part receives two operations. The first operation takes an exponential time with a mean of 15 minutes. The second operation is done immediately after the first one and it takes an exponential time with a mean of 20 minutes.

- (i) Show that the LST of the production lead time (waiting time plus processing time) in hours is given by

$$\tilde{S}(s) = \frac{5}{4} \cdot \frac{1}{1+s} - \frac{1}{4} \cdot \frac{1}{5+s}.$$

- (ii) Determine the distribution of the production lead time and the mean production lead time.
- (iii) One uses 3 hours as a norm for the production lead time. When the production lead time of a part exceeds this norm, it costs 100 dollars. Calculate the mean cost per hour.

EXERCISE 6.10 (Warehouse). In a warehouse for small items orders arrive according to a Poisson stream with a rate of 6 orders per hour. An order is a list with the quantities of products requested by a customer. The orders are picked one at a time by one order picker. For a quarter of the orders the pick time is exponentially distributed with a mean of 10 minutes and for the other orders the pick time is exponentially distributed with a mean of 5 minutes.

- (i) Show that the LST of the pick time in minutes of an arbitrary order is given by

$$\tilde{B}(s) = \frac{1}{4} \cdot \frac{4+35s}{(1+10s)(1+5s)}.$$

- (ii) Show that the LST of the lead time (waiting time plus pick time) in minutes of an arbitrary order is given by

$$\tilde{S}(s) = \frac{5}{32} \cdot \frac{3}{3+20s} + \frac{27}{32} \cdot \frac{1}{1+20s}.$$

- (iii) Determine the fraction of orders for which the lead time is longer than half an hour.
 (iv) Determine the mean lead time.

SOLUTION. The time unit is 1 minute: $\lambda = \frac{1}{10}$.

(i)

$$\tilde{B}(s) = \frac{1}{4} \cdot \frac{1}{1+10s} + \frac{3}{4} \cdot \frac{1}{1+5s} = \frac{4+35s}{4(1+10s)(1+5s)}.$$

(ii)

$$\mathbb{E}[B] = \frac{1}{4} \cdot 10 + \frac{3}{4} \cdot 5 = \frac{25}{4}.$$

Then $\rho = \lambda \mathbb{E}[B] = \frac{5}{8}$.

$$\begin{aligned} \tilde{S}(s) &= \frac{(1-\rho)\tilde{B}(s)s}{\lambda\tilde{B}(s) + s - \lambda} \\ &= \frac{(1-\frac{5}{8}) \frac{(4+35s)s}{4(1+10s)(1+5s)}}{\frac{1}{10} \cdot \frac{4+35s}{4(1+10s)(1+5s)} + s - \frac{1}{10}} \\ &= \frac{5}{32} \cdot \frac{3}{3+20s} + \frac{27}{32} \cdot \frac{1}{1+20s}. \end{aligned}$$

- (iii) The distribution function of the sojourn time is given by

$$F_S(x) = \frac{5}{32} \left(1 - e^{-\frac{3}{20}x}\right) + \frac{27}{32} \left(1 - e^{-\frac{1}{20}x}\right).$$

Hence, the fraction of orders for which the lead time is longer than half an hour is given by

$$1 - F_S(30) = \frac{5}{32} \cdot e^{-\frac{9}{2}} + \frac{27}{32} \cdot e^{-\frac{3}{2}} \approx 0.190.$$

(iv)

$$\mathbb{E}[B^2] = \frac{1}{4} \cdot \frac{2}{\left(\frac{1}{10}\right)^2} + \frac{3}{4} \cdot \frac{2}{\left(\frac{1}{5}\right)^2} = \frac{175}{2},$$

$$\mathbb{E}[R] = \frac{\mathbb{E}[B^2]}{2\mathbb{E}[B]} = \frac{\frac{175}{2}}{2 \cdot \frac{25}{4}} = 7,$$

$$\mathbb{E}[S] = \mathbb{E}[W] + \mathbb{E}[B] = \frac{\rho}{1-\rho} \mathbb{E}[R] + \mathbb{E}[B] = \frac{\frac{5}{8}}{1-\frac{5}{8}} \cdot 7 + \frac{25}{4} = \frac{215}{12}.$$

□

EXERCISE 6.11 (Machine with Breakdowns). Consider a machine processing parts. Per hour arrive according to a Poisson process on average 5 orders for the production of a part. The processing time of a part is exactly 10 minutes. During processing, however, tools can break down. When this occurs, the machine immediately stops, broken tools are replaced by new ones and then the machine resumes production again. Replacing tools takes exactly 2 minutes. The time that elapses between two breakdowns is exponentially distributed with a mean of 20 minutes. Hence, the total production time of a part, B , consists of the processing time of 10 minutes plus a random number of interruptions of 2 minutes to replace broken tools.

- (i) Determine the mean and variance of the production time B .
- (ii) Determine the mean lead time (waiting time plus production time) of an order.

EXERCISE 6.12.

EXERCISE 6.13.

$M/G/1$ Queue with Priorities

We consider an $M/G/1$ queue with r types of customers. The type i customers arrive according to a Poisson stream with rate λ_i , $i = 1, \dots, r$. The service time and residual service of a type i customer is denoted by B_i and R_i , respectively. Occupation rate of type i is $\rho_i = \lambda_i E(B_i)$, for stability $\sum_{i=1}^r \rho_i < 1$. The type 1 customers have the highest priority, type 2 customers the second highest priority and so on.

SECTION 7.1

| Non-Preemptive Priority

The mean waiting time of a type i customer is denoted by $E(W_i)$ and $E(L_i^q)$ is the number of type i customers waiting in the queue. For the highest priority customers it holds that

$$E(W_1) = E(L_1^q)E(B_1) + \sum_{j=1}^r \rho_j E(R_j).$$

According to Little's law we have

$$E(L_1^q) = \lambda_1 E(W_1).$$

Combining the two equations yields

$$E(W_1) = \frac{\sum_{j=1}^r \rho_j E(R_j)}{1 - \rho_1}.$$

The mean waiting time for a type 2 customer thus consists of several parts: X_1 is the amount of work upon arrival of customers in service and in queue with the same or higher priority; X_{k+1} is the portion of work arriving during the work that is done during portion X_k . In total we have that

$$E(W_2) = \sum_{k=1}^{\infty} E(X_k).$$

For the first portion

$$E(X_1) = E(L_1^q)E(B_1) + E(L_2^q)E(B_2) + \sum_{j=1}^r \rho_j E(R_j).$$

We therefore condition on the length of X_k . Denote the density of X_k by $f_k(x)$. Then it follows that

$$\begin{aligned} E(X_{k+1}) &= \int_0^\infty E(X_{k+1}|X_k = x) f_k(x) dx \\ &= \int_0^\infty (\lambda_1 x E(B_1) + \cdots + \lambda_{i-1} x E(B_{i-1})) f_k(x) dx \\ &= (\rho_1 + \cdots + \rho_{i-1}) E(X_k). \end{aligned}$$

By recursion, we have

$$E(X_{k+1}) = \rho_1^k E(X_1)$$

We thus have that

$$\begin{aligned} E(W_2) &= \sum_{k=1}^{\infty} \rho_1^{k-1} E(X_1) \\ &= \frac{E(X_1)}{1 - \rho_1} \\ &= \frac{E(L_1^q)E(B_1) + E(L_2^q)E(B_2) + \sum_{j=1}^r \rho_j E(R_j)}{1 - \rho_1}. \end{aligned}$$

Apply Little's law to customers of type $i = 1, 2$, $E(L_i^q) = \lambda_i E(W_i)$, to obtain

$$E(W_2) = \frac{\sum_{j=1}^r \rho_j E(R_j)}{(1 - \rho_1)(1 - \rho_1 - \rho_2)}.$$

The mean waiting time for a type i customer is given by

$$E(W_i) = \sum_{k=1}^{\infty} E(X_k).$$

We have

$$E(X_1) = \sum_{j=1}^i E(L_j^q)E(B_j) + \sum_{j=1}^r \rho_j E(R_j).$$

To determine $E(X_{k+1})$, conditioning on the length of X_k , with $f_k(x)$ the density of X_k ,

$$\begin{aligned} E(X_{k+1}) &= \int_0^\infty E(X_{k+1}|X_k = x) f_k(x) dx \\ &= \int_0^\infty \sum_{j=1}^{i-1} \lambda_j x E(B_j) f_k(x) dx = \sum_{j=1}^{i-1} \rho_j E(X_k). \end{aligned}$$

By recursion, we have

$$E(X_{k+1}) = \left(\sum_{j=1}^{i-1} \rho_j \right)^k E(X_1).$$

Using $E(W_i) = \sum_{k=1}^{\infty} E(X_k)$, we get

$$E(W_i) = \frac{\sum_{j=1}^i E(L_j^q)E(B_j) + \sum_{j=1}^r \rho_j E(R_j)}{1 - \sum_{j=1}^{i-1} \rho_j}.$$

Apply Little's law to customers of type $j = 1, 2, \dots, i$, $E(L_j^q) = \lambda_j E(W_j)$. After rewriting and simplification, we get that

$$E(W_i) = \frac{\sum_{j=1}^r \rho_j E(R_j)}{(1 - \sum_{j=1}^{i-1} \rho_j)(1 - \sum_{j=1}^i \rho_j)}.$$

SECTION 7.2

| Preemptive-Resume Priority

When a higher-priority type customer arrives, it forces a lower-priority class in service back into the queue and goes into service itself. Ideas for the derivation of $\mathbb{E}[W_i]$ are almost identical to non-preemptive priority.

Let W_i be the time until type i customer goes into service for the first time. For preemptive resume priority, we have similar formulas with a slight adaption

$$\mathbb{E}[W_i] = \frac{\sum_{j=1}^i \rho_j \mathbb{E}[R_j]}{\left(1 - \sum_{j=1}^{i-1} \rho_j\right) \left(1 - \sum_{j=1}^i \rho_j\right)}.$$

Note that the numerator is now $\sum_{j=1}^i \rho_j E(R_j)$ instead of $\sum_{j=1}^r \rho_j E(R_j)$. For type i , customers of lower priority do not exist.

SECTION 7.3

| A Conservation Law

Assume that customers enter service in an order independent of their service times and they may not be interrupted during their service. The amount of work V in the system does not depend on the order in which the customers are served, and

$$E(V) = \sum_{i=1}^r E(L_i^q)E(B_i) + \sum_{i=1}^r \rho_i E(R_i).$$

Using Little's law $E(L_i^q) = \lambda_i E(W_i)$, we have

$$\sum_{i=1}^r \rho_i E(W_i) = E(V) - \sum_{i=1}^r \rho_i E(R_i) = \text{constant}.$$

In addition, assume customer of type i causes a cost c_i per time unit while in the system, and server works according to non-preemptive priority discipline. We aim to determine the optimal order to minimise average cost per time unit.

 $c\mu$ Rule

Start with determining the total average cost rate C :

$$C = \sum_{i=1}^r c_i E(L_i) = \sum_{i=1}^r c_i \lambda_i E(S_i).$$

Hence,

$$\begin{aligned} C &= \sum_{i=1}^r c_i \lambda_i (E(W_i) + E(B_i)) \\ &= \sum_{i=1}^r c_i \rho_i \frac{1}{E(B_i)} (E(W_i) + E(B_i)) = \tilde{C} + \sum_{i=1}^r c_i \rho_i, \end{aligned}$$

with

$$\tilde{C} = \sum_{i=1}^r c_i \rho_i \frac{1}{E(B_i)} E(W_i).$$

We thus need to minimise

$$\tilde{C} = \sum_{i=1}^r c_i \rho_i \frac{1}{E(B_i)} E(W_i) = \sum_{i=1}^r f_i g_i,$$

with $f_i = \frac{c_i}{E(B_i)}$ and $g_i = \rho_i E(W_i)$. Observe that f_i and ρ_i are constant and independent of the service order.

$$\sum_{i=1}^r g_i = \sum_{i=1}^r \rho_i E(W_i)$$

is constant, but individual $E(W_i)$ can be influenced. Prioritise customers with highest f_i in order to minimise $\tilde{C}$, then highest f_i corresponds with lowest $E(W_i)$.

This is a heuristic derivation of the $c\mu$ rule: prioritise customer types based on $f_i = \frac{c_i}{E(B_i)}$; μ stands for the mean service rate $\frac{1}{E(B_i)}$; the higher f_i , the higher the priority.

Remark. The higher the cost, the higher the priority. The longer the mean service time, the lower the priority. Higher moments of the service time influence mean waiting times, but do not play a role in this derivation.

Shortest Processing Time First

Assume that service requirements can be observed. We consider the SPTF service discipline: after each service completion, the customer with the shortest service time is served next. This is a special case of a model with non-preemptive priority and infinitely many priority classes. A type- x customer has a service time between x and $x + dx$. Let $E(W(x))$ denote the mean waiting time of a type x customer. Let $\rho(x) = \lambda x f_B(x)$ be fraction of time type x customers are served.

For the non-preemptive priority discipline, we have

$$E(W_i) = \frac{\sum_{j=1}^r \rho_j E(R_j)}{(1 - \sum_{j=1}^{i-1} \rho_j)(1 - \sum_{j=1}^i \rho_j)}.$$

Instead of r types, view the customer in service as an arbitrary customer. Replace $\sum_{j=1}^r \rho_j E(R_j)$ by $\rho \mathbb{E}[R]$. Now, modify the denominator: instead of

$$(1 - \sum_{j=1}^{i-1} \rho_j)(1 - \sum_{j=1}^i \rho_j),$$

we need to consider type x whereas we have infinitely many types; replace individual ρ_i 's by fraction of time serving type x , i.e., $\rho(x) = \lambda x f_B(x)$; The denominator should be equal to

$$\left(1 - \lambda \int_{y=0}^x y f_B(y) dy\right)^2.$$

Combining, the mean waiting time of a type x customer is

$$E(W(x)) = \frac{\rho \mathbb{E}[R]}{\left(1 - \lambda \int_{y=0}^x y f_B(y) dy\right)^2}.$$

So, the mean waiting time for an arbitrary customer is

$$\mathbb{E}[W] = \int_{x=0}^{\infty} E(W(x)) f_B(x) dx = \rho \mathbb{E}[R] \int_{x=0}^{\infty} \frac{f_B(x)}{\left(1 - \lambda \int_{y=0}^x y f_B(y) dy\right)^2} dx.$$

Consider simplest case: suppose that $B \sim \exp(\mu)$, try to determine $E(W(x))$ and $\mathbb{E}[W]$.

$$E(W(x)) = \frac{\rho \mathbb{E}[R]}{\left(1 - \lambda \int_{y=0}^x y f_B(y) dy\right)^2}, \quad \mathbb{E}[W] = \int_{x=0}^{\infty} \frac{\rho \mathbb{E}[R] f_B(x)}{\left(1 - \lambda \int_{y=0}^x y f_B(y) dy\right)^2} dx.$$

The SPTF rule is optimal in the class of all non-preemptive priority rules when we want to minimise the mean queue length at an arbitrary time. Motivation: for r types, the $c\mu$ rule is optimal when minimising

$$C = \sum_{i=1}^r c_i E(L_i).$$

Take $c_i = 1$ to minimise mean number of customers, thus prioritise according to $1/E(B_i)$. Favor customers with smallest service time.

Shortest Remaining Processing Time First

Similar to SPTF, but allow for service interruptions. Job in service with a higher remaining service time is put back in the queue. At any moment, job in service is the one with the smallest remaining processing time.

Adjust formula derived for the non-preemptive priority case (SPTF rule) to the preemptive case. Under SPTF we have

$$E(W(x)) = \frac{\rho \mathbb{E}[R]}{\left(1 - \lambda \int_{y=0}^x y f_B(y) dy\right)^2}.$$

Rewrite the numerator to

$$\rho \mathbb{E}[R] = \lambda \mathbb{E}[B] \mathbb{E}[B^2] / (2 \mathbb{E}[B]) = \lambda \mathbb{E}[B^2] / 2.$$

Under SRPT, type x not influenced by jobs with remaining service of at least x . Replace the factor $\mathbb{E}[B^2]$ by $E(\min\{B, x\}^2)$. Under SRPT, the mean waiting time until type x is first taken into service

$$E(W(x)) = \frac{\lambda}{2} \frac{E(\min\{B, x\}^2)}{\left(1 - \lambda \int_{y=0}^x y f_B(y) dy\right)^2}.$$

The mean waiting time for an arbitrary customer is

$$\mathbb{E}[W] = \int_{x=0}^{\infty} \frac{\lambda}{2} \frac{E(\min\{B, x\}^2)}{\left(1 - \lambda \int_{y=0}^x y f_B(y) dy\right)^2} f_B(x) dx.$$

SRPT minimises the number of jobs at any point in time and is thus optimal in that sense. Suppose a scheduling discipline that behaves like SRPT except it exchanges the order of two jobs j and k . Which is best?

At some point t , jobs j and k are in the system with remaining service times b_j and b_k with $b_j < b_k$. Under SRPT job j is in service, while under the alternative, job k is in service at time t and remains in service for some interval $(t, t + v)$ with $v > 0$. The alternative service rule thus works for some time on job k while job j is in the queue.

Suppose a scheduling discipline that behaves like SRPT except it exchanges the order of two jobs j and k . The remaining total service time of jobs j and k combined is $b_j + b_k$, independent of the service order. If we prioritise job j , it would finish earlier than job k , as $b_j < b_k$. Under SRPT, we have a lower queue length for some period of time than under the alternative. Any alternative can be improved, so SRPT is optimal when minimising the queue length.

| Exercises

EXERCISE 7.1. Customers arrive to a single-server queue according to a Poisson process with a rate of 10 customers per hour. Half of the customers has an exponential service time with a mean of 3 minutes. The other half has an exponential service time with a mean of 6 minutes. The first half are called type 1 customers, the second half type 2 customers. Determine in each of the following three cases the mean waiting time:

- (i) Service in order of arrival (FCFS).
- (ii) Non-preemptive priority for type 1 customers.
- (iii) Non-preemptive priority for type 2 customers.

EXERCISE 7.2.

EXERCISE 7.3. Consider a machine where jobs arrive according to a Poisson stream with a rate of 4 jobs per hour. Half of the jobs have a processing time of exactly 10 minutes, a quarter of the jobs have a processing time of exactly 15 minutes and the remaining quarter have a processing time of 20 minutes. The jobs with a processing time of 10 minutes are called type 1 jobs, the ones with a processing time of 15 minutes type 2 jobs and the rest type 3 jobs. The jobs are processed in order of arrival.

- (i) Determine the mean sojourn time (waiting time plus processing time) of a type 1, 2 and 3 job and also of an arbitrary job.

One decides to process smaller jobs with priority. So type 1 orders have highest priority, type 2 orders second highest priority and type 3 orders lowest priority. Answer question (i) for the following two cases.

- (ii) Jobs processed at the machine may not be interrupted.
- (iii) Type 1 and type 2 jobs may interrupt the processing of a type 3 job. Type 1 jobs may not interrupt the processing of a type 2 job.

EXERCISE 7.4. A machine produces a specific part type. Orders for the production of these parts arrive according to a Poisson stream with a rate of 10 orders per hour. The number of parts that has to be produced for an order is equal to n with probability 0.5^n , $n = 1, 2, \dots$. The production of one part takes exactly 2 minutes. Orders are processed in order of arrival.

- (i) Denote by B the production time in minutes of an arbitrary order. Show that

$$\mathbb{E}[B] = 4, \quad \sigma^2(B) = 8.$$

- (ii) Determine the mean production lead time (waiting time plus production time) of an arbitrary order.

The management decides to give orders for the production of 1 part priority over the other orders. But the production of an order may not be interrupted.

- (iii) Determine the mean production lead of an order for the production of 1 part, and of an order for the production of at least 2 parts.

(iv) Determine the mean production lead time of an arbitrary order.

EXERCISE 7.5. Consider a machine for mounting electronic components on printed circuit boards. Per hour arrive according to a Poisson process on average 30 printed circuit boards. The time required to mount all components on a printed circuit board is uniformly distributed between 1 and 2 minutes.

(i) Calculate the mean sojourn time of an arbitrary printed circuit board.

One decides to give printed circuit boards with a mounting time between 1 and x ($1 \leq x \leq 2$) minutes non-preemptive priority over printed circuit boards the mounting time of which is greater than x minutes.

(ii) Determine the mean sojourn time of an arbitrary printed circuit board as a function of x .
 (iii) Determine the value of x for which the mean sojourn time of an arbitrary printed circuit board is minimised, and calculate for this specific x the relative improvement with respect to the mean sojourn time calculated in (i).

EXERCISE 7.6.

EXERCISE 7.7.

EXERCISE 7.8. Consider a queue where customers arrive according to Poisson process with rate $\lambda = 1$ and assume that each customer requires a service time that is uniformly distributed on $(0, 1)$.

(i) Assume that the service order is according to the SPTF service discipline. Compute the mean waiting time for a customer of size x (and for an arbitrary customer).
 (ii) Assume that the service order is according to the SRPT service discipline. Compute the mean waiting time for a customer of size x (and for an arbitrary customer).

EXERCISE 7.9. Consider a queueing system with 3 types of customers. The service times of customers in class 1 are exponentially distributed with parameter 3, for customers in class 2 they are distributed according to an Erlang distribution with parameters 2 and 3 and for customers in class 3 they are deterministic with value 1. The arrival process for customers of class i is a Poisson process with rate λ_i with $\lambda_1 = \frac{1}{6}$, $\lambda_2 = \frac{1}{6}$ and $\lambda_3 = \frac{1}{2}$. The holding cost c_i , incurred per unit of time, for customers in class i are $c_1 = 1$, $c_2 = 2$ and $c_3 = 3$. Compute the cost

$$C = \sum_{i=1}^3 c_i \mathbb{E}[L_i]$$

for all possible non-preemptive priority schedules. Relate your answer to the $c\mu$ rule.

Variations of the $M/G/1$ Model

SECTION 8.1

| Machine with Setup Times

Consider a single machine where jobs are being processed in order of arrival and suppose that it is expensive to keep the machine in operation while there are no jobs. Therefore the machine is turned off as soon as the system is empty. When a new job arrives the machine is turned on again, but it takes some setup time till the machine is ready for processing. So turning off the machine leads to longer production leadtimes.

Suppose that the jobs arrive according to a Poisson stream with rate λ . The first and second moment of the processing time are denoted by $\mathbb{E}[B]$ and $\mathbb{E}[B^2]$. $E(T)$ and $E(T^2)$ are the first and second moment of the setup time. For stability we require that $\rho = \lambda\mathbb{E}[B] < 1$. The mean waiting time $\mathbb{E}[W]$ of a job satisfies

$$\begin{aligned}\mathbb{E}[W] = & \mathbb{E}[L^q]\mathbb{E}[B] + P(\text{Machine is working on arrival})E(R_B) \\ & + P(\text{Machine is off on arrival})E(T) \\ & + P(\text{Machine is in setup phase on arrival})E(R_T),\end{aligned}$$

where the mean residual processing and mean residual setup time are given by

$$E(R_B) = \frac{\mathbb{E}[B^2]}{2\mathbb{E}[B]}, \quad E(R_T) = \frac{E(T^2)}{2E(T)}.$$

Furthermore

$$\begin{aligned}P(\text{Machine is working on arrival}) &= \rho, \\ P(\text{Machine is off on arrival}) &= (1 - \rho) \frac{1/\lambda}{1/\lambda + E(T)}, \\ P(\text{Machine is in setup phase on arrival}) &= (1 - \rho) \frac{E(T)}{1/\lambda + E(T)}.\end{aligned}$$

This yields, using Little's law,

$$\mathbb{E}[W] = \frac{\rho E(R_B)}{1 - \rho} + \frac{1/\lambda}{1/\lambda + E(T)} E(T) + \frac{E(T)}{1/\lambda + E(T)} E(R_T).$$

A natural extension to the setup policy in the previous section is the one in which the machine is switched on when the number of jobs in the system reaches some threshold value N . The

mean waiting time now satisfies

$$\begin{aligned}\mathbb{E}[W] = \mathbb{E}[L^q]\mathbb{E}[B] + \rho E(R_B) + \sum_{i=1}^N P(\text{Arriving job is number } i) \left(\frac{N-i}{\lambda} + E(T) \right) \\ + P(\text{Machine is in setup phase on arrival})E(R_T),\end{aligned}$$

with

$$\begin{aligned}P(\text{Arriving job is number } i) &= (1 - \rho) \frac{1/\lambda}{N/\lambda + E(T)}, \\ P(\text{Machine is in setup phase on arrival}) &= (1 - \rho) \frac{E(T)}{N/\lambda + E(T)}.\end{aligned}$$

This yields, using Little's law,

$$\mathbb{E}[W] = \frac{\rho E(R_B)}{1 - \rho} + \frac{N/\lambda}{N/\lambda + E(T)} \left(\frac{N-1}{2\lambda} + E(T) \right) + \frac{E(T)}{N/\lambda + E(T)} E(R_T).$$

SECTION 8.2

| Unreliable Machine

Single machine on which jobs are processed in order of arrival. Machine can break down at any time it is operational, even when it is not processing jobs. Arrival process is Poisson process with rate λ . Processing times: first and second moment $\mathbb{E}[B]$ and $\mathbb{E}[B^2]$. Time till next breakdown: exponentially distributed with mean $1/\eta$. Down times: first and second moment $E(D)$ and $E(D^2)$. Fraction of time machine is up:

$$\frac{1/\eta}{1/\eta + E[D]} = \frac{1}{1 + \eta E[D]}.$$

For stability we require:

$$\lambda \mathbb{E}[B] < \frac{1}{1 + \eta E(D)}.$$

Generalised processing time G : processing time plus the down times occurring in that processing time.

$$G = B + \sum_{i=1}^{N(B)} D_i,$$

where $N(B)$ is the number of break-downs during the processing time B and D_i is the i -th down time. One can show

$$\mathbb{E}[G] = \mathbb{E}[B] + \mathbb{E}[B]\eta E(D),$$

and

$$\mathbb{E}[G^2] = \mathbb{E}[B^2] (1 + \eta E(D))^2 + \mathbb{E}[B]\eta E(D^2).$$

For the mean waiting time it holds that

$$\begin{aligned} \mathbb{E}[W] &= \mathbb{E}[L^q]\mathbb{E}[G] + \rho_G E(R_G) \\ &\quad + P[\text{Arrival in a down time in a non-processing period}]E(R_D), \end{aligned}$$

where ρ_G is the fraction of time the machine works on generalised jobs and $E(R_G)$ and $E(R_D)$ denote the mean residual generalised processing time and the mean residual down time, so

$$\rho_G = \lambda \mathbb{E}[G], \quad E(R_G) = \frac{\mathbb{E}[G^2]}{2\mathbb{E}[G]}, \quad E(R_D) = \frac{E(D^2)}{2E(D)}.$$

The probability that a job arriving in a non-processing period finds that the machine is down, is given by

$$\frac{E(D)\eta/(\lambda + \eta)}{1/(\lambda + \eta) + E(D)\eta/(\lambda + \eta)} = \frac{E(D)\eta}{1 + E(D)\eta}.$$

(Non-processing periods consist of up-times with average length $1/(\lambda + \eta)$ followed with probability $\eta/(\lambda + \eta)$ by down-times with average length $E(D)$). Since a job arrives with probability $1 - \rho_G$ in a non-processing period, we have

$$P(\text{Arrival in a down time in a non-processing period}) = (1 - \rho_G) \frac{E(D)\eta}{1 + E(D)\eta}.$$

Substitution all relations in the arrival relation and using Little's law yields

$$\mathbb{E}[W] = \frac{\rho_G E(R_G)}{1 - \rho_G} + \frac{E(D)\eta}{1 + E(D)\eta} E(R_D),$$

and the mean production lead time finally follows from

$$\mathbb{E}[S] = \mathbb{E}[W] + \mathbb{E}[G].$$

SECTION 8.3

| Exceptional First Customer in a Busy Period

B_f and R_f denote service time and residual service time of first customer in busy period. B and R denote service time and residual service time of all other customers in busy period. ρ_f denotes fraction of time server works on first customers. ρ denotes fraction of time server works on other (ordinary) customers. For the mean waiting time we have

$$\mathbb{E}[W] = \mathbb{E}[L^q]\mathbb{E}[B] + \rho_f E(R_f) + \rho \mathbb{E}[R],$$

Together with Little's law we then obtain that

$$\mathbb{E}[W] = \frac{\rho_f E(R_f) + \rho \mathbb{E}[R]}{1 - \lambda \mathbb{E}[B]}.$$

The probability that an arriving customer is the first one in a busy cycle is $1 - \rho_f - \rho$. The probability that an arriving customer is an other customer is $\rho_f + \rho$. Hence, for the mean sojourn time we get

$$\mathbb{E}[S] = \mathbb{E}[W] + (1 - \rho_f - \rho)E(B_f) + (\rho_f + \rho)\mathbb{E}[B].$$

It remains to determine ρ_f and ρ . Number of first customers arriving per time unit is $\lambda(1 - \rho_f - \rho)$. Number of other customers arriving per time unit is $\lambda(\rho_f + \rho)$. Thus ρ_f and ρ satisfy

$$\begin{aligned}\rho_f &= \lambda(1 - \rho_f - \rho)E(B_f), \\ \rho &= \lambda(\rho_f + \rho)\mathbb{E}[B],\end{aligned}$$

from which it follows that

$$\rho_f = \frac{\lambda E(B_f)(1 - \lambda \mathbb{E}[B])}{1 + \lambda E(B_f) - \lambda \mathbb{E}[B]}, \quad \rho = \frac{\lambda E(B_f)\lambda \mathbb{E}[B]}{1 + \lambda E(B_f) - \lambda \mathbb{E}[B]}.$$

SECTION 8.4

| Group Arrivals

$M/G/1$ queue where customers arrive in groups. Mean processing times and mean residual processing times of customers are denoted with $\mathbb{E}[B]$ and $\mathbb{E}[R]$ respectively. Groups arrive according to a Poisson process with rate λ . Group size is denoted by the random variable G with probability distribution

$$g_k = P(G = k), \quad k = 0, 1, 2, \dots$$

The mean waiting time is given by

$$\mathbb{E}[W] = \mathbb{E}[L^q]\mathbb{E}[B] + \rho\mathbb{E}[R] + \sum_{k=1}^{\infty} r_k(k-1)\mathbb{E}[B],$$

where $\rho = \lambda\mathbb{E}[G]\mathbb{E}[B]$ is the server utilisation, and r_k is the probability that our customer is the k -th customer served in his group.

To find r_k we first determine the probability h_n that our customer is a member of a group of size n . Since it is more likely that our customer belongs to a large group than to a small one, it follows that h_n is proportional to the group size n . Since it is more likely that our customer belongs to a group that happens more often than to a group that happens rarely, it follows that h_n is proportional to the probability g_n a certain group size n occurs. Thus we can write

$$h_n = Cng_n,$$

where C is a constant to normalise this distribution. So

$$C^{-1} = \sum_{n=1}^{\infty} ng_n = \mathbb{E}[G].$$

Hence

$$h_n = \frac{ng_n}{\mathbb{E}[G]}, \quad n = 1, 2, \dots$$

Given that our customer is a member of a group of size n , he will be with probability $1/n$ the k -th customer in his group going into service (of course, $n \geq k$). So we obtain

$$r_k = \sum_{n=k}^{\infty} h_n \cdot \frac{1}{n} = \frac{1}{\mathbb{E}[G]} \sum_{n=k}^{\infty} g_n,$$

furthermore,

$$\begin{aligned} \sum_{k=1}^{\infty} r_k(k-1)\mathbb{E}[B] &= \frac{1}{\mathbb{E}[G]} \sum_{k=1}^{\infty} \sum_{n=k}^{\infty} g_n(k-1)\mathbb{E}[B] \\ &= \frac{1}{\mathbb{E}[G]} \sum_{n=1}^{\infty} \sum_{k=1}^n g_n(k-1)\mathbb{E}[B] \\ &= \frac{1}{\mathbb{E}[G]} \sum_{n=1}^{\infty} g_n \sum_{k=1}^n (k-1)\mathbb{E}[B] \\ &= \frac{1}{\mathbb{E}[G]} \sum_{n=1}^{\infty} g_n \frac{1}{2}n(n-1)\mathbb{E}[B] \end{aligned}$$

$$= \frac{\mathbb{E}[G^2] - \mathbb{E}[G]}{2\mathbb{E}[G]} \mathbb{E}[B].$$

Combining all these results and using *Little's law* $\mathbb{E}[L^q] = \lambda \mathbb{E}[G] \mathbb{E}[W]$, we finally obtain

$$\mathbb{E}[W] = \frac{\rho}{1-\rho} \mathbb{E}[R] + \frac{\mathbb{E}[G^2] - \mathbb{E}[G]}{2\mathbb{E}[G](1-\rho)} \mathbb{E}[B].$$

The first term at the right-hand side is equal to the mean waiting time in the system where customers arrive one by one according to a Poisson process with rate $\lambda \mathbb{E}[G]$. Clearly, the second term indicates the extra mean delay due to clustering of arrivals.

EXAMPLE 8.1 (Uniform Group Sizes). In case the group size is uniformly distributed over $1, 2, \dots, n$, so

$$g_k = \frac{1}{n}, \quad k = 1, \dots, n,$$

we find

$$\mathbb{E}[G] = \sum_{k=1}^n \frac{k}{n} = \frac{n+1}{2}, \quad \mathbb{E}(G^2 - G) = \sum_{k=1}^n \frac{k^2 - k}{n} = \frac{(n-1)(n+1)}{3}.$$

Hence

$$\mathbb{E}[W] = \frac{\rho \mathbb{E}[R]}{1-\rho} + \frac{(n-1)\mathbb{E}[B]}{3(1-\rho)}.$$

| Exercises

EXERCISE 8.1.

EXERCISE 8.2.

EXERCISE 8.3.

EXERCISE 8.4.

EXERCISE 8.5. A machine produces products in two phases. The first phase is standard and the same for all products. The second phase is customer specific (the finishing touch). The first (resp. second) phase takes an exponential time with a mean of 10 (resp. 2) minutes. Orders for the production of one product arrive according to a Poisson stream with a rate of 3 orders per hour. Orders are processed in order of arrival.

- (i) Determine the mean production lead time (waiting time plus production time) of an order.

The machine is switched off when the system is empty and it is switched on again as soon as the first order arrives. A fixed cost of £20 is incurred each time the machine is switched on (the time needed to switch the machine on or off is negligible).

- (ii) Determine the average switch-on cost per hour.

To reduce the production lead time one decides to start already with the production of phase 1 when the system is empty. If upon completion of phase 1 no order has arrived yet, the production stops and the machine is switched off. When the first order arrives the machine is switched on again and can directly start with phase 2.

- (iii) Determine the reduction in the mean production lead time.
- (iv) Determine the average switch-on cost per hour.

EXERCISE 8.6. In a bank there is a special office for insurances. Here arrive according to a Poisson process on average 8 customers per hour. The service times are exponentially distributed with a mean of 5 minutes. As soon as all customers are served and the office is empty again, the clerk also leaves to get some coffee. He returns after an exponential time with a mean of 5 minutes and then starts servicing the waiting customers (if there are any) or patiently waits for the first customer to arrive.

- (i) What is the mean sojourn time and the mean number of customers in the office?

Now suppose that when the clerk finds an empty office upon his return, he immediately leaves again to get another cup of coffee. This takes again an exponential time with a mean of 5 minutes. The clerk repeats to leave until he finds a waiting customer upon return.

- (ii) How many cups of coffee the clerk drinks on average before he starts servicing again?
- (iii) Determine the mean sojourn time and the mean number of customers in the office.

EXERCISE 8.7.

EXERCISE 8.8. Consider a queueing system where on average 3 groups of customers arrive per hour. The mean group size is 10 customers. The service time is exactly 1 minute for each customer. Determine the mean sojourn time of the first customer in a group, the last customer in a group and an arbitrary one in the following two cases:

- (i) The group size is Poisson distributed.
- (ii) The group size is geometrically distributed.

EXERCISE 8.9. Customers arrive in groups at a server. The groups consist of 1 or 3 customers, both with equal probability, and they arrive according to a Poisson stream with a rate of 2 groups per hour. Customers are served one by one, and they require an exponential service time with a mean of 10 minutes.

- (i) Determine the mean sojourn time of an arbitrary customer.
- (ii) Determine the mean number of customers in the system.

EXERCISE 8.10. Passengers are brought with small vans from the airport to hotels nearby. At one of those hotels on average 6 vans per hour arrive according to a Poisson process. With probability $\frac{1}{4}$ a van brings 2 guests for the hotel, with probability $\frac{1}{4}$ only one guest and with probability $\frac{1}{2}$ no guests at all. At the reception of the hotel there is always one receptionist present. It takes an exponential time with a mean of 5 minutes to check in a guest.

- (i) Determine the distribution of the number of guests at the reception.
- (ii) Determine the mean waiting time of an arbitrary guest at the reception.

EXERCISE 8.11.

G/M/1 Queue

In this system customers arrive one by one with interarrival times *i.i.d.* according to an arbitrary distribution function $F_A(\cdot)$ with density $f_A(\cdot)$. As usual let $\lambda := \frac{1}{\mathbb{E}[A]}$ be the arrival rate of customers. The service times are exponentially distributed with mean $\frac{1}{\mu}$. For stability we again require that the occupation rate $\rho = \frac{\lambda}{\mu}$ is less than one.

Interarrival times are not memoryless anymore. Embedded discrete-time Markov chain: number of customers in system just before arrival instants of customers. Instead of two-dimensional continuous-time Markov chain with continuous state space, we get one-dimensional discrete-time Markov chain with discrete state space.

Remark. $a_n = d_n$ for all n still holds. $a_n = p_n$ for all n does not hold anymore. Arrival process is not a Poisson process (hence no PASTA in this case).

SECTION 9.1

| Arrival Distribution

$$a_n = P(L^a = n) = \lim_{k \rightarrow \infty} P(L_k^a = n).$$

Denote with D_{k+1} the number of customers served during interarrival time between customer k and customer $k + 1$. Then we have

$$L_{k+1}^a = L_k^a + 1 - D_{k+1}.$$

Process $\{L_k^a\}_{k=0}^\infty$ is discrete-time Markov chain with transition probabilities

$$p_{i,j} = \begin{cases} 0, & j > i + 1 \\ P(D_{k+1} = i + 1 - j), & j \leq i + 1 \end{cases}.$$

Denote with

$$\beta_n = \int_{t=0}^{\infty} \frac{(\mu t)^n}{n!} e^{-\mu t} f_A(t) dt,$$

which is the probability of serving n customers during an interarrival time when the server is *always serving* during this interval (thus there are always customers present). Then from

$$p_{i,j} = \begin{cases} 0, & j > i + 1 \\ \beta_{i+1-j}, & j \leq i + 1 \end{cases}.$$

we conclude

$$p_{i,j} = \begin{cases} 0, & j > i+1 \\ \beta_{i+1-j}, & 0 < j \leq i+1 \\ \sum_{h=i+1}^{\infty} \beta_h, & j = 0 \end{cases}$$

Hence the matrix P of transition probabilities of $\{L_k^a\}_{k=0}^{\infty}$ takes the form

$$P = \begin{pmatrix} p_{0,0} & \beta_0 & 0 & 0 & \cdots \\ p_{1,0} & \beta_1 & \beta_0 & 0 & \cdots \\ p_{2,0} & \beta_2 & \beta_1 & \beta_0 & 0 & \cdots \\ p_{3,0} & \beta_3 & \beta_2 & \beta_1 & \beta_0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix},$$

with

$$p_{i,0} = 1 - \sum_{h=0}^i \beta_h = \sum_{h=i+1}^{\infty} \beta_h.$$

The limiting probabilities a_n satisfy the equilibrium equations:

$$a_0 = \sum_{i=0}^{\infty} a_i p_{i,0}, \quad (9.1)$$

$$a_n = \sum_{i=0}^{\infty} a_{n-1+i} \beta_i, \quad n = 1, 2, \dots \quad (9.2)$$

To find the solution of the equilibrium equations we try to find solution of the form

$$a_n = \sigma^n, \quad n = 0, 1, 2, \dots \quad (9.3)$$

Substitution of this form into equation (9.2) and dividing by the common power σ^{n-1} yields

$$\sigma = \sum_{i=0}^{\infty} \sigma^i \beta_i.$$

Using the fact that

$$\beta_i = \int_{t=0}^{\infty} \frac{(\mu t)^i}{i!} e^{-\mu t} f_A(t) dt,$$

we obtain

$$\begin{aligned} \sigma &= \sum_{i=0}^{\infty} \sigma^i \int_{t=0}^{\infty} \frac{(\mu t)^i}{i!} e^{-\mu t} f_A(t) dt \\ &= \int_{t=0}^{\infty} e^{-(\mu - \mu\sigma)t} f_A(t) dt \\ &= \tilde{A}(\mu - \mu\sigma). \end{aligned} \quad (9.4)$$

It can be shown that as long as $\rho < 1$ equation (9.4) has a unique solution in the range $0 < \sigma < 1$ (see [Exercise 9.1](#)), and this is the solution we take. Note that the remaining equilibrium equation (9.1) is also satisfied by (9.3) since the equilibrium equations are dependent. We finally have to normalise solution (9.3) yielding

$$a_n = (1 - \sigma)\sigma^n, \quad n = 0, 1, 2, \dots \quad (9.5)$$

We conclude that the queue length distribution just before an arrival of a customer is *geometric* with parameter σ , where σ is the unique solution of the equation

$$\sigma = \tilde{A}(\mu - \mu\sigma)$$

in the interval $(0, 1)$.

EXAMPLE 9.1 ($M/M/1$ Queue). Here we have $\tilde{A}(s) = \frac{\lambda}{\lambda+s}$. Hence, we find

$$\sigma = \tilde{A}(\mu - \mu\sigma) = \frac{\lambda}{\lambda + \mu - \mu\sigma},$$

so

$$\sigma(\lambda + \mu - \mu\sigma) - \lambda = (\sigma - 1)(\lambda - \mu\sigma) = 0.$$

Thus the desired solution in $(0, 1)$ is $\sigma = \rho$ and hence we have

$$a_n = (1 - \rho)\rho^n, \quad n = 0, 1, 2, \dots$$

EXAMPLE 9.2 ($E_2/M/1$ Queue). Suppose that the interarrival times are Erlang-2 distributed with mean $\frac{2}{3}$, so

$$\tilde{A}(s) = \left(\frac{3}{3+s} \right)^2.$$

Further assume that $\mu = 4$ (so $\rho = \frac{3}{2} \cdot \frac{1}{4} = \frac{3}{8} < 1$). Then we have

$$\sigma = \tilde{A}(\mu - \mu\sigma) = \left(\frac{3}{7 - 4\sigma} \right)^2.$$

Thus

$$\sigma(7 - 4\sigma)^2 - 9 = (\sigma - 1)(4\sigma - 9)(4\sigma - 1) = 0.$$

Hence the desired solution is $\sigma = \frac{1}{4}$ and

$$a_n = \frac{3}{4} \left(\frac{1}{4} \right)^n, \quad n = 0, 1, 2, \dots$$

EXAMPLE 9.3.

SECTION 9.2

| Distribution of the Sojourn Time and the Waiting Time

$$\begin{aligned}
\tilde{S}(s) &= E(e^{-sS}) \\
&= \sum_{n=0}^{\infty} a_n \left(\frac{\mu}{\mu + s} \right)^{n+1} \\
&= \sum_{n=0}^{\infty} (1 - \sigma) \sigma^n \left(\frac{\mu}{\mu + s} \right)^{n+1} \\
&= \frac{\mu(1 - \sigma)}{\mu + s} \sum_{n=0}^{\infty} \left(\frac{\mu \sigma}{\mu + s} \right)^n \\
&= \frac{\mu(1 - \sigma)}{\mu(1 - \sigma) + s}.
\end{aligned}$$

From this we can conclude that the sojourn time S is exponentially distributed with parameter $\mu(1 - \sigma)$, i.e.,

$$P(S \leq t) = 1 - e^{-\mu(1 - \sigma)t}, \quad t \geq 0.$$

$$\begin{aligned}
\tilde{W}(s) &= E(e^{-sW}) \\
&= \sum_{n=0}^{\infty} a_n \left(\frac{\mu}{\mu + s} \right)^n \\
&= \sum_{n=0}^{\infty} (1 - \sigma) \sigma^n \left(\frac{\mu}{\mu + s} \right)^n \\
&= (1 - \sigma) + (1 - \sigma) \sum_{n=1}^{\infty} \left(\frac{\sigma \mu}{\mu + s} \right)^n \\
&= (1 - \sigma) + (1 - \sigma) \frac{\frac{\sigma \mu}{\mu + s}}{1 - \frac{\sigma \mu}{\mu + s}} \\
&= (1 - \sigma) + (1 - \sigma) \cdot \frac{\sigma \mu}{\mu + s} \cdot \frac{1}{1 - \frac{\sigma \mu}{\mu + s}} \\
&= (1 - \sigma) + \sigma \cdot \frac{\mu(1 - \sigma)}{\mu(1 - \sigma) + s}.
\end{aligned}$$

From this we can conclude that the waiting time distribution is given by

$$P(W \leq t) = 1 - \sigma e^{-\mu(1 - \sigma)t}, \quad t \geq 0.$$

Note that the probability that a customer does not have to wait is given by $1 - \sigma$ (and *not* by $1 - \rho$).

SECTION 9.3

| Mean Sojourn Time

Let us try to determine the mean sojourn time directly by the mean value approach. For an arriving customer we have

$$\mathbb{E}[S] = \mathbb{E}[L^a] \frac{1}{\mu} + \frac{1}{\mu}, \quad (9.6)$$

where the random variable L^a denotes the number of customers in the system found on arrival. Unfortunately, we do not have Poisson arrivals, so

$$\mathbb{E}[L^a] \neq \mathbb{E}[L].$$

Hence the mean value approach does not work here, since we end up with only two equations for three unknowns.

Additional information is needed in the form of (9.5), yielding

$$\mathbb{E}[L^a] = \sum_{n=0}^{\infty} n a_n = \sum_{n=0}^{\infty} n(1-\sigma)\sigma^n = \frac{\sigma}{1-\sigma}.$$

Then it follows from (9.6) and Little's law $\mathbb{E}[L] = \lambda \mathbb{E}[S]$ that

$$\begin{aligned} \mathbb{E}[S] &= \frac{\sigma}{(1-\sigma)\mu} + \frac{1}{\mu} = \frac{1}{(1-\sigma)\mu}, \\ \mathbb{E}[L] &= \frac{\lambda}{(1-\sigma)\mu} = \frac{\rho}{1-\sigma}. \end{aligned}$$

The limiting distribution of the number of customers at arbitrary time points is given by

$$\begin{aligned} p_0 &= 1 - \rho, \\ p_n &= \rho a_{n-1} = \rho(1-\sigma)\sigma^{n-1}, \quad n = 1, 2, \dots \end{aligned}$$

Proof. λa_{n-1} : upcrossing rate from level $n-1$ to level n ; μp_n : downcrossing rate from level n to level $n-1$; These two rates should be equal, hence $p_n = \rho a_{n-1}$. $\square$

| Exercises

EXERCISE 9.1.

EXERCISE 9.2. In a record shop customers arrive according to a hyperexponential arrival process. The interarrival time is with probability $\frac{1}{3}$ exponentially distributed with a mean of 1 minute and with probability $\frac{2}{3}$ it is exponentially distributed with a mean of 3 minutes. The service times are exponentially distributed with a mean of 1 minute.

- (i) Calculate the distribution of the number of customers found in the record shop by an arriving customer.
- (ii) Calculate the mean number of customers in the record shop found on arrival.
- (iii) Determine $\tilde{S}(s)$.

- (iv) Determine the mean time a customer spends in the record shop.
- (v) Calculate the mean number of customers in the record shop (now at an arbitrary point in time).

SOLUTION. As time unit we choose 1 minute: $\mu = 1$.

- (i) The LST of the interarrival time distribution is given by

$$\tilde{A}(s) = \frac{1}{3} \cdot \frac{1}{1+s} + \frac{2}{3} \cdot \frac{1}{1+3s}.$$

Solving $\sigma = \tilde{A}(\mu - \mu\sigma)$, we obtain the solution in $(0, 1)$:

$$\sigma = \frac{21 - \sqrt{153}}{18} \approx 0.48.$$

Then

$$a_n = (1 - \sigma)\sigma^n \approx 0.52 \cdot 0.48^n.$$

- (ii)

$$\mathbb{E}[L^a] = \frac{\sigma}{1 - \sigma} \approx 0.92.$$

- (iii)

$$\tilde{S}(s) = \frac{\mu(1 - \sigma)}{\mu(1 - \sigma) + s} = \frac{1 - \sigma}{1 - \sigma + s} \approx \frac{0.52}{0.52 + s}.$$

- (iv)

$$\mathbb{E}[S] = \frac{1}{(1 - \sigma)\mu} = \frac{1}{1 - \sigma} \approx 1.92.$$

- (v)

$$\lambda = \frac{1}{\mathbb{E}[A]} = \frac{1}{\frac{1}{3} \cdot 1 + \frac{2}{3} \cdot 3} = \frac{3}{7},$$

$$\mathbb{E}[L] = \lambda \mathbb{E}[S] \approx 0.82.$$

□

EXERCISE 9.3.

EXERCISE 9.4.

EXERCISE 9.5.

EXERCISE 9.6. At a small river cars are brought from the left side to the right side of the river by a ferry. On average 15 cars per hour arrive according to a Poisson process. It takes the ferry an exponentially distributed time with a mean of 3 minutes to cross the river and return. The capacity of the ferry is equal to 2 cars. The ferry only takes off when there are two or more cars waiting.

- (i) What is the fraction of time that the ferry is on its way between the two river sides?
- (ii) Determine the distribution of the number of cars that are waiting for the ferry.
- (iii) Determine the mean waiting time of a car.

Insensitive Systems

Insensitive systems are queueing systems for which the equilibrium queue length distribution is *insensitive* to the distribution of the service time, but only depends on the mean of the service time.

SECTION 10.1

| M/G/∞ Queue

Customers arrive according to a Poisson process with rate λ . Their service times are *i.i.d.* with some general distribution function $F_B(x) = \mathbb{P}(B \leq x)$. The number of servers is infinite. So there is always a server available for each arriving customer. Hence, the waiting time of each customer is zero and the sojourn time is equal to the service time. Thus by *Little's law* we immediately obtain that

$$\mathbb{E}[L] = \rho,$$

where $\rho = \lambda \mathbb{E}[B]$ denotes the mean amount of work that arrives per unit time. In the sequel we want to determine the equilibrium distribution of L , i.e., the equilibrium probabilities p_n that there are n customers in the system.

EXAMPLE 10.1 (*M/M/∞ Queue*). In this model the service times are exponentially distributed with mean $\frac{1}{\mu}$. Equating flow from $n - 1$ to n and flow from n to $n - 1$ gives

$$p_{n-1}\lambda = p_n n \mu$$

and thus that

$$p_n = \frac{\lambda}{n\mu} p_{n-1} = \frac{\rho}{n} p_{n-1} = \frac{\rho^2}{n(n-1)} p_{n-2} = \cdots = \frac{\rho^n}{n!} p_0.$$

Since the probabilities p_n have to add up to one, it follows that

$$p_0 = \left(\sum_{n=0}^{\infty} \frac{\rho^n}{n!} \right)^{-1} = e^{-\rho}.$$

Summarising, we have found that

$$p_n = \frac{\rho^n}{n!} e^{-\rho}.$$

Number of customers in system has Poisson distribution with mean $\rho = \frac{\lambda}{\mu}$.

EXAMPLE 10.2 ($M/D/\infty$ Queue).

EXAMPLE 10.3 ($M/G/\infty$ Queue with Discrete Service Time Distribution).

EXAMPLE 10.4 ($M/G/\infty$ Queue with Arbitrary Service Time Distribution).

$M/G/\infty$ model is useful due to nice properties.

SECTION 10.2

| $M/G/c/c$ Queue

Customers arrive according to a Poisson process with rate λ . Service times are independent and identically distributed with general distribution function. There are c servers available. Each newly arriving customer immediately goes into service if there is a server available, and that customer is lost if all servers are occupied. This system is also referred to as the $M/G/c$ loss system.

Of special interest is the probability p_c , which, according to the PASTA property, describes the fraction of customers that are lost.

Equating flow from $n - 1$ to n and flow from n to $n - 1$ gives $p_{n-1}\lambda = p_n n\mu$, and thus that

$$p_n = \frac{\lambda}{n\mu} p_{n-1} = \frac{\rho}{n} p_{n-1} = \frac{\rho^2}{n(n-1)} p_{n-2} = \cdots = \frac{\rho^n}{n!} p_0,$$

where $\rho = \lambda\mathbb{E}[B]$. Since the probabilities p_n have to add up to one, we find that

$$p_n = \frac{\rho^n/n!}{\sum_{n=0}^c \rho^n/n!}, \quad n = 0, 1, \dots, c,$$

hence, the so-called *blocking probability* $B(c, \rho)$ is given by

$$B(c, \rho) = p_c = \frac{\rho^c/c!}{\sum_{n=0}^c \rho^n/n!}.$$

This formula is also known as *Erlang's loss formula*. Furthermore, note that

$$\mathbb{E}[S] = \mathbb{E}[B](1 - B(c, \rho))$$

and hence by *Little's law* we obtain that

$$\mathbb{E}[L] = \rho(1 - B(c, \rho)).$$

SECTION 10.3

| Stable Recursion for $B(c, \rho)$

Fortunately it is possible to derive a simple and numerically stable recursion for the blocking probabilities $B(c, \rho)$. We have that

$$B(c, \rho) = \frac{\rho^c / c!}{\sum_{n=0}^{c-1} \rho^n / n! + \rho^c / c!}.$$

Dividing the numerator and denominator by $\sum_{n=0}^{c-1} \rho^n / n!$ yields

$$B(c, \rho) = \frac{\rho B(c-1, \rho) / c}{1 + \rho B(c-1, \rho) / c} = \frac{\rho B(c-1, \rho)}{c + \rho B(c-1, \rho)}.$$

Starting with $B(0, \rho) = 1$ we can use relation this formula to subsequently compute the blocking probabilities $B(c, \rho)$ for $c = 1, 2, 3, \dots$

There is a simple relation between the delay probability Π_W in the $M/M/c$ with server utilisation ρ and the blocking probability $B(c-1, c\rho)$. We have

$$\Pi_W = \frac{\rho B(c-1, c\rho)}{1 - \rho + \rho B(c-1, c\rho)}.$$

By first computing $B(c-1, c\rho)$ using the recursion on the previous slide we can use the relation above to determine Π_W .

| Exercises

EXERCISE 10.1.

EXERCISE 10.2.

EXERCISE 10.3. For a certain type of article there is a stock of at most 5 articles to satisfy customer demand directly from the shelf. Customers arrive according to a Poisson process at a rate of 2 customers per week. Each customer demands 1 article. The ordering policy is as follows. Each time an article is sold to a customer, an order for 1 article is immediately placed at the supplier. The lead time (the time that elapses from the moment the order is placed until the order arrives) is exponentially distributed with a mean of 1 week. If on arrival of a customer the shelf is empty, the customer demand will be lost. The inventory costs are 20 guilders per article per week. Each time an article is sold, this yields a reward of 100 guilders.

- (i) Calculate the probability distribution of the number of outstanding orders.
- (ii) Determine the mean number of articles on stock.
- (iii) What is the average profit (reward-inventory costs) per week?

EXERCISE 10.4. In our library there are 4 VUBIS terminals. These terminals can be used to obtain information about the available literature. If all terminals are occupied when someone wants information, then that person will not wait but leave immediately (to look for the required information somewhere else). A user session on a VUBIS terminal takes on average 2.5 minutes. Since the number of potential users is large, it is reasonable to assume that users arrive according to a Poisson stream. On average 72 users arrive per hour.

- (i) Determine the probability that i terminals are occupied, $i = 0, 1, \dots, 4$.
- (ii) What is the fraction of arriving users finding all terminals occupied?
- (iii) How many VUBIS terminals are required such that at most 5% of the arriving users find all terminals occupied?

EXERCISE 10.5. Consider a machine continuously processing parts (there is always raw material available). The processing time of a part is exponentially distributed with a mean of 20 seconds. A finished part is transported immediately to an assembly cell by an automatic conveyor system. The transportation time is exactly 3 minutes.

- (i) Determine the mean and variance of the number of parts on the conveyor.

To prevent that too many parts are simultaneously on the conveyor one decides to stop the machine as soon as there are N parts on the conveyor. The machine is turned on again as soon as this number is less than N .

- (ii) Determine the throughput of the machine as a function of N .
- (iii) Determine the smallest N for which the throughput is at least 100 parts per hour.

EXERCISE 10.6. A small company renting cars has 6 cars available. The costs (depreciation, insurance, maintenance, etc.) are 60 guilders per car per day. Customers arrive according to a Poisson process with a rate of 5 customers per day. A customer rents a car for an exponential time with a mean of 1.5 days. Renting a car costs 110 guilders per day. Arriving customers for which no car is available are lost (they will go to another company).

- (i) Determine the fraction of arriving customers for which no car is available.
- (ii) Determine the mean profit per day.
- (iii) The company is considering to buy extra cars. How many cars should be bought to maximise the mean profit per day?

SOLUTION. (i)

$$\rho = \frac{\lambda}{\mu} = \frac{5}{\frac{2}{3}} = 7.5.$$

Then

$$B(6, 7.5) = \frac{\frac{\rho^6}{6!}}{\sum_{n=0}^6 \frac{\rho^n}{n!}} = \frac{\frac{7.5^6}{6!}}{\sum_{n=0}^6 \frac{7.5^n}{n!}} \approx 0.3615.$$

(ii)

$$110 \cdot \mathbb{E}[L^q] - 60 \cdot 6 = 110\rho(1 - B(6, 7.5)) - 360 \approx 166.8.$$

(iii) When the company has c cars, the mean profit per day equals

$$825 \cdot (1 - B(c, 7.5)) - 60c.$$

So, if the company buys 1 extra car, the mean profit becomes 174.7 guilders, if the company buys 2 extra cars, it becomes 173.8 guilders, if the company buys 3 extra cars, it becomes 163.4 guilders, and so on. The mean profit per day is maximised when the company buys 1 extra car.

□

Appendix

SECTION A.1

| Final Exams

Semester 2025–2026

EXERCISE A.1. Three computer mechanics are in the business of repairing laptops. They work in shifts, each being available for 8 hours per day so that at any time, one mechanic is available. When a mechanic takes over from his colleague while a laptop is being repaired, this does not influence the repair process in any way. Assume that broken laptops arrive to their workshop day and night, according to a Poisson process with constant rate λ per hour. The repair times are independent exponential random variables with mean μ^{-1} hours (which is less than λ^{-1}). The laptops are repaired in order of arrival. So when the current mechanic finishes working on a laptop, he starts working on the laptop which arrived earliest. Let $\rho = \frac{\lambda}{\mu}$ and let $P_L(z)$ be the probability generating function of L , the stationary number of unrepaired laptops in the workshop.

- (i) Give two different interpretations for ρ .
- (ii) When a mechanic at the end of his shift has been working on one laptop for $\frac{1}{2}\mu^{-1}$ hours, what is the expected time his colleague needs to finish this repair?
- (iii) Give an expression for $P_L(z)$ in terms of λ , μ and z , and use it to find an expression for $\mathbb{E}[L]$.
- (iv) The mechanics know from experience that $\rho = 0.8$, and that the expected busy period is 4 hours. Give the values of λ and μ .
- (v) How would your answers to (ii) and (iii) above change if the service discipline were not FCFS but ROS, i.e., after finishing a repair, the mechanic selects a laptop at random to be repaired next, independently of all else.

EXERCISE A.2. An art gallery offers a service where people can bring their paintings to have them valued by an expert. Customers arrive according to a Poisson process with rate λ and are served in order of arrival, possibly after some waiting time. Service times for each painting are *i.i.d.* exponential with mean μ^{-1} , and you may assume the system is in stationarity. If you wish, you may (but need not) use the values $\mu^{-1} = \frac{1}{5}$ hour and $\lambda = 2$ per hour. Determine both the probability that an arriving customer needs to wait and the expected waiting time, in the following cases.

- (i) Each customer brings one painting, and there are two experts available.
- (ii) Each customer brings two paintings, and there is one expert available.

- (iii) As the service becomes more popular, the arrival rate increases to $\lambda_{\text{new}} = 5\lambda$ (=10 customers per hour). Also the experts decide they do not want any waiting customers in their shop, so arriving customers are declined when all experts are busy. Are three experts enough to ensure that at least 90% of the arriving customers can have their painting valued? (Assume one painting per customer.)

EXERCISE A.3. A cloud computing provider operates a single powerful server that processes incoming computational jobs. Jobs arrive according to a Poisson process with rate $\lambda > 0$ and are served in order of arrival. The service times of the jobs are given to be independent and distributed according to some positive random variable B , so that $\lambda\mathbb{E}[B] \in (0, 1)$.

- (i) Use a formula to show that $\mathbb{P}(L = 0) = 1 - \lambda\mathbb{E}[B]$, where L is the stationary number of jobs in the system.
- (ii) Assume for this subquestion only that service times are exponentially distributed with mean μ^{-1} . Derive a formula for the LST of the busy period under this assumption.

Suppose now that jobs are not necessarily served in the order of arrival. Instead, we assume there are three types of jobs. Indeed, the provider processes real-time fraud detection jobs (job type 1), which take preemptive priority over jobs of the other two types. Then, there are simulation-based jobs (job type 2), which take non-preemptive priority over data-processing jobs (job type 3). An arriving job is assumed to be of any of these types with probability $\frac{1}{3}$.

- (iii) By means of mean-value analysis, derive expressions for $\mathbb{E}[W_1]$ and $\mathbb{E}[W_2]$, the mean waiting time of type-1 jobs and type-2 jobs (before they are taken into service for the first time), in terms of λ and moments of B .
- (iv) Comment on how the distribution of the server's busy period duration changes when moving from the current priority policy to a policy where jobs have no priority over one another and are thus served in a FCFS fashion.

EXERCISE A.4. We consider a $G/M/1$ queueing system where the expectation of the service times $B_1, B_2, \dots$ is μ^{-1} for some $\mu > 0$. The inter-arrival time distribution A satisfies

$$\mathbb{P}(A \leq t) = \frac{1}{2}\left(1 - e^{-\frac{t}{2}}\right) + \frac{1}{2}\left(1 - e^{-\frac{t}{4}}\right).$$

The number of customers that an arriving customer finds already present in the system is represented by L^a .

- (i) Take the formula for the $G/M/1$ queue for the value σ and adapt it to this particular queue, i.e., which equation does σ satisfy in this particular case?
- (ii) For which values of μ is the system stable?
- (iii) Using the equation $W = \sum_{i=1}^{L^a} B_i$, argue that the LST of the waiting time distribution is, for $s > 0$, given by

$$\tilde{W}(s) = \frac{(1 - \sigma)(\mu + s)}{\mu(1 - \sigma) + s}.$$

- (iv) Use the expression in (iii) to derive an LST for the sojourn time distribution in terms of μ and σ . What known distribution does the sojourn time have?

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EXERCISE A.5. A service station consists of three parallel servers. Customers arrive according to a Poisson process with a rate of 20 customers per hour. The time needed to help a customer is exponentially distributed with a mean of 6 minutes.

- (i) Determine the limiting distribution of the number of customers in the system.
- (ii) Determine the probability that a customer has to wait.
- (iii) Use the mean value technique to derive an expression for the mean waiting time of a customer in terms of the probability that a customer has to wait.
- (iv) Now assume that arriving customers that have to wait leave the station immediately and are lost. Determine the long-run fraction of customers that are lost. Use this expression to determine the mean number of servers that are occupied.

SOLUTION. (i) The system can be modelled by an $M/M/3$ queue with $\lambda = \frac{1}{3}$ and $\mu = \frac{1}{6}$. From the equations

$$\frac{1}{6}p_1 = \frac{1}{3}p_0, \quad \frac{1}{3}p_2 = \frac{1}{3}p_1, \quad \frac{1}{2}p_n = \frac{1}{3}p_{n-1}$$

for $n = 3, 4, \dots$, using $\sum_{n=0}^{\infty} p_n = 1$ we obtain

$$p_0 = \frac{1}{9}, \quad p_1 = \frac{2}{9}, \quad p_n = \frac{2}{9} \left(\frac{2}{3}\right)^{n-2}, \quad n = 2, 3, \dots$$

(ii) Due to PASTA, we have $\Pi_W = 1 - p_0 - p_1 - p_2 = \frac{4}{9}$.

(iii) The arrival relation is

$$\mathbb{E}[W] = \mathbb{E}[L^q] \frac{1}{3\mu} + \Pi_W \frac{1}{3\mu},$$

as $\frac{1}{3\mu}$ is, for all servers occupied, the expected time until the next service completion as well as the expected time between two completions. Substituting *Little's law* $\mathbb{E}[L^q] = \lambda \mathbb{E}[W]$ and rewriting yields

$$\mathbb{E}[W] = \Pi_W \frac{1}{3\mu(1-\rho)}.$$

Using $\rho = \frac{2}{3}$, $\mu = \frac{1}{6}$, and $\Pi_W = \frac{4}{9}$ we obtain $\mathbb{E}[W] = \frac{8}{3}$.

(iv) This corresponds to the $M/M/3/3$ queue with offered load $\rho = \frac{\lambda}{\mu} = 2$. By similar relations as in (i),

$$p_3 = \frac{2}{3}p_2 = \frac{2}{3}p_1 = \frac{4}{3}p_0,$$

so the long-run fraction of customers lost is $B(3, 2) = \frac{4}{19}$. *Little's law* then gives the mean number of occupied servers as

$$\frac{\lambda(1 - B(3, 2))}{\mu} = \frac{30}{19}.$$

□

EXERCISE A.6. Customers arrive at a single server according to a Poisson process with a rate of 24 customers per hour. With probability $(\frac{1}{2})^i$, $i = 1, 2, \dots$, the service time of a customer consists of i independent, exponential phases of work, each with a mean of 1 minute. Customers are served in order of arrival.

- (i) Show that the LST of the total service time of an arbitrary customer in minutes is given by

$$\tilde{B}(s) = \frac{1}{1+2s}.$$

- (ii) Determine the mean waiting time of a customer and the probability that the waiting time is larger than 10 minutes.

Now assume that customers for which the service time consists of only one exponential phase of work have non-preemptive priority over customers for which the service time consists of two or more exponential phases of work. Furthermore, denote with B_2 the service time of a customer for which the service time consists of two or more exponential phases of work.

- (iii) Show that the LST of the total service time of an arbitrary customer for which the service time consists of two or more exponential phases of work in minutes is given by

$$\tilde{B}_2(s) = \frac{1}{1+2s} \cdot \frac{1}{1+s}.$$

- (iv) Determine the mean waiting time of a customer for which the service time consists of only one exponential phase of work.
 (v) Determine the mean waiting time of a customer for which the service time consists of two or more exponential phases of work.
 (vi) Formulate the conservation law for this system and use it to verify that your answers in parts (iv) and (v) are consistent with your answer in part (ii).

SOLUTION. (i) Conditioning on the number of exponential phases of work yields

$$\tilde{B}(s) = \sum_{i=1}^{\infty} \left(\frac{1}{2}\right)^i \left(\frac{1}{1+s}\right)^i = \frac{\frac{1}{2(1+s)}}{1 - \frac{1}{2(1+s)}} = \frac{1}{1+2s}.$$

- (ii) Observe that, due to (i), the service time is exponential with rate $\frac{1}{2}$, so the system is $M/M/1$ with $\rho = \frac{2}{5} \cdot 2 = \frac{4}{5}$. With time in minutes we have

$$\mathbb{E}[W] = \frac{\frac{4}{5}}{\frac{1}{2}(1 - \frac{4}{5})} = 8, \quad P(W > 10) = \frac{4}{5} e^{-\frac{1}{2}(1 - \frac{4}{5})10} = \frac{4}{5} e^{-1}.$$

- (iii) The probability that at least two phases are required is $\frac{1}{2}$. One way is to note

$$\tilde{B}(s) = \frac{1}{2} \frac{1}{1+s} + \frac{1}{2} \tilde{B}_2(s).$$

Alternatively,

$$\tilde{B}_2(s) = \frac{1}{2} \sum_{i=2}^{\infty} \left(\frac{1}{2}\right)^i \left(\frac{1}{1+s}\right)^i = 2 \frac{\left(\frac{1}{2(1+s)}\right)^2}{1 - \frac{1}{2(1+s)}} = \frac{1}{1+2s} \frac{1}{1+s}.$$

- (iv) The residual service time R is exponential with mean 2. Hence

$$\rho \mathbb{E}[R] = \frac{4}{5} \cdot 2 = \frac{8}{5}, \quad \rho_1 = \frac{1}{2} \cdot \frac{2}{5} \cdot 1 = \frac{1}{5}.$$

Thus

$$E(W_1) = \frac{\rho \mathbb{E}[R]}{1 - \rho_1} = 2.$$

- (v) We have

$$E(W_2) = \frac{\rho \mathbb{E}[R]}{(1 - \rho_1)(1 - \rho_1 - \rho_2)} = 10.$$

(vi) The conservation law $\sum_{k=1}^2 \rho_k E(W_k) = \rho \mathbb{E}[W]$ holds independent of discipline. From (iv)-(v):

$$\sum_{k=1}^2 \rho_k E(W_k) = \frac{1}{5} \cdot 2 + \frac{3}{5} \cdot 10 = \frac{32}{5}.$$

From (ii):

$$\rho \mathbb{E}[W] = \frac{4}{5} \cdot 8 = \frac{32}{5},$$

as required. □

EXERCISE A.7. Jobs at a single machine arrive according to a hyperexponential arrival process. The interarrival time is with probability $\frac{2}{3}$ exponentially distributed with a mean of 2 minutes and with probability $\frac{1}{3}$ it is exponentially distributed with a mean of 6 minutes. The service times are exponentially distributed with a mean of 2 minutes.

- (i) Determine the equilibrium distribution of the number of jobs in the system just before an arrival of a new job.
- (ii) Use this equilibrium distribution to determine the LST of the waiting time. What is the probability that the duration of the waiting time exceeds 5 minutes?
- (iii) Determine the mean number of jobs in the system at arrival instants and at arbitrary instants.

SOLUTION. (i) The system corresponds to an $H_2/M/1$ queue with $\mu = \frac{1}{2}$ and

$$\tilde{A}(s) = \frac{2}{3} \cdot \frac{\frac{1}{2}}{\frac{1}{2} + s} + \frac{1}{3} \cdot \frac{\frac{1}{6}}{\frac{1}{6} + s}.$$

Solving $\sigma = \tilde{A}(\mu(1 - \sigma))$ we obtain $\sigma = \frac{2}{3}$ and hence

$$a_n = (1 - \sigma)\sigma^n = \frac{1}{3} \cdot \left(\frac{2}{3}\right)^n.$$

- (ii) Conditioning on the number of customers found upon arrival gives

$$\tilde{W}(s) = \sum_{n=0}^{\infty} a_n \left(\frac{\frac{1}{2}}{\frac{1}{2} + s} \right)^n = \frac{1}{3} + \frac{2}{3} \cdot \frac{\frac{1}{6}}{\frac{1}{6} + s}.$$

Observe that the LST shows that the waiting time, given it is non-zero, is exponentially distributed with rate $\frac{1}{6}$. Thus, with the time unit in minutes,

$$\mathbb{P}(W > 5) = \frac{2}{3} e^{-\frac{5}{6}}.$$

- (iii) The mean number of jobs at arrival instants is

$$\mathbb{E}[L^a] = \sum_{n=0}^{\infty} n a_n = \frac{\frac{2}{3}}{\frac{1}{3}} = 2.$$

For the mean number of jobs at arbitrary instants $\mathbb{E}[L]$, note that

$$\mathbb{E}[S] = \mathbb{E}[L^a] \cdot \frac{1}{\mu} + \frac{1}{\mu} = 6.$$

The expected interarrival time equals

$$\mathbb{E}[A] = \frac{2}{3} \cdot 2 + \frac{1}{3} \cdot 6 = \frac{10}{3},$$

such that

$$\lambda = \frac{3}{10}.$$

Using *Little's law* yields

$$\mathbb{E}[L] = \lambda \mathbb{E}[S] = \frac{9}{5}.$$

□

EXERCISE A.8. At a single server groups of customers arrive according to a Poisson process with a rate of 10 groups per hour. The group sizes are *i.i.d.* random variables. Each group contains either 2 or 3 customers, both with probability $\frac{1}{2}$. The service times of the individual customers are uniformly distributed between 1 and 3 minutes.

- (i) Give the two linear relations between the mean number of customers in the queue, $\mathbb{E}[L^q]$, and the mean waiting time of a customer in the queue, $\mathbb{E}[W]$, from which both of these two performance measures can be obtained.
- (ii) What is the probability that an arbitrary customer is served as i -th customer in his group, $i = 1, 2, 3$?
- (iii) Calculate the mean number of customers in the queue and the mean waiting time of a customer in the queue.

SOLUTION. (i) The system corresponds to an $M/G/1$ queue with arrival rate $\lambda = \frac{1}{6}$ and group arrivals, with mean group size $\mathbb{E}[G] = \frac{5}{2}$. The relations involve *Little's law*

$$\mathbb{E}[L^q] = \lambda \mathbb{E}[G] \mathbb{E}[W]$$

and the arrival relation

$$\mathbb{E}[W] = \mathbb{E}[L^q] \mathbb{E}[B] + \rho \mathbb{E}[R] + (r_2 + 2r_3) \mathbb{E}[B],$$

where r_k is the probability that a customer is served as k -th within the group.

(ii) We have

$$r_1 = r_2 = \frac{2}{5}, \quad r_3 = \frac{1}{5}.$$

(iii) First, verify that $\mathbb{E}[B] = 2$ and

$$\mathbb{E}[B^2] = \frac{1}{2} \int_1^3 u^2 du = \frac{13}{3},$$

yielding

$$\mathbb{E}[R] = \frac{\mathbb{E}[B^2]}{2\mathbb{E}[B]} = \frac{13}{12}.$$

The load equals

$$\rho = \lambda \mathbb{E}[G] \mathbb{E}[B] = \frac{5}{6}.$$

Using *Little's law* in the equation above yields

$$\mathbb{E}[W] = \frac{5}{6} \mathbb{E}[W] + \frac{5}{6} \cdot \frac{13}{12} + \frac{4}{5} \cdot 2.$$

Solving for $\mathbb{E}[W]$, we obtain

$$\mathbb{E}[W] = \frac{901}{60}.$$

Moreover,

$$\mathbb{E}[L^q] = \lambda \mathbb{E}[G] \mathbb{E}[W] = \frac{901}{144}.$$

□

EXERCISE A.9. Jobs arrive at a machine according to a Poisson process with a rate of 4 jobs per hour. The machine processes the jobs at an exponential rate of 4 jobs per hour. Whenever the total number of jobs in the system becomes larger than 3, a second identical machine is immediately turned on and also starts processing the jobs at an exponential rate of 4 jobs per hour. Whenever the total number of jobs in the system is reduced to 3 again, the machine just finishing a job is turned off.

- (i) Compute the limiting distribution of the number of jobs in the system.
- (ii) Compute the mean number of jobs waiting in the queue and the mean waiting time of a job in the queue.
- (iii) What is the expected length of periods during which, uninterruptedly, both machines are processing jobs?

EXERCISE A.10. Consider a single server serving jobs. Jobs arrive according to a Poisson process with a rate of $\lambda = \frac{1}{6}$ jobs per minute. The service times of jobs (in minutes) are independent, Erlang-2 distributed with a mean of 2 minutes (i.e., every job consists of two exponential phases of work, each with a mean service time of 1 minute).

- (i) Show that the limiting distribution of the number of exponential phases of work in the system is given by

$$p_n = \frac{2}{5} \left(\frac{1}{2} \right)^n + \frac{4}{15} \left(-\frac{1}{3} \right)^n, \quad n = 0, 1, 2, \dots$$

- (ii) Calculate the mean sojourn time of a job using mean value analysis.
- (iii) Show that the generating function $g(z)$ of the number of customers served in a busy period satisfies the equation

$$g(z) = z \tilde{B}(\lambda(1 - g(z))).$$

Here $\tilde{B}(s)$ denotes the LST of the service time.

- (iv) What is the expected number of customers served in a busy period?

EXERCISE A.11. A doctor's office receives patients by appointment. Exactly every 12 minutes a patient is scheduled to arrive, but due to various reasons 20% of the patients simply never show up, i.e., at every 12-minute point a patient only shows up with probability $\frac{4}{5}$, independent of all other 12-minute points. The amount of time it takes for the doctor to meet with a patient and provide possible treatment is exponentially distributed with a mean of 12 minutes.

- (i) Derive the LST of the interarrival time between two patients.
- (ii) Determine the distribution of the number of patients in the doctor's office at the arrival instant of an arbitrary patient.

- (iii) Show how the LST of the sojourn time of an arbitrary patient, i.e., the total amount of time that a patient spends in the doctor's office before leaving again, can be derived from the distribution determined in part (ii) above.
- (iv) Now suppose that patients arrive according to a Poisson process with a rate of 4 patients per hour instead of according to the arrival process described before. Compare the mean waiting time of a patient before seeing the doctor in this new situation with the mean waiting time in the case where patients arrived according to the arrival process described before.

EXERCISE A.12. At a server customers arrive according to a Poisson process with a rate of 30 customers per hour. The service times of customers are uniformly distributed with a minimum of 1 minute and a maximum of 2 minutes. As soon as there are no customers anymore in the system the server leaves to do another activity. After an Erlang-2 distributed time with a mean of 4 minutes he returns. If at that time there are customers present, the server starts serving the customers. However, if at that time the system is still empty, the server waits, while doing nothing, until the next customer arrives.

- (i) Show that the probability that no customers arrive during a single period in which the server does another activity is equal to $\frac{1}{4}$.
- (ii) Determine the fraction of time the server serves customers, the fraction of time the server does another activity and the fraction of time the server is doing nothing, respectively.
- (iii) Determine the mean waiting time of a customer.

EXERCISE A.13. The arrival process of persons at a taxi+bus stop can be modelled by a Poisson process with rate $\lambda = 6$ persons per hour. Taxis arrive at the stop according to a Poisson process with rate $\mu = 7$ taxis per hour. Each time a taxi arrives one waiting person, if there is any, will enter the taxi on a FCFS basis, and the taxi will leave the stop immediately. The time it takes for a person to enter the taxi is negligible. Buses arrive at the stop according to a Poisson process with rate $\nu = 4$ buses per hour. Buses take all waiting persons at the taxi+bus stop (the capacity of the bus is always large enough to take all waiting persons) and leave immediately. We also assume that the time it takes for persons to enter the bus is negligible.

- $$p_n = \frac{4}{7} \cdot \left(\frac{3}{7}\right)^n, \quad n = 0, 1, 2, 3, \dots$$

- by directly solving the system of balance equations;
- by using generating functions.

- (ii) Calculate the probability that an arbitrary person will use a taxi and a bus, respectively.
- (iii) What is the mean time a person spends at the taxi+bus stop?
- (iv) Determine the distribution of the time a person spends at the taxi+bus stop.
- (v) Also answer the questions in parts (ii), (iii) and (iv) for a person who sees at arrival k waiting persons at the taxi+bus stop, $k = 0, 1, 2, \dots$

The diagram shows a directed graph with nodes labeled 0, 1, 2, ..., n-1, n. Nodes 0, 1, and 2 are arranged horizontally. Nodes n-1 and n are also arranged horizontally, with an ellipsis (...) between node 2 and node n-1. There are directed edges from 0 to 1 and 1 to 2, both labeled λ . There is a directed edge from n-1 to n, labeled λ . There are three long curved edges from node n back to nodes 0, 1, and 2. The edge from n to 0 is labeled ν . The edge from n to 1 is labeled ν . The edge from n to 2 is labeled $\nu + \nu$.

From the flow diagram, we can, by equating the flow out of state n and the flow into state n , $n = 0, 1, \dots$, derive the following balance equations:

$$\begin{aligned} \lambda p_0 &= (\mu + \nu) p_1 + \nu \sum_{n=2}^{\infty} p_n \\ (\lambda + \mu + \nu) p_n &= \lambda p_{n-1} + \mu p_{n+1}, \quad n = 1, 2, \dots \end{aligned} \quad (\text{A.1})$$

and we note that from the normalisation equation $\sum_{n=0}^{\infty} p_n = 1$ we can simplify equation (A.1) as

$$\lambda p_0 = (\mu + \nu)p_1 + \nu(1 - p_0 - p_1)$$

that is,

$$(\lambda + \nu)p_0 = \mu p_1 + \nu.$$

By plugging in the values $\lambda = 6$, $\mu = 7$ and $\nu = 4$ we obtain the final form of the balance equations,

$$10p_0 = 7p_1 + 4 \quad (\text{A.2})$$

$$17p_n = 6p_{n-1} + 7p_{n+1}, \quad n = 1, 2, \dots \quad (\text{A.3})$$

We start by finding a solution for the system of balance equations (A.2) and (A.3) by solving the equations directly. The equations are a second-order recurrence relation with constant coefficients. Its general solution is of the form

$$p_n = c_1 x_1^n + c_2 x_2^n, \quad n = 0, 1, 2, \dots$$

where x_1 and x_2 are roots of the quadratic equation

$$6 - 17x + 7x^2 = 0.$$

The left hand side of this quadratic equation can be factored as

$$(x - 2)(7x - 3) = 0$$

and thus we obtain the roots $x = 2$ and $x = \frac{3}{7}$. So all solutions to (A.3) are of the form

$$p_n = c_1 2^n + c_2 \left(\frac{3}{7}\right)^n, \quad n = 0, 1, 2, \dots$$

From the normalisation equation $\sum_{n=0}^{\infty} p_n = 1$ it is clear that we must have $c_1 = 0$. By plugging $p_1 = \frac{3}{7}c_2$ into equation (A.2) and solving for c_2 , we obtain that $c_2 = \frac{4}{7}$. Therefore we conclude that

$$p_n = \frac{4}{7} \left(\frac{3}{7}\right)^n, \quad n = 0, 1, 2, \dots \quad (\text{A.4})$$

Now we want to solve the balance equations again, using generating functions. Let the random variable L denote the number of customers in the system. Its generating function is given by

$$P_L(z) = \sum_{n=0}^{\infty} p_n z^n, \quad |z| \leq 1 \quad (\text{A.5})$$

By multiplying equation (A.3) with z^n , and then summing the balance equations over all n , we obtain the equation

$$10p_0 + \sum_{n=1}^{\infty} 17p_n z^n = 7p_1 + 4 + \sum_{n=1}^{\infty} (6p_{n-1} z^n + 7p_{n+1} z^n)$$

which can be rewritten as

$$-7p_0 + 17 \sum_{n=0}^{\infty} p_n z^n = 4 - \frac{7}{z} p_0 + 6z \sum_{n=0}^{\infty} p_n z^n + \frac{7}{z} \sum_{n=0}^{\infty} p_n z^n$$

that is,

$$-7p_0 + 17P_L(z) = 4 - \frac{7}{z} p_0 + 6zP_L(z) + \frac{7}{z} P_L(z)$$

that is,

$$\left(17 - 6z - \frac{7}{z}\right)P_L(z) = 4 + \left(7 - \frac{7}{z}\right)p_0.$$

Now since $|P_L(z)| = 1 < \infty$ for all $|z| \leq 1$, we get for $z = \frac{1}{2}$ that

$$\left(17 - 6 \cdot \frac{1}{2} - \frac{7}{\frac{1}{2}}\right) P_L\left(\frac{1}{2}\right) = 4 + \left(7 - \frac{7}{\frac{1}{2}}\right) p_0$$

that is, $0 = 4 - 7p_0$, or $p_0 = \frac{4}{7}$. Hence,

$$\left(17 - 6z - \frac{7}{z}\right) P_L(z) = 4 + \frac{4}{7} \left(7 - \frac{7}{z}\right).$$

then

$$\begin{aligned} P_L(z) &= \frac{8 - \frac{4}{z}}{17 - 6z - \frac{7}{z}} \\ &= \frac{8z - 4}{17z - 6z^2 - 7} \\ &= \frac{4(2z - 1)}{-(2z - 1)(3z - 7)} \\ &= \frac{4}{7} \cdot \frac{1}{1 - \frac{3}{7}z} \end{aligned} \tag{A.6}$$

$$= \sum_{n=0}^{\infty} \frac{4}{7} \left(\frac{3}{7}\right)^n z^n. \tag{A.7}$$

By equating the coefficients of z^n in (A.5) and (A.7), we recover the solution (A.4).

- (ii) In steady state, we can write the expected number of people that a taxi will transport in the next step as

$$\sum_{n=1}^{\infty} p_n \frac{\mu}{\lambda + \mu + \nu} = \frac{\mu}{\lambda + \mu + \nu} (1 - p_0) = \frac{7}{6 + 7 + 4} \left(1 - \frac{4}{7}\right) = \frac{3}{17}.$$

Likewise, we can write the expected number of people that a bus will transport in the next step in steady state as

$$\sum_{n=1}^n n p_n \frac{\nu}{\lambda + \mu + \nu} = P'_L(1) \cdot \frac{\nu}{\lambda + \mu + \nu} = \frac{4}{17} P'_L(1)$$

since $P'_L(z) = \sum_{n=1}^{\infty} n p_n z^{n-1}$. Now, by using (A.6) we obtain

$$P'_L(z) = \frac{12}{49} \frac{1}{\left(1 - \frac{3}{7}z\right)^2}$$

and thus $P'_L(1) = \frac{3}{4}$. Hence,

$$\sum_{n=1}^n n p_n \frac{\nu}{\lambda + \mu + \nu} = \frac{4}{17} \cdot \frac{3}{4} = \frac{3}{17}.$$

Now, the probability that an arbitrary person will use a taxi, is equal to the expected number of people that a taxi will transport in the next step in steady state, divided by the total expected number of people that a taxi or a bus will transport in the next step in steady state,

$$\frac{\frac{3}{17}}{\frac{3}{17} + \frac{3}{17}} = \frac{1}{2}.$$

Likewise, the probability that an arbitrary person will use a bus, is equal to $\frac{1}{2}$.

- (iii) In the system, the service time of a person can be regarded as zero, since when a person is serviced, that is, he/she enters a taxi or a bus, he/she leaves the system. Therefore, the time a person spends at the taxi+bus stop is just the time the person waits in the queue. Hence the time a person spends at the taxi+bus stop, the sojourn time S , is equal to the waiting time.

The mean number of customers in the system is

$$\mathbb{E}[L] = \sum_{n=0}^{\infty} n p_n = P'_L(1) = \frac{3}{4}.$$

By applying *Little's law*, we obtain that

$$\mathbb{E}[S] = \frac{1}{\lambda} \mathbb{E}[L] = \frac{1}{6} \cdot \frac{3}{4} = \frac{1}{8} \text{ hours.}$$

- (iv) As discussed in (iii), the time a person spends at the taxi+bus stop is just the time the person waits in the queue. This implies that the sojourn time S of a person in the system equals the waiting time of that person. The sojourn time of a person at the taxi+bus stop is determined by whether the person has to wait in queue for a taxi or if a bus will arrive at the stop before the person will enter a taxi. Let L^a denote the number of customers in the system just before the arrival of a customer. Then we can write

$$S = \min \left\{ X, \sum_{j=1}^{L^a+1} Y_j \right\}$$

where $X \sim \exp(\nu)$ denotes the arrival of a bus, and $Y_j \sim \exp(\mu)$ denotes the arrival of a taxi at the stop, $j = 1, \dots, L^a + 1$. The sum $\sum_{j=1}^{L^a+1} Y_j$ then denotes the arrival of $L^a + 1$ taxi. We note that $X, Y_1, \dots, Y_{L^a+1}$ are all independent and thus X and $\sum_{j=1}^{L^a+1} Y_j$ are also independent. We also note that by *PASTA*, we have

$$P(L^a = n) = p_n = \frac{4}{7} \left(\frac{3}{7} \right)^n.$$

Now, we have for $t \geq 0$,

$$\begin{aligned} P(S > t) &= P \left(\min \left\{ X, \sum_{j=1}^{L^a+1} Y_j \right\} > t \right) \\ &= P \left(X > t, \sum_{j=1}^{L^a+1} Y_j > t \right) \\ &= P(X > t) P \left(\sum_{j=1}^{L^a+1} Y_j \right) \quad (\text{independence}) \\ &= P(X > t) \sum_{n=0}^{\infty} P \left(\sum_{j=1}^{L^a+1} Y_j \mid L^a = n \right) P(L^a = n) \\ &= P(X > t) \sum_{n=0}^{\infty} P \left(\sum_{j=1}^{n+1} Y_j \right) p_n \\ &= e^{-\nu t} \sum_{n=0}^{\infty} \sum_{j=0}^n \frac{(\mu t)^j}{j!} e^{-\mu t} p_n \quad (\sum_{j=1}^{n+1} Y_j \sim \text{Erlang}-(n+1)) \\ &= e^{-11t} \sum_{j=0}^{\infty} \sum_{n=j}^{\infty} \frac{(7t)^j}{j!} \cdot \frac{4}{7} \left(\frac{3}{7} \right)^n \quad (\mu = 7 \text{ and } \nu = 4) \\ &= e^{-11t} \sum_{j=0}^{\infty} \frac{(7 \cdot \frac{3}{7} t)^j}{j!} \\ &= e^{-11t} \cdot e^{3t} = e^{-8t}. \end{aligned}$$

Hence, S is exponentially distributed with parameter 8.

- (v) Now in (ii), we want to compute the probability that an arbitrary person who sees at arrival k waiting persons at the taxi+bus stop, $k = 0, 1, 2, \dots$, will use a taxi and a bus, respectively. Let the random variable X denote the arrival of a bus, and the random variable $Y = \sum_{j=1}^{k+1} Y_j$ denote the arrival of $k + 1$ taxis, similar to our definition in d). We have $X \sim \exp(\nu)$ and $Y \sim \text{Erlang}(k + 1, \mu)$ and $Y_j \sim \exp(\mu)$, $j = 1, \dots, k + 1$. The probability that an arbitrary person who sees at arrival k waiting persons will use a taxi is then $P(X > Y)$, that is, the probability that a $k + 1$ taxi will arrive before a single bus arrives. We have,

$$\begin{aligned}
 P(X > Y) &= \int_0^\infty P(X > t) f_Y(t) dt \\
 &= \int_0^\infty e^{-\nu t} \cdot \mu \cdot \frac{(\mu t)^k}{k!} e^{-\mu t} dt \\
 &= \frac{\mu^{k+1}}{k!} \int_0^\infty t^k e^{-(\nu+\mu)t} dt \\
 &= \frac{\mu^{k+1}}{k!} \int_0^\infty \left(\frac{u}{\nu + \mu} \right)^k e^{-u} \cdot \frac{1}{\nu + \mu} du && \text{(substitute } u = (\nu + \mu)t) \\
 &= \left(\frac{\mu}{\nu + \mu} \right)^{k+1} \cdot \frac{1}{k!} \int_0^\infty u^k e^{-u} du \\
 &= \left(\frac{\mu}{\nu + \mu} \right)^{k+1} \cdot \frac{1}{k!} \cdot \Gamma(k + 1) \\
 &= \left(\frac{\mu}{\nu + \mu} \right)^{k+1} \cdot \frac{1}{k!} \cdot k! = \left(\frac{\mu}{\nu + \mu} \right)^{k+1}
 \end{aligned}$$

and by inserting the numerical values for μ and ν into this equation, we obtain that the probability that an arbitrary person at the taxi+bus stop who sees at arrival k waiting persons will enter a taxi is,

$$P(X > Y) = \left(\frac{7}{11} \right)^{k+1}, \quad k = 0, 1, 2, \dots$$

Then the probability that an arbitrary person at the taxi+bus stop who sees at arrival k waiting persons will enter a bus is,

$$P(Y > X) = 1 - P(X > Y) = 1 - \left(\frac{7}{11} \right)^{k+1}, \quad k = 0, 1, 2, \dots$$

In (iv) we want to determine the distribution of the time a person spends at the taxi+bus stop, given that there are k waiting persons at the stop when the person arrives, $k = 0, 1, 2, \dots$. Hence our computations from d) become slightly easier, since here it is known that $L^a = k$. We write the sojourn time of the person as

$$S = \min \left\{ X, \sum_{j=1}^{k+1} Y_j \right\}$$

where X and Y_j are defined as in d). Then we have for $t \geq 0$,

$$\begin{aligned}
 P(S > t) &= P \left(\min \left\{ X, \sum_{j=1}^{k+1} Y_j \right\} > t \right) \\
 &= P \left(X > t, \sum_{j=1}^{k+1} Y_j > t \right) \\
 &= P(X > t) P \left(\sum_{j=1}^{k+1} Y_j > t \right) && \text{(independence)}
 \end{aligned}$$

$$\begin{aligned}
&= e^{-vt} \sum_{j=0}^k \frac{(\mu t)^j}{j!} e^{-\mu t} \\
&= e^{-(v+\mu)t} \sum_{j=0}^k \frac{(\mu t)^j}{j!} = e^{-11t} \sum_{j=0}^k \frac{(7t)^j}{j!}.
\end{aligned}$$

Thus,

$$P(S \leq t) = \begin{cases} 1 - e^{-11t} \sum_{j=0}^k \frac{(7t)^j}{j!}, & t \geq 0 \\ 0, & t < 0 \end{cases}.$$

Finally in (iii) we have from (iv) the cumulative distribution function (with μ and ν) instead of the numerical values),

$$F_S(t) = 1 - P(S > t) = 1 - e^{-(\nu+\mu)t} \sum_{j=0}^k \frac{(\mu t)^j}{j!}, \quad t \geq 0.$$

Then we can calculate its density,

$$\begin{aligned}
f_S(t) &= \frac{d}{dt} F_S(t) \\
&= \frac{d}{dt} \left(1 - e^{-(\nu+\mu)t} \sum_{j=0}^k \frac{(\mu t)^j}{j!} \right) \\
&= (\nu + \mu) e^{-(\nu+\mu)t} \sum_{j=0}^k \frac{(\mu t)^j}{j!} - \mu e^{-(\nu+\mu)t} \sum_{j=1}^k \frac{(\mu t)^{j-1}}{(j-1)!}
\end{aligned}$$

Therefore, the mean time a person spends at the taxi+bus stop is

$$\begin{aligned}
\mathbb{E}[S] &= \int_0^\infty t f_S(t) dt \\
&= \int_0^\infty t \left((\nu + \mu) e^{-(\nu+\mu)t} \sum_{j=0}^k \frac{(\mu t)^j}{j!} - \mu e^{-(\nu+\mu)t} \sum_{j=1}^k \frac{(\mu t)^{j-1}}{(j-1)!} \right) dt \\
&= (\nu + \mu) \sum_{j=0}^k \frac{\mu^j}{j!} \int_0^\infty t^{j+1} e^{-(\nu+\mu)t} dt - \mu \sum_{j=1}^k \frac{\mu^{j-1}}{(j-1)!} \int_0^\infty t^j e^{-(\nu+\mu)t} dt \\
&= (\nu + \mu) \sum_{j=0}^k \frac{\mu^j}{j!} \int_0^\infty \left(\frac{u}{\nu + \mu} \right)^{j+1} e^{-u} \frac{du}{\nu + \mu} - \mu \sum_{j=1}^k \frac{\mu^{j-1}}{(j-1)!} \int_0^\infty \left(\frac{u}{\nu + \mu} \right)^j e^{-u} \frac{du}{\nu + \mu} \\
&\quad \text{(substitute } u = (\nu + \mu)t) \\
&= \frac{1}{\mu} \sum_{j=0}^k \left(\frac{\mu}{\nu + \mu} \right)^{j+1} \cdot \frac{1}{j!} \int_0^\infty u^{j+1} e^{-u} du - \frac{1}{\nu + \mu} \sum_{j=1}^k \left(\frac{\mu}{\nu + \mu} \right)^j \cdot \frac{1}{(j-1)!} \int_0^\infty u^j e^{-u} du \\
&= \frac{1}{\mu} \sum_{j=0}^k \left(\frac{\mu}{\nu + \mu} \right)^{j+1} \cdot \frac{1}{j!} \cdot \Gamma(j+2) - \frac{1}{\nu + \mu} \sum_{j=1}^k \left(\frac{\mu}{\nu + \mu} \right)^j \cdot \frac{1}{(j-1)!} \cdot \Gamma(j+1) \\
&= \frac{1}{\mu} \sum_{j=0}^k \left(\frac{\mu}{\nu + \mu} \right)^{j+1} \cdot \frac{1}{j!} \cdot (j+1)! - \frac{1}{\nu + \mu} \sum_{j=1}^k \left(\frac{\mu}{\nu + \mu} \right)^j \cdot \frac{1}{(j-1)!} \cdot j! \\
&= \frac{1}{\mu} \sum_{j=0}^k \left(\frac{\mu}{\nu + \mu} \right)^{j+1} (j+1) - \frac{1}{\nu + \mu} \sum_{j=1}^k \left(\frac{\mu}{\nu + \mu} \right)^j \cdot j
\end{aligned}$$

$$= \frac{1}{7} \sum_{j=0}^k \left(\frac{7}{11}\right)^{j+1} (j+1) - \frac{1}{11} \sum_{j=1}^k \left(\frac{7}{11}\right)^j \cdot j.$$

□

EXERCISE A.14. Customers arrive according to a Poisson process with a rate of 4 customers per hour at a service station in which there are two servers. Service times of customers are exponentially distributed with a mean of 30 minutes. Customers are served in order of arrival. Whenever there are customers waiting in the queue, the last customer in the queue is impatient and leaves the system after an exponentially distributed time with a mean of 15 minutes.

- (i) Determine the limiting distribution of the number of customers in the system.
- (ii) Determine the fraction of customers that leave the system without getting service.
- (iii) The system alternates between periods during which there are customers waiting in the queue (and hence there is an impatient customer in the system) and periods during which there are no customers in the queue (and hence also no impatient customers in the system). Determine the expected durations of these two types of periods.

SOLUTION. As time unit we take 1 hour. The arrival rate is $\lambda = 4$ and the service rate is $\mu = 2$. For $n \geq 2$, the combined occupation rate is $\rho = \frac{\lambda}{2\mu + \nu} = \frac{1}{2}$, where $\nu = 4$ is the rate at which an impatient customer leaves the system.

- (i) A flow diagram for the system is shown in Figure A.2.

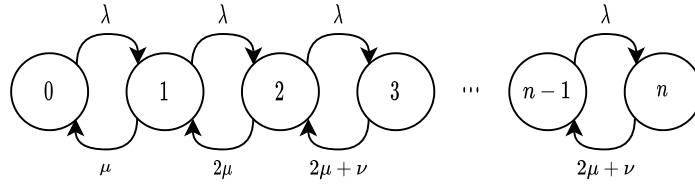


FIGURE A.2. Flow Diagram for Exercise 2

From the flow diagram, we can, by equating the flow out of state n and the flow into state n , $n = 0, 1, \dots$, derive the following balance equations:

$$\begin{aligned} \lambda p_0 &= \mu p_1 \\ \lambda p_1 &= 2\mu p_2 \\ \lambda p_n &= (2\mu + \nu)p_{n+1}, \quad n = 2, 3, \dots, \end{aligned}$$

and by inserting the numerical values for λ, μ and ν we get the balance equations:

$$4p_0 = 2p_1 \tag{A.8}$$

$$4p_1 = 4p_2 \tag{A.9}$$

$$4p_n = 8p_{n+1}, \quad n = 2, 3, \dots \tag{A.10}$$

From the normalisation equation $\sum_{n=0}^{\infty} p_n = 1$ and equation (A.10), we have that

$$\begin{aligned} 1 - p_0 - p_1 &= \sum_{n=2}^{\infty} p_n \\ &= \sum_{n=2}^{\infty} 2p_{n+1} = 2 \left(\sum_{n=2}^{\infty} p_n - p_2 \right) = 2(1 - p_0 - p_1 - p_2). \end{aligned}$$

Now, from equations (A.8) and (A.9), we have that $p_0 = \frac{1}{2}p_1 = \frac{1}{2}p_2$, and $p_1 = p_2$. By inserting this into the equation above, we obtain that

$$1 - \frac{1}{2}p_2 - p_2 = 2\left(1 - \frac{1}{2}p_2 - p_2 - p_2\right)$$

which can be solved for p_2 to obtain $p_2 = \frac{2}{7}$. Then again from equations (A.8) and (A.9), we have that $p_1 = \frac{2}{7}$ and $p_0 = \frac{1}{2} \cdot \frac{2}{7} = \frac{1}{7}$. From equation (A.10), we have that $p_{n+1} = \frac{1}{2}p_n$ for $n \geq 2$. Thus, $p_{n+2} = p_2 \cdot (\frac{1}{2})^n$, for $n \geq 0$, that is, $p_{n+2} = \frac{2}{7}(\frac{1}{2})^n$, for $n \geq 0$. So the solution to the system of balance equations is,

$$p_0 = \frac{1}{7}, \quad p_1 = \frac{2}{7}, \quad p_{n+2} = \frac{2}{7}\left(\frac{1}{2}\right)^n, \quad n = 0, 1, 2, \dots$$

- (ii) The probability that there is an impatient customer in the system is $\sum_{n=3}^{\infty} p_n$, and the rate at which he/she leaves is ν , together: $\nu \sum_{n=3}^{\infty} p_n$. The probability that only a single server is working is p_1 and the service rate of that single server is μ ; the probability that both servers are working is $\sum_{n=2}^{\infty} p_n$ and the service rate is 2μ : together $\mu p_1 + 2\mu \sum_{n=2}^{\infty} p_n$. Thus the fraction of customers that leave the system without getting service is

$$\frac{\nu \sum_{n=3}^{\infty} p_n}{\mu p_1 + 2\mu \sum_{n=2}^{\infty} p_n + \nu \sum_{n=3}^{\infty} p_n} = \frac{\nu(1 - p_0 - p_1 - p_2)}{\mu p_1 + 2\mu(1 - p_0 - p_1) + \nu(1 - p_0 - p_1 - p_2)} = \frac{2}{7}.$$

- (iii) The period during which there are no customers in the queue, the idle period IP , has expected duration $\frac{1}{\lambda} = \frac{1}{4}$ hours, due to the memoryless property of the exponential distribution. We denote the period during which there are customers waiting in the queue, the busy period, with BP . The fraction of time the server is working is

$$\frac{1}{2}p_1 + \sum_{n=2}^{\infty} p_n = \frac{1}{2}p_1 + 1 - p_0 - p_1 = 1 - \frac{1}{7} - \frac{1}{2} \cdot \frac{2}{7} = \frac{5}{7}.$$

We have that the expected duration of a busy period divided by the mean cycle length, that is the expected duration of a busy period plus the expected duration of an idle period, is equal to the fraction of time the server is working. Thus,

$$\frac{E(BP)}{E(BP) + E(IP)} = \frac{E(BP)}{E(BP) + \frac{1}{4}} = \frac{5}{7}$$

Hence,

$$E(BP) = \frac{5}{8} \text{ hours.}$$

□

EXERCISE A.15. Customers arrive according to a Poisson process with a rate of 4 customers per hour at a service station in which there are two servers. Service times of customers are exponentially distributed with a mean of 20 minutes if they are served by the experienced server and with a mean of 30 minutes if they are served by the less-experienced server. Customers are served in order of arrival. Customers arriving when both servers are idle are served by the less-experienced (i.e., the slowest) server. Otherwise they are served by the server who is or becomes available first.

- Determine the limiting distribution of the number of customers in the system.
- Determine the fraction of customers served by the experienced and the less-experienced server, respectively.
- Determine the LST of the waiting time and the sojourn time of an arbitrary customer.

SOLUTION. As time unit we take 1 hour. The arrival rate of customers is $\lambda = 4$. The service time of customers served by the experienced (fast) server is 20 minutes, or $\frac{1}{3}$ hour, thus the service rate is $\mu_f = 3$. The service time of customers served by the inexperienced (slow) server is 30 minutes, or $\frac{1}{2}$ hour, thus the service rate of this server is $\mu_s = 2$. Furthermore let $\mu = \mu_f + \mu_s$.

As state description we use the number of jobs in the system, and if this number of jobs is equal to 1, we distinguish between state $(1, f)$ in which the fast server is working and state $(1, s)$ in which the slow server is working. Also let $p_{1,f}$ and $p_{1,s}$ be the probability that there is one customer in the system and he is served by the fast server and slow server, respectively. Then we put $p_1 = p_{1,f} + p_{1,s}$.

(i) A flow diagram for the system is shown in Figure A.3.

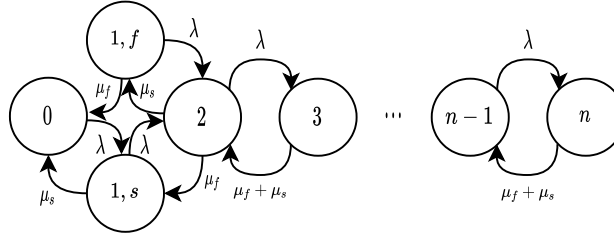


FIGURE A.3. Flow Diagram for Exercise 3

From the flow diagram, we can, by equating the flow out of state n and the flow into state n , $n = 0, 1, \dots$, derive the following balance equations:

$$\begin{aligned}\lambda p_0 &= \mu_f p_{1,f} + \mu_s p_{1,s} \\ (\lambda + \mu_s) p_{1,s} &= \lambda p_0 + \mu_f p_2 \\ (\lambda + \mu_f) p_{1,f} &= \mu_s p_2 \\ \lambda p_n &= \mu p_{n+1} \quad n = 1, 2, \dots\end{aligned}$$

and after we plug in the numerical values for λ , μ_f and μ_s they become

$$4p_0 = 3p_{1,f} + 2p_{1,s} \quad (\text{A.11})$$

$$6p_{1,s} = 4p_0 + 3p_2 \quad (\text{A.12})$$

$$7p_{1,f} = 2p_2 \quad (\text{A.13})$$

$$4p_n = 5p_{n+1} \quad n = 1, 2, \dots \quad (\text{A.14})$$

From equation (A.14) we have that $p_2 = \frac{4}{5}p_1 = \frac{4}{5}(p_{1,f} + p_{1,s})$. By substituting for p_2 in equations (A.12) and (A.13) and solving for $p_{1,f}$ and $p_{1,s}$ we obtain

$$p_{1,f} = \frac{16}{39}p_0, \quad p_{1,s} = \frac{18}{13}p_0$$

and thus $p_1 = \frac{70}{39}p_0$. Then from equation (A.14) we have that

$$p_n = \frac{4}{5}p_{n-1} = \left(\frac{4}{5}\right)^2 p_{n-2} = \dots = \left(\frac{4}{5}\right)^{n-1} p_1 = \left(\frac{4}{5}\right)^{n-1} \cdot \frac{70}{39}p_0.$$

Finally, from the normalisation equation we obtain that

$$1 = p_0 + \frac{70}{39} \sum_{n=1}^{\infty} \left(\frac{4}{5}\right)^{n-1} p_0$$

or, $p_0 = \frac{39}{389}$. Hence the solution to the balance equations is,

$$p_0 = \frac{39}{389}, \quad p_{1,f} = \frac{16}{389}, \quad p_{1,s} = \frac{54}{389}, \quad p_n = \frac{70}{389} \left(\frac{4}{5}\right)^{n-1}, \quad n = 1, 2, \dots$$

(ii) The amount of work per hour done by the experienced server is

$$\begin{aligned}\mu_f \left(p_{1,f} + \sum_{n=2}^{\infty} p_n \right) &= \mu_f (p_{1,f} + 1 - p_1 - p_0) \\ &= 3 \left(\frac{16}{389} + 1 - \frac{70}{389} - \frac{39}{389} \right) = \frac{888}{389}\end{aligned}$$

likewise, the amount of work per hour done by the less-experienced server is

$$\begin{aligned}\mu_s \left(p_{1,s} + \sum_{n=2}^{\infty} p_n \right) &= \mu_s (p_{1,s} + 1 - p_1 - p_0) \\ &= 2 \left(\frac{54}{389} + 1 - \frac{70}{389} - \frac{39}{389} \right) = \frac{668}{389}.\end{aligned}$$

Then the fraction of customers served by the experienced server is equal to

$$\frac{\frac{888}{389}}{\frac{888}{389} + \frac{668}{389}} = \frac{222}{389} \approx 0.57,$$

and the fraction of customers served by the less-experienced server is equal to

$$\frac{\frac{668}{389}}{\frac{888}{389} + \frac{668}{389}} = \frac{167}{389} \approx 0.43.$$

(iii) Let the random variable L^q denote the number of customers in the queue before the arrival of a customer. Then we can write the waiting time of a customer as

$$W = \sum_{j=1}^{L^q} A_j$$

where $A_j \sim \exp(\mu_f + \mu_s) = \exp(5)$. Then the LST of W is

$$\begin{aligned}\tilde{W}(s) &= E(e^{-sW}) \\ &= \sum_{n=0}^{\infty} P(L^q = n) E(e^{-s \sum_{j=1}^{L^q} A_j} \mid L^q = n) \\ &= \sum_{n=0}^{\infty} P(L^q = n) E(e^{-s \sum_{j=1}^n A_j}) \\ &= P(L^q = 0) E(e^0) + \sum_{n=1}^{\infty} P(L^q = n) E(e^{-s \sum_{j=1}^n A_j}) \\ &= (p_0 + p_1 + p_2) + \sum_{n=1}^{\infty} p_{n+2} E(e^{-sA_1}) \cdots E(e^{-sA_n}) \\ &= \frac{165}{389} + \sum_{n=1}^{\infty} \frac{70}{389} \left(\frac{4}{5} \right)^{n+1} \left(\frac{s}{s+5} \right)^n \\ &= \frac{165}{389} + \frac{56}{389} \sum_{n=1}^{\infty} \left(\frac{4}{5} \cdot \frac{s}{s+5} \right)^n \\ &= \frac{165}{389} + \frac{56}{389} \left(\frac{1}{1 - \frac{4}{5} \cdot \frac{s}{s+5}} - 1 \right) \\ &= \frac{165}{389} + \frac{56}{389} \left(\frac{5(s+5)}{5(s+5) - 20} - 1 \right) \\ &= \frac{165}{389} + \frac{224}{389} \cdot \frac{1}{s+1}.\end{aligned}$$

The sojourn time is the sum of the waiting time W and the service time B , and we define

$$B = \begin{cases} X \sim \exp(\mu_f), & z = 1 \\ Y \sim \exp(\mu_s), & z = 0 \end{cases}$$

where $z \sim \text{Ber}(p)$, and $p = \frac{222}{389}$ is the fraction of customers served by the experienced server, computed in b). Then the LST of S is

$$\begin{aligned} \tilde{S}(s) &= E\left(e^{-s(W+B)}\right) \\ &= E\left(e^{-sW}\right) \cdot E\left(e^{-sB}\right) \\ &= E\left(e^{-sW}\right) \left(P(z=1) \frac{\mu_f}{\mu_f + s} + P(z=0) \frac{\mu_s}{\mu_s + s} \right) \\ &= \left(\frac{165}{389} + \frac{224}{389} \cdot \frac{1}{s+1} \right) \left(\frac{222}{389} \cdot \frac{3}{3+s} + \frac{167}{389} \cdot \frac{2}{2+s} \right). \end{aligned}$$

□

EXERCISE A.16. Consider an $M/M/1$ queueing system in which there are three types of customers: very important customers (type 1), important customers (type 2) and less important customers (type 3). Customers of type i , $i = 1, 2, 3$, arrive according to a Poisson process with rate $\lambda_i = 2i$ customers per hour. The service times of all customers are exponentially distributed with a mean of 4 minutes.

- (i) Determine the mean sojourn time of type 1, type 2 and type 3 customers when customers are served according to the following mixed priority rule. Type 1 and type 2 customers may interrupt the processing of a type 3 customer. However, type 1 customers may not interrupt type 2 customers. Compare the performance of this system also with the performance of the system in which all customers are served according to the first come first served rule.
- (ii) Also calculate in both cases (i.e., the mixed priority rule and the first come first served rule) the mean total amount of unfinished work in the system.
- (iii) Which part of this mean total amount of unfinished work in the system is of type 1, of type 2 and of type 3 customers in both cases (i.e., in the system with the mixed priority rule and in the system with the first come first served rule, respectively)?

SOLUTION. As time unit we take 1 minute. Given that the mean service time is 4 minutes, the service rate is $\mu = \frac{1}{4}$. The arrival rates are $\lambda_1 = \frac{1}{30}$, $\lambda_2 = \frac{1}{15}$, and $\lambda_3 = \frac{1}{10}$, respectively.

- (i) Since type 1 and type 2 customers can interrupt the processing of a type 3 customer, but type 1 customers cannot interrupt type 2 customers, the effective arrival rate is $\lambda_{1,2} = \lambda_1 + \lambda_2 = \frac{1}{10}$, while type 3's arrival rate remains $\lambda_3 = \frac{1}{10}$. Then the occupation rates are $\rho_{1,2} = \rho_3 = \frac{2}{5}$. Therefore, the preemptive-resume mean sojourn time for type 3 customers is

$$E(S_3) = \frac{1}{\mu(1 - \rho_{1,2})(1 - \rho_{1,2} - \rho_3)} = \frac{100}{3} \approx 33.33 \text{ minutes.}$$

For type 1 and type 2 customers, we have $\rho_1 = \frac{2}{15}$ and $\rho_2 = \frac{4}{15}$. We calculate the non-preemptive mean sojourn time as follows:

$$\begin{aligned} E(S_1) &= \frac{1 + \rho_2}{\mu(1 - \rho_1)} = \frac{76}{13} \approx 5.85 \text{ minutes,} \\ E(S_2) &= \frac{1 - \rho_1(1 - \rho_1 - \rho_2)}{\mu(1 - \rho_1)(1 - \rho_1 - \rho_2)} = \frac{92}{13} \approx 7.08 \text{ minutes.} \end{aligned}$$

For the first come first served rule, the total arrival rate is $\lambda = \lambda_1 + \lambda_2 + \lambda_3 = \frac{1}{5}$, and the system occupation rate is $\rho = \frac{\lambda}{\mu} = \frac{4}{5}$. Then the mean sojourn time is

$$\mathbb{E}[S] = \frac{1}{\mu(1-\rho)} = 20 \text{ minutes.}$$

(ii) For the priority rule, the total amount of unfinished work is given by

$$\frac{\mathbb{E}[L]}{\mu} = \frac{\rho_{1,2} + \rho_3}{\mu(1-\rho_{1,2}-\rho_3)} = 16.$$

And for the first come first served rule, by the *Little's law*, we have

$$\frac{\mathbb{E}[L]}{\mu} = \frac{\lambda \mathbb{E}[S]}{\mu} = 16.$$

(iii) It follows from (i) that for the priority rule:

$$\begin{aligned} E(L_1) &= \frac{(1+\rho_2)\rho_1}{1-\rho_1} = \frac{38}{195} \approx 0.19, \\ E(L_2) &= \frac{(1-\rho_1(1-\rho_1-\rho_2))\rho_2}{(1-\rho_1)(1-\rho_1-\rho_2)} = \frac{92}{195} \approx 0.47, \\ E(L_3) &= \frac{\rho_3}{(1-\rho_{1,2})(1-\rho_{1,2}-\rho_3)} = \frac{10}{3} \approx 3.33. \end{aligned}$$

Then the mean total amount of unfinished work for the type 1 customer is $\frac{E(L_1)}{\mu} = \frac{152}{195} \approx 0.78$, for the type 2 customer it is $\frac{E(L_2)}{\mu} = \frac{368}{195} \approx 1.89$, and finally it is $\frac{E(L_3)}{\mu} = \frac{40}{3} \approx 13.33$ for the type 3 customer.

For the first come first served rule, we distribute the unfinished work according to proportions:

$$E(L_i) = \mathbb{E}[L] \frac{\lambda_i}{\lambda}, \quad i = 1, 2, 3$$

where $\mathbb{E}[L] = 16\mu = 4$, as in (ii). Then

$$E(L_1) = \frac{2}{3} \approx 0.67, \quad E(L_2) = \frac{4}{3} \approx 1.33, \quad E(L_3) = 2.$$

Then by dividing the mean number of customers in the system with μ , we have the mean total amount of unfinished work to be $\frac{8}{3} \approx 2.67$ for the type 1 customer, $\frac{16}{3} \approx 5.33$ for the type 2 customer and 8 for the type 3 customer.

□

EXERCISE A.17. Consider an $M/G/1$ queue with arrival rate λ and let B denote a generic service time. When the server becomes idle, it automatically switches to stand-by mode. When the server is in standby mode and a customer arrives, there is a setup time T required before the server becomes available again to start servicing. Denote the LST of the service and setup times by $\tilde{B}(s)$ and $\tilde{T}(s)$, respectively. Our aim is to derive the generating function of the number of customers L^d left behind by a departing customer, denoted by $P_{L^d}(z)$.

(i) Make a sketch to verify that L_k^d , the number of customers left behind by the k -th departing customer, satisfies the following recursive relation:

$$L_{k+1}^d = \begin{cases} L_k^d - 1 + A_{k+1}, & \text{if } L_k^d > 0, \\ A_{k+1}^T + A_{k+1}, & \text{if } L_k^d = 0, \end{cases}$$

where A_k and A_k^T are the number of arriving customers during service time B and setup time T , respectively.¹

- (ii) Use the recursive relation in (i) to determine $P_{L^d}(z)$ in terms of the probability of a customer leaving behind an empty system $d_0 = P(L^d = 0)$.

Hint. You may apply similar arguments as in the derivation of $P_{L^d}(z)$ for the regular $M/G/1$ queue. For instance, you may use that the generating function of A_k is $P_A(z) = \tilde{B}(\lambda(1-z))$.

- (iii) Use (ii) and the fact that $P_{L^d}(1) = 1$ to determine d_0 .

SOLUTION. (i) Let L_k^d denote the number of customers in the system just *after* the k -th departure, we make two sketches to illustrate the recursive relation: one that illustrates the case when $L_k^d > 0$ and one where we have $L_k^d = 0$.

In the case where $L_k^d > 0$, we have no setup time. In the figure below, we have $L_k^d = 2$ and $L_{k+1}^d = 2$. During the service time of the $k+1$ -st customer, B_{k+1} , there is one arrival, $A_{k+1} = 1$. Thus we have $L_{k+1}^d = 2 = 2 - 1 + 1 = L_k^d - 1 + A_{k+1}$. In the case where $L_k^d = 0$, we have a positive setup

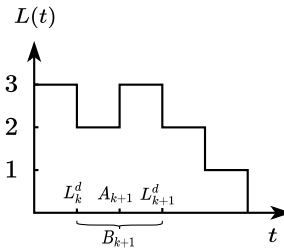


FIGURE A.4. $L_{k+1}^d = L_k^d - 1 + A_{k+1}$ if $L_k^d > 0$

time. In the figure below, we have one arrival during the setup time of the $k+1$ -st customer, $A_{k+1}^T = 1$, and one arrival during the service time of the $k+1$ -st customer, $A_{k+1} = 1$. The $(k+1)$ -st customer leaves behind exactly those two customers when he/she leaves the system. Thus we have $L_{k+1}^d = 2 = 1 + 1 = A_{k+1}^T + A_{k+1}$.

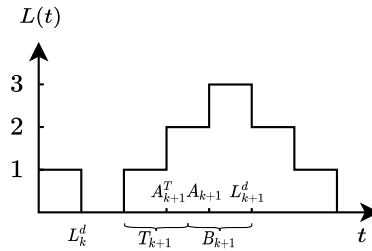


FIGURE A.5. $L_{k+1}^d = A_{k+1}^T + A_{k+1}$ if $L_k^d = 0$

- (ii) We define the probabilities α_n , β_n and γ_n in the following way:

α_n : the probability that during a service time, exactly n customers arrive;

β_n : the probability that during a setup time, exactly n customers arrive;

γ_n : the probability that during a setup plus service time, exactly n customers arrive.

Hence, we have

$$\alpha_n = \int_{t=0}^{\infty} \frac{(\lambda t)^n}{n!} e^{-\lambda t} f_B(t) dt$$

¹ Note that A_k^T can be more formally defined as the number of arrivals during the setup time right before customer k , if there is any setup time, and 0 otherwise.

$$\beta_n = \int_{t=0}^{\infty} \frac{(\lambda t)^n}{n!} e^{-\lambda t} f_T(t) dt$$

and

$$\gamma_n = \sum_{k=0}^n \alpha_k \beta_{n-k}.$$

The transition probability diagram for $\{L_k^d\}_{k \geq 1}$ can be viewed in the figure below. The limiting

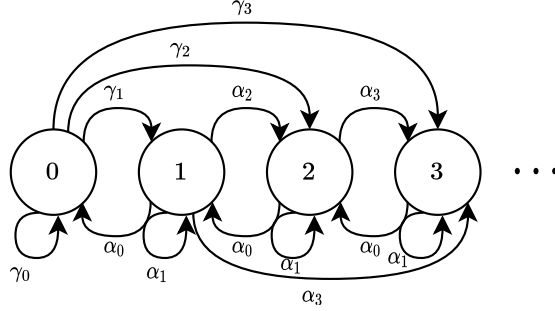


FIGURE A.6. Transition probability diagram for $\{L_k^d\}_{k \geq 1}$

probabilities d_n satisfy the equilibrium equations

$$\begin{aligned} d_n &= d_{n+1}\alpha_0 + d_n\alpha_1 + \cdots + d_1\alpha_n + d_0\gamma_n \\ &= \sum_{k=0}^n d_{n+1-k}\alpha_k + d_0\gamma_n. \end{aligned} \quad (\text{A.15})$$

Let A and A^T denote the number of arrivals during service time B and setup time T , respectively. We introduce the probability generating functions

$$P_{L^d}(z) = \sum_{n=0}^{\infty} d_n z^n, \quad P_A(z) = \sum_{n=0}^{\infty} \alpha_n z^n, \quad P_{A^T}(z) = \sum_{n=0}^{\infty} \beta_n z^n,$$

and

$$\begin{aligned} P_{A+A^T}(z) &= P_A(z)P_{A^T}(z) \\ &= \left(\sum_{n=0}^{\infty} \alpha_n z^n \right) \cdot \left(\sum_{n=0}^{\infty} \beta_n z^n \right) \\ &= \sum_{n=0}^{\infty} \gamma_n z^n. \end{aligned}$$

Now, by multiplying (A.15) by z^n and summing over all n we obtain

$$\begin{aligned} P_{L^d}(z) &= \sum_{n=0}^{\infty} \left(\sum_{k=0}^n d_{n+1-k}\alpha_k + d_0\gamma_n \right) z^n \\ &= \sum_{n=0}^{\infty} \sum_{k=0}^{\infty} d_{n+1-k}\alpha_k + \sum_{n=0}^{\infty} d_0\gamma_n z^n \\ &= z^{-1} \sum_{k=0}^{\infty} \alpha_k z^k \sum_{n=k}^{\infty} d_{n+1-k} z^{n+1-k} + d_0 P_A(z) P_{A+A^T}(z) \\ &= z^{-1} P_A(z) (P_{L^d}(z) - d_0) + d_0 P_A(z) P_{A+A^T}(z). \end{aligned}$$

Hence

$$P_{L^d}(z)(1 - z^{-1}P_A(z)) = d_0 P_A(z) (P_{A^T}(z) - z^{-1})$$

that is,

$$P_{L^d}(z) = \frac{d_0 P_A(z)(P_{A^T}(z) - z^{-1})}{1 - z^{-1} P_A(z)}.$$

Furthermore, by multiplying the numerator and denominator by $-z$ we obtain

$$P_{L^d}(z) = \frac{d_0 P_A(z)(1 - z P_{A^T}(z))}{P_A(z) - z}. \quad (\text{A.16})$$

From the lecture notes, the probability generating function $P_A(z)$ can be written as $P_A(z) = \tilde{B}(\lambda - \lambda z)$. In the same way, the probability generating function $P_{A^T}(z)$ can be written as $P_{A^T} = \tilde{T}(\lambda - \lambda z)$. By substituting these expressions into (A.16), we finally obtain

$$P_{L^d}(z) = \frac{d_0 \tilde{B}(\lambda - \lambda z)(1 - z \tilde{T}(\lambda - \lambda z))}{\tilde{B}(\lambda - \lambda z) - z}.$$

(iii) We have

$$\begin{aligned} 1 &= P_{L^d}(1) \\ &= \lim_{z \rightarrow 1} P_{L^d}(z) \\ &= \lim_{z \rightarrow 1} \frac{d_0 \tilde{B}(\lambda - \lambda z)(1 - z \tilde{T}(\lambda - \lambda z))}{\tilde{B}(\lambda - \lambda z) - z} && (\text{from b)}) \\ &= \lim_{z \rightarrow 1} \frac{(d_0 \tilde{B}(\lambda - \lambda z)(1 - z \tilde{T}(\lambda - \lambda z)))'}{(\tilde{B}(\lambda - \lambda z) - z)'} && (\text{by L'Hôpital's rule}) \\ &= \lim_{z \rightarrow 1} \frac{-\lambda d_0 \tilde{B}'(\lambda - \lambda z) - d_0(\tilde{B}(\lambda - \lambda z)(\tilde{T}(\lambda - \lambda z) - \lambda z \tilde{T}'(\lambda - \lambda z)) - \lambda z \tilde{T}(\lambda - \lambda z) \tilde{B}'(\lambda - \lambda z))}{-\lambda \tilde{B}'(\lambda - \lambda z) - 1} \\ &= \frac{-\lambda d_0 \tilde{B}'(0) - d_0(\tilde{B}(0)(\tilde{T}(0) - \lambda \tilde{T}'(0)) - \lambda \tilde{T}(0) \tilde{B}'(0))}{-\lambda \tilde{B}'(0) - 1} \\ &= \frac{\lambda d_0 \mathbb{E}[B] - d_0(1 + \lambda E(T) + \lambda \mathbb{E}[B])}{\lambda \mathbb{E}[B] - 1} \\ &= \frac{-d_0(1 + \lambda E(T))}{\lambda \mathbb{E}[B] - 1}. \end{aligned}$$

Hence,

$$d_0 = \frac{1 - \lambda \mathbb{E}[B]}{1 + \lambda E(T)}.$$

□

EXERCISE A.18. Consider an $M/G/1$ queue with two types of customers. Customers of type 1 and 2 arrive according to independent Poisson processes with rates $\lambda_1 = 1$ and $\lambda_2 = \frac{1}{2}$. The service times of type 1 customers consist of just a single part which is exponentially distributed with a mean of $\frac{1}{3}$. The service times of type 2 customers consist of two independent parts, the first part is exponentially distributed with a mean of $\frac{1}{2}$, the second part is exponentially distributed with a mean of $\frac{1}{3}$. Assume in this exercise that orders are processed in the order of arrival (FCFS).

- (i) Determine the LST of the waiting time of an arbitrary customer.
- (ii) Determine the waiting time distribution of an arbitrary customer using the LST found in a). Do you find the number of exponential terms of the waiting time distribution surprising? Motivate your answer.

Consider the setting with the two types of customers, but now under a priority scheme. The system imposes the following cost structure: each time unit that a type 1 or type 2 customer spends in the system incurs a cost of 12 or 24, respectively.

- (iii) Determine the mean waiting time of type 1 and type 2 customers, and calculate the long-run average cost, in case type 1 has non-preemptive priority over type 2.

Now, consider the following mixed preemptive/non-preemptive service discipline. Customers of type 1 may interrupt a service time of a type 2 customer during the first part of the service time of the type 2 customer, but not anymore during the second part of the service time of the type 2 customer.

- (iv) Determine the mean sojourn time of type 1 and type 2 customers, and calculate the long-run average cost, for this mixed preemptive/non-preemptive service discipline.
- (v) The operator of the system asks you which policy to use for taking customers into service. Argue what the policy is that minimises the long-run average cost. (It is not required to give a formal proof.)

SOLUTION. (i) We note that the fractions of type 1 and type 2 customers are as follows:

$$\frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{2}{3}, \quad \frac{\lambda_2}{\lambda_1 + \lambda_2} = \frac{1}{3},$$

respectively. To obtain the LST of the waiting time W of an arbitrary customer, we have the relation

$$\tilde{S}(s) = \tilde{W}(s)\tilde{B}(s),$$

that is

$$\tilde{W}(s) = \frac{\tilde{S}(s)}{\tilde{B}(s)}.$$

Hence, to obtain the LST of the waiting time, we need the LST of the sojourn time S and service time B .

Let B_1 and B_2 denote the service times of arbitrary type 1 and type 2 customers, respectively. We have $B_1 \sim \exp(3)$ and $B_2 = X + Y$, where $X \sim \exp(2)$ and $Y \sim \exp(3)$ are the two independent parts of the service time of type 2 customers. Hence we have the LST:

$$\tilde{B}_1(s) = \frac{3}{3+s}, \quad \text{and} \quad \tilde{B}_2(s) = \tilde{X}(s)\tilde{Y}(s) = \frac{2}{2+s} \cdot \frac{3}{3+s}.$$

Now, let B denote the service time of an arbitrary customer,

$$B = \begin{cases} B_1 & \text{if customer is type 1 customer} \\ B_2 & \text{if customer is type 2 customer} \end{cases},$$

that is,

$$B = \begin{cases} B_1 & \text{with probability } \frac{2}{3} \\ B_2 & \text{with probability } \frac{1}{3} \end{cases}.$$

Now, for $t \in \mathbb{R}$, the distribution function of B is given by

$$F_B(t) = \frac{2}{3}F_{B_1}(t) + \frac{1}{3}F_{B_2}(t),$$

where F_{B_1} and F_{B_2} are the distribution functions of B_1 and B_2 , respectively. Likewise, for $t \in \mathbb{R}$, the density function of B is given by

$$f_B(t) = \frac{2}{3}f_{B_1}(t) + \frac{1}{3}f_{B_2}(t),$$

where f_{B_1} and f_{B_2} are the density functions of B_1 and B_2 , respectively. Thus we have

$$\begin{aligned}
 \widetilde{B}(s) &= \int_{x=0}^{\infty} e^{-sx} f_B(x) dx \\
 &= \int_{x=0}^{\infty} e^{-sx} \left(\frac{2}{3} f_{B_1}(x) + \frac{1}{3} f_{B_2}(x) \right) dx \\
 &= \frac{2}{3} \int_{x=0}^{\infty} e^{-sx} f_{B_1}(x) dx + \frac{1}{3} \int_{x=0}^{\infty} e^{-sx} f_{B_2}(x) dx \\
 &= \frac{2}{3} \widetilde{B}_1(s) + \frac{1}{3} \widetilde{B}_2(s) \\
 &= \frac{2}{3} \cdot \frac{3}{3+s} + \frac{1}{3} \cdot \frac{2}{2+s} \cdot \frac{3}{3+s} \\
 &= \frac{2}{3+s} + \frac{2}{(2+s)(3+s)} = \frac{2(2+s) + 2}{(2+s)(3+s)} = \frac{2}{2+s}.
 \end{aligned}$$

Now, by Pollaczek-Khinchin formula (7.7) in the lecture notes, we have

$$\widetilde{S}(s) = \frac{(1-\rho)\widetilde{B}(s)s}{\lambda\widetilde{B}(s) + s - \lambda}.$$

We have $\lambda = \lambda_1 + \lambda_2 = \frac{3}{2}$ and $\widetilde{B}(s)$ is as above. To ensure stability we must have $\rho = \lambda\mathbb{E}[B] = -\lambda\widetilde{B}'(0)$. We have $\widetilde{B}'(s) = -\frac{2}{(2+s)^2}$ and thus $\widetilde{B}'(0) = -\frac{1}{2}$. So $\rho = \frac{3}{4}$. Hence,

$$\begin{aligned}
 \widetilde{S}(s) &= \frac{\frac{1}{4} \cdot \frac{2}{2+s} \cdot s}{\frac{3}{2} \cdot \frac{2}{2+s} + s - \frac{3}{2}} \\
 &= \frac{\frac{s}{2(2+s)}}{\frac{3+s(2+s) - \frac{3}{2}(2+s)}{2+s}} \\
 &= \frac{s}{2(3+s(2+s) - \frac{3}{2}(2+s))} = \frac{1}{2(2+s) - 3} = \frac{\frac{1}{2}}{\frac{1}{2} + s}.
 \end{aligned}$$

Now, we finally obtain,

$$\begin{aligned}
 \widetilde{W}(s) &= \frac{\widetilde{S}(s)}{\widetilde{B}(s)} \\
 &= \frac{\frac{\frac{1}{2}}{\frac{1}{2} + s}}{\frac{2}{2+s}} \\
 &= \frac{\frac{1}{2}}{\frac{1}{2} + s} \cdot \frac{2+s}{2} \\
 &= \frac{\frac{1}{2}}{\frac{1}{2} + s} + \frac{\frac{1}{4}s}{\frac{1}{2} + s}
 \end{aligned}$$

A partial fraction decomposition of the second term of the right hand side of the last equation yields

$$\frac{\frac{1}{4}s}{\frac{1}{2} + s} = \frac{1}{4} \left(1 - \frac{\frac{1}{2}}{s + \frac{1}{2}} \right).$$

Hence we finally obtain

$$\begin{aligned}
 \widetilde{W}(s) &= \frac{\frac{1}{2}}{\frac{1}{2} + s} + \frac{1}{4} \left(1 - \frac{\frac{1}{2}}{s + \frac{1}{2}} \right) \\
 &= \frac{1}{4} + \frac{3}{4} \cdot \frac{\frac{1}{2}}{\frac{1}{2} + s} = \frac{2+s}{2(1+2s)}.
 \end{aligned} \tag{3}$$

(ii) By following *Example 7.4.1* in the lecture notes and using (3), the inverse transform yields

$$F_W(t) = \frac{1}{4} + \frac{3}{4} \left(1 - e^{-\frac{1}{2}t}\right), \quad t \geq 0.$$

Hence, with probability $\frac{1}{4}$, the waiting time is zero and the system is empty on arrival. Given that the waiting time is positive, the waiting time is exponentially distributed with parameter $\frac{1}{2}$. Given the complexity of the queueing model and the fact that the service time of a type 2 customer is the sum of two independent exponentially distributed parts, the simplicity of the waiting-time distribution of an arbitrary customer may initially be surprising. However, we notice that under the assumption of FCFS, we can blend the two types of customers, use the effective service rate and get

$$\begin{aligned} \mathbb{E}[B] &= E(B_1) + E(B_2) \\ &= \frac{\lambda_1}{\lambda_1 + \lambda_2} \cdot \left(\frac{1}{\mu_1}\right) + \frac{\lambda_2}{\lambda_1 + \lambda_2} \cdot \left(\frac{1}{\mu_{2,1}} + \frac{1}{\mu_{2,2}}\right) \\ &= \frac{2}{3} \cdot \left(\frac{1}{3}\right) + \frac{1}{3} \cdot \left(\frac{1}{2} + \frac{1}{3}\right) \\ &= \frac{3}{18} + \frac{5}{18} = \frac{1}{2}, \end{aligned}$$

which is a simplified mean service time ($\mu_{2,1}$ and $\mu_{2,2}$ represent the service rates of the first and second parts of a type 2 service time). This indicates that the superposition of Poisson processes and exponential service distributions tends to yield a simplified aggregate behaviour, which aligns with our finding of the smaller number of exponential terms of the waiting time distribution.

(iii) From (9.1) and (9.3) in the lecture notes we have

$$E(W_1) = \frac{\rho_1 E(R_1) + \rho_2 E(R_2)}{1 - \rho_1}, \quad E(W_2) = \frac{\rho_1 E(R_1) + \rho_2 E(R_2)}{(1 - \rho_1 - \rho_2)(1 - \rho_1)}.$$

We have $\rho_1 = \lambda_1 E(B_1) = -\lambda_1 \tilde{B}'_1(0)$. Now, $\tilde{B}'_1(s) = -\frac{3}{(3+s)^2}$ and thus $\tilde{B}'_1(0) = -\frac{1}{3}$. So $\rho_1 = 1 \cdot \frac{1}{3} = \frac{1}{3}$. Likewise, $\rho_2 = \lambda_2 E(B_2) = -\lambda_2 \tilde{B}'_2(0)$. We have $\tilde{B}'_2(s) = -\frac{6(2s+5)}{(2+s)^2(3+s)^2}$ and thus $\tilde{B}'_2(0) = -\frac{5}{6}$. Thus $\rho_2 = \frac{1}{2} \cdot \frac{5}{6} = \frac{5}{12}$.

Since $B_1 \sim \exp(3)$, we have $E(R_1) = \frac{1}{3}$ by the memoryless property of the exponential distribution. For the residual service time of type 2 customers, we have $E(R_2) = \frac{E(B_2^2)}{2E(B_2)}$. We have $E(B_2) = \frac{5}{6}$ from our computations above. We obtain the second moment in a similar way, $E(B_2^2) = \tilde{B}''_2(0)$. We have

$$\tilde{B}''_2(s) = \frac{12}{(3+s)^3(2+s)} + \frac{12}{(3+s)^2(2+s)^2} + \frac{12}{(3+s)(2+s)^3},$$

and thus $\tilde{B}''_2(0) = \frac{12}{3^3 \cdot 2} + \frac{12}{3^2 \cdot 2^2} + \frac{12}{3 \cdot 2^3} = \frac{19}{18}$. So $E(R_2) = \frac{\frac{19}{18}}{2 \cdot \frac{5}{6}} = \frac{19}{30}$.

Thus we finally obtain

$$\begin{aligned} E(W_1) &= \frac{\frac{1}{3} \cdot \frac{1}{3} + \frac{5}{12} \cdot \frac{19}{30}}{1 - \frac{1}{3}} = \frac{9}{16} \text{ time units,} \\ E(W_2) &= \frac{\frac{1}{3} \cdot \frac{1}{3} + \frac{5}{12} \cdot \frac{19}{30}}{\left(1 - \frac{1}{3} - \frac{5}{12}\right)\left(1 - \frac{1}{3}\right)} = \frac{9}{4} \text{ time units.} \end{aligned}$$

Now, we will compute the long-run average cost per time unit in the system. We let c_1 and c_2 denote the long-run average cost per type 1 and type 2 customer, respectively. Since the length of the queue does not grow towards infinity, we have that the long-run average cost per time unit equals the average number of type 1 customers that arrive per time unit times c_1 , plus the average number of type 2 customers that arrive per time unit times c_2 , that is

$$\text{Long-run average cost per time unit} = \lambda_1 c_1 + \lambda_2 c_2.$$

To compute c_1 and c_2 , we need the mean sojourn time of both types of customers. We have

$$\begin{aligned} E(S_1) &= E(W_1) + E(B_1) = \frac{9}{16} + \frac{1}{3} = \frac{43}{48} \text{ time units,} \\ E(S_2) &= E(W_2) + E(B_2) = \frac{9}{4} + \frac{5}{6} = \frac{37}{12} \text{ time units.} \end{aligned}$$

The average cost of a type 1 customer is then $c_1 = E(S_1) \cdot 12 = 10.75$ and the average cost of a type 2 customer is $c_2 = E(S_2) \cdot 24 = 74$. The long-run average cost per time unit is therefore

$$\lambda_1 c_1 + \lambda_2 c_2 = 1 \cdot 10.75 + \frac{1}{2} \cdot 74 = 47.75.$$

- (iv) We have $B_2 = X + Y$, where $X \sim \exp(2)$ and $Y \sim \exp(3)$ are the two independent parts of the service time of type 2 customers. Then for type 2 customers, we have the occupation rates:

$$\rho_{2,x} = \lambda_2 E(X) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}, \quad \rho_{2,y} = \lambda_2 E(Y) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}.$$

Again, from (9.1) and (9.3) in the lecture notes, we calculate:

$$\begin{aligned} E(W_1) &= \frac{\rho_1 E(R_1) + \rho_{2,y} E(R_{2,y})}{1 - \rho_1} \\ &= \frac{\rho_1 E(R_1) + \rho_{2,y} \frac{E(Y^2)}{2E[Y]}}{1 - \rho_1} \\ &= \frac{\frac{1}{3} \cdot \frac{1}{3} + \frac{1}{6} \cdot \frac{\frac{2}{3}}{2 \cdot \frac{1}{3}}}{1 - \frac{1}{3}} \\ &= \frac{1}{4} \text{ time units,} \\ E(W_2) &= \frac{\rho_1 E(R_1) + \rho_2 E(R_2)}{(1 - \rho_1 - \rho_2)(1 - \rho_1)} = \frac{9}{4} \text{ time units.} \end{aligned}$$

Then by (9.5) in the lecture notes, we have the mean sojourn time:

$$\begin{aligned} E(S_1) &= E(W_1) + E(B_1) = \frac{1}{4} + \frac{1}{3} = \frac{7}{12} \text{ time units,} \\ E(S_2) &= E(W_2) + \frac{E(X)}{1 - \rho_1} + E(Y) = \frac{9}{4} + \frac{\frac{1}{2}}{1 - \frac{1}{3}} + \frac{1}{3} = \frac{10}{3} \text{ time units.} \end{aligned}$$

Therefore, the long-run average cost per customer is

$$C = 12\lambda_1 \cdot E(S_1) + 24\lambda_2 \cdot E(S_2) = 12 \cdot \left(\frac{7}{12} + \frac{10}{3} \right) = 47.$$

- (v) Under the non-preemptive-priority policy for type 1 customers, the calculated long-run average cost per time unit is 47.75, whereas under the mixed policy it is 47, which is more cost-effective. From the $c\mu$ rule, we have

$$f_1 = \frac{c_1}{E(B_1)} = \frac{12}{\frac{1}{3}} = 36, \quad f_2 = \frac{c_2}{E(B_2)} = \frac{24}{\frac{6}{5}} = 20,$$

where $f_1 > f_2$, and we should prioritise customer types with the highest f_i in order to minimise C . Thus, the operator of the system should consider implementing the mixed preemptive/non-preemptive service discipline to minimise costs. $\square$

EXERCISE A.19. File requests arrive at a transmission link (which acts as a single server) according to a Poisson process with rate $\frac{1}{3}$. The time required to process a file follows the following hyperexponential distribution

$$\mathbb{P}(B < x) = \frac{1+a}{2} \left(1 - e^{-\frac{1+a}{2}x}\right) + \frac{1-a}{2} \left(1 - e^{-\frac{1-a}{2}x}\right), \quad x \geq 0,$$

with $0 \leq a < 1$.

- (i) Suppose that files are served in the order of arrival. Determine the expected waiting time in terms of a . Give the expected waiting time for $a = 0$, $a = \sqrt{\frac{1}{2}}$, and $a = \sqrt{\frac{3}{4}}$. What happens when $a \rightarrow 1$. Give an intuitive explanation for the differences.
- (ii) Suppose that the service order is according to the SPTF service discipline. Consider a file with transmission time x . Determine its mean waiting time $E(W(x))$ in terms of a (and x).
- (iii) Let $a = 0$. Numerically compute the mean overall waiting time for the FCFS and SPTF disciplines under (i) and (ii), and compare the results. Do the same for $a = \{0.5, 0.8, 0.95\}$ and elaborate on the differences.

SOLUTION. (i) Given the distribution of the service time

$$\mathbb{P}(B < x) = \frac{1+a}{2} \left(1 - e^{-\frac{1+a}{2}x}\right) + \frac{1-a}{2} \left(1 - e^{-\frac{1-a}{2}x}\right), \quad x \geq 0,$$

with $0 \leq a < 1$ and the arrival rate $\lambda = \frac{1}{3}$, we calculate the mean service time

$$\mathbb{E}[B] = \frac{1+a}{2} \cdot \frac{2}{1+a} + \frac{1-a}{2} \cdot \frac{2}{1-a} = 2.$$

Since for an exponentially distributed random variable X with rate ξ^2 , the second moment is given by $E(X^2) = \frac{2}{\xi^2}$, for the hyperexponential distribution we have

$$E(B^2) = \frac{1+a}{2} \cdot \frac{2}{\left(\frac{1+a}{2}\right)^2} + \frac{1-a}{2} \cdot \frac{2}{\left(\frac{1-a}{2}\right)^2} = \frac{4}{1+a} + \frac{4}{1-a}.$$

Then the occupation rate is $\rho = \lambda \mathbb{E}[B] = \frac{2}{3}$, and the mean residual service time

$$\mathbb{E}[R] = \frac{E(B^2)}{2\mathbb{E}[B]} = \frac{\frac{4}{1+a} + \frac{4}{1-a}}{4} = \frac{1}{1+a} + \frac{1}{1-a}.$$

So by Pollaczek-Khinchin mean value formula (7.14), the expected waiting time is determined by

$$\mathbb{E}[W] = \frac{\rho \mathbb{E}[R]}{1-\rho} = \frac{\frac{2}{3} \left(\frac{1}{1+a} + \frac{1}{1-a} \right)}{1 - \frac{2}{3}} = \frac{4}{1-a^2}.$$

For specific values of a , we have

$$\begin{aligned} a = 0 : \quad \mathbb{E}[W] &= 4, \\ a = \sqrt{\frac{1}{2}} : \quad \mathbb{E}[W] &= 8, \\ a = \sqrt{\frac{3}{4}} : \quad \mathbb{E}[W] &= 16. \end{aligned}$$

Therefore, as $a \rightarrow 1$, the variance of the service times increases substantially and $\mathbb{E}[W] \Rightarrow \infty$, as systems with high variability in service times typically have longer queues and waiting times.

- (ii) Given that

$$F_B(x) = \frac{1+a}{2} \left(1 - e^{-\frac{1+a}{2}x}\right) + \frac{1-a}{2} \left(1 - e^{-\frac{1-a}{2}x}\right), \quad x \geq 0,$$

with $0 \leq a < 1$, we differentiate $F_B(x)$ with respect to x and get

$$f_B(x) = \frac{(1+a)^2}{4} e^{-\frac{1+a}{2}x} + \frac{(1-a)^2}{4} e^{-\frac{1-a}{2}x}, \quad x \geq 0.$$

Then we compute

$$\begin{aligned} \int_{y=0}^{y=x} y f_B(y) dy &= \int_{y=0}^{y=x} y \left(\frac{(1+a)^2}{4} e^{-\frac{1+a}{2}y} + \frac{(1-a)^2}{4} e^{-\frac{1-a}{2}y} \right) dy \\ &= \frac{(1+a)^2}{4} \int_{y=0}^{y=x} y e^{-\frac{1+a}{2}y} dy + \frac{(1-a)^2}{4} \int_{y=0}^{y=x} y e^{-\frac{1-a}{2}y} dy \\ &= \frac{(1+a)^2}{4} \cdot \frac{-2e^{-\frac{x(1+a)}{2}}(ax+x+2)+4}{(1+a)^2} + \frac{(1-a)^2}{4} \cdot \frac{-2e^{-\frac{x(1-a)}{2}}(-ax+x+2)+4}{(1-a)^2} \\ &= \frac{\left(-2e^{-\frac{x(1+a)}{2}}(ax+x+2)+4\right) + \left(-2e^{-\frac{x(1-a)}{2}}(-ax+x+2)+4\right)}{4} \\ &= -\frac{e^{-\frac{x(1+a)}{2}}(ax+x+2) + e^{-\frac{x(1-a)}{2}}(-ax+x+2)}{2} + 2 \end{aligned}$$

for $0 \leq a < 1$. Since the service order is according to the SPTF service discipline, we have

$$\begin{aligned} E(W(x)) &= \frac{\rho \mathbb{E}[R]}{\left(1 - \lambda \int_{y=0}^{y=x} y f_B(y) dy\right)^2} \\ &= \frac{\frac{2}{3} \left(\frac{1}{1+a} + \frac{1}{1-a}\right)}{\left(1 - \frac{1}{3} \left(-\frac{e^{-\frac{x(1+a)}{2}}(ax+x+2)+e^{-\frac{x(1-a)}{2}}(-ax+x+2)}{2} + 2\right)\right)^2} \\ &= \frac{\frac{2}{3} \left(\frac{1}{1+a} + \frac{1}{1-a}\right)}{\left(\frac{1}{3} + \frac{e^{-\frac{x(1+a)}{2}}(ax+x+2)+e^{-\frac{x(1-a)}{2}}(-ax+x+2)}{6}\right)^2} \\ &= \frac{48}{(1-a^2) \left(e^{-\frac{x(1+a)}{2}}(ax+x+2) + e^{-\frac{x(1-a)}{2}}(-ax+x+2) + 2\right)^2}, \end{aligned}$$

which is the mean waiting time $E(W(x))$ in terms of $0 \leq a < 1$ and transmission time x .

(iii) For the SPTF service discipline, the mean overall waiting time is given by

$$\begin{aligned} \mathbb{E}[W] &= \rho \mathbb{E}[R] \int_{x=0}^{\infty} \frac{f_B(x)}{\left(1 - \lambda \int_{y=0}^{y=x} y f_B(y) dy\right)^2} dx \\ &= \frac{2}{3} \left(\frac{1}{1+a} + \frac{1}{1-a}\right) \int_{x=0}^{\infty} \frac{\frac{(1+a)^2}{4} e^{-\frac{1+a}{2}x} + \frac{(1-a)^2}{4} e^{-\frac{1-a}{2}x}}{\left(1 - \frac{1}{3} \left(-\frac{e^{-\frac{x(1+a)}{2}}(ax+x+2)+e^{-\frac{x(1-a)}{2}}(-ax+x+2)}{2} + 2\right)\right)^2} dx \\ &= \frac{4}{3(1-a^2)} \int_{x=0}^{\infty} \frac{\frac{(1+a)^2}{4} e^{-\frac{1+a}{2}x} + \frac{(1-a)^2}{4} e^{-\frac{1-a}{2}x}}{\left(\frac{1}{3} + \frac{e^{-\frac{x(1+a)}{2}}(ax+x+2)+e^{-\frac{x(1-a)}{2}}(-ax+x+2)}{6}\right)^2} dx \\ &= \frac{12}{1-a^2} \int_{x=0}^{\infty} \frac{e^{-\frac{x(1+a)}{2}}(1+a)^2 + e^{-\frac{x(1-a)}{2}}(1-a)^2}{\left(e^{-\frac{x(1+a)}{2}}(ax+x+2) + e^{-\frac{x(1-a)}{2}}(-ax+x+2) + 2\right)^2} dx. \end{aligned}$$

We use the following Python code to numerically compute the mean overall waiting time for the FCFS and SPTF disciplines for $a = \{0, 0.5, 0.8, 0.95\}$.

```

import numpy as np
from scipy.integrate import quad

def fcfs(a):
    return 4 / (1 - a**2)

def integrand(x, a):
    numerator = (np.exp(-x * (1 + a) / 2) * (1 + a)**2 + np.exp(-x * (1 -
↪ a) / 2) * (1 - a)**2)
    denominator = (np.exp(-x * (1 + a) / 2) * (a * x + x + 2) + np.exp(-x *
↪ (1 - a) / 2) * (-a * x + x + 2) + 2)**2
    return numerator / denominator

def sptf(a):
    integral, _ = quad(integrand, 0, np.inf, args=(a,))
    return 12 / (1 - a**2) * integral

a_values = [0, 0.5, 0.8, 0.95]

for a in a_values:
    print(f"For a = {a}: FCFS Result = {fcfs(a)}, "
          f"SPTF Result = {sptf(a)}")

```

And one output as follows.

```

For a = 0: FCFS Result = 4.0, SPTF Result = 2.359629796373805
For a = 0.5: FCFS Result = 5.333333333333333, SPTF Result =
↪ 2.8835873520275483
For a = 0.8: FCFS Result = 11.111111111111114, SPTF Result =
↪ 5.223171632393134
For a = 0.95: FCFS Result = 41.025641025641015, SPTF Result =
↪ 17.45433539022651

```

□

References

- [1] Ivo Adan and Jacques Resing. Queueing systems. Course notes, Eindhoven University of Technology, Eindhoven, The Netherlands, March 2015. Department of Mathematics and Computing Science, version dated March 26, 2015.